

KSTAR Diagnostics Data User Guide

4 Mar 2026

Before read this guide...

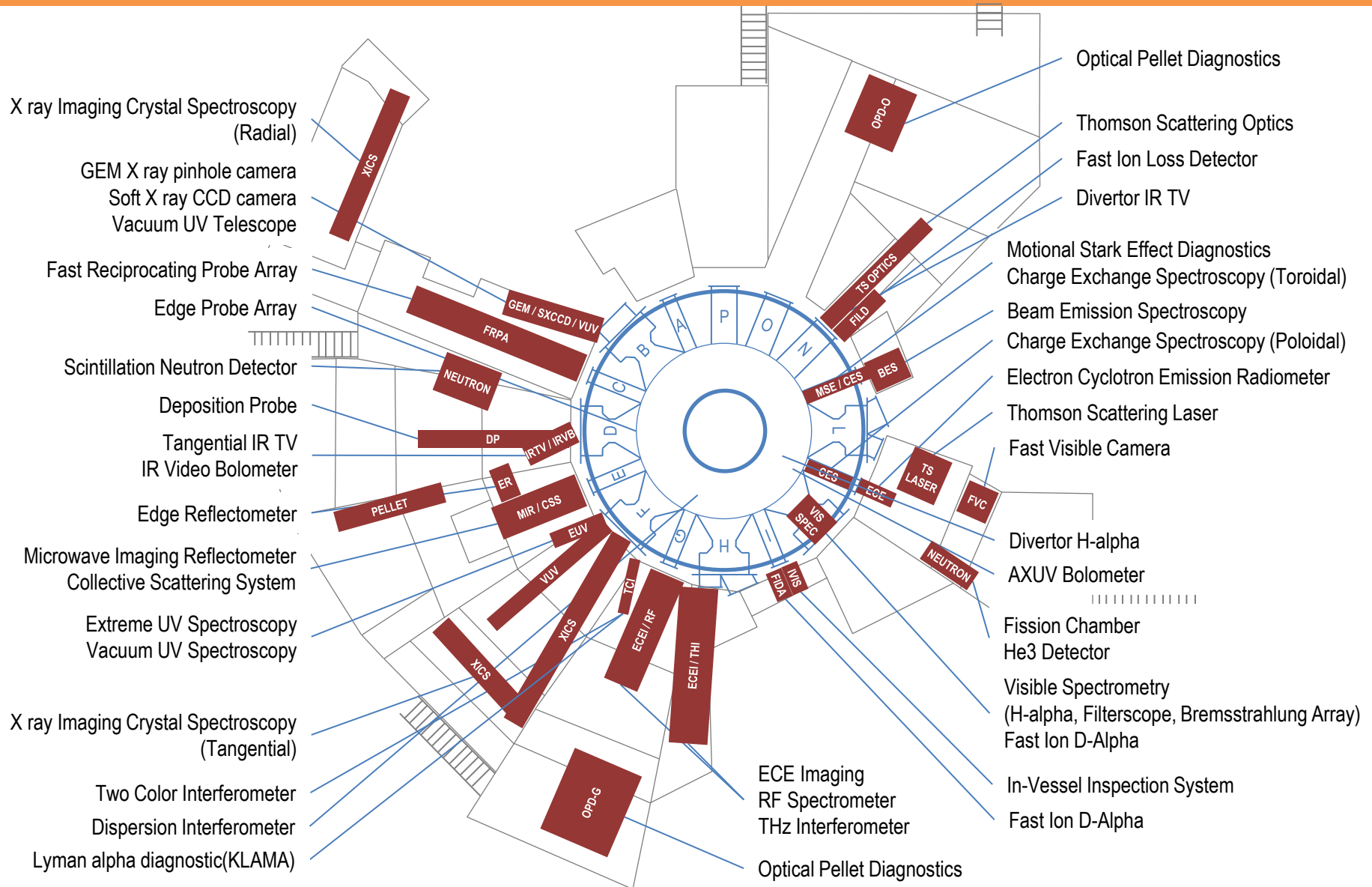
General Information

- Diagnostics data uploaded on KSTAR MDS+ should be used for the preliminary analysis only. Some of data are automatically uploaded by data acquisition systems without final confirmation from person in charge.
- Please consult to the contact person in this guide for the detail information on the diagnostics and for the in-depth analysis.
- Before a publication using KSTAR experimental results, all data included on the publication **MUST** be confirmed by person in charge.

- Magnetic Diagnostics
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- Electric Probe Diagnostics - [FRPA/ELPA](#), [TCD LP](#)
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- [Fusion Neutron Diagnostics](#) / [HXR Monitor](#)
- [ECEI](#) / [RF spectrometer](#) / [MIR](#) / [CSS](#) / [ER](#)
- [KLAMA](#)

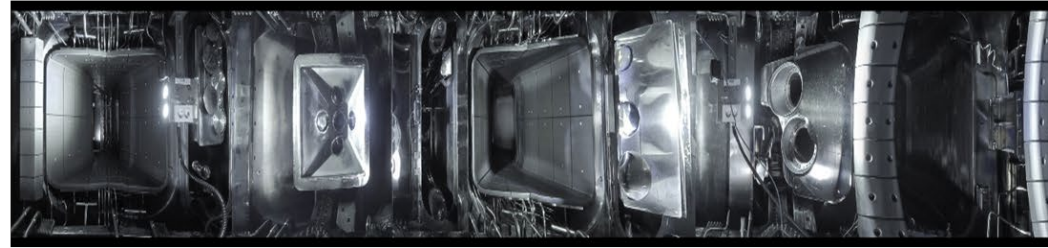
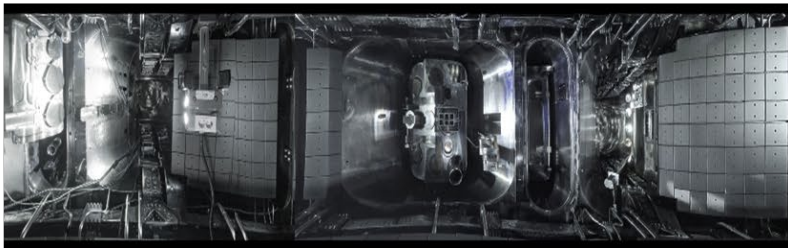
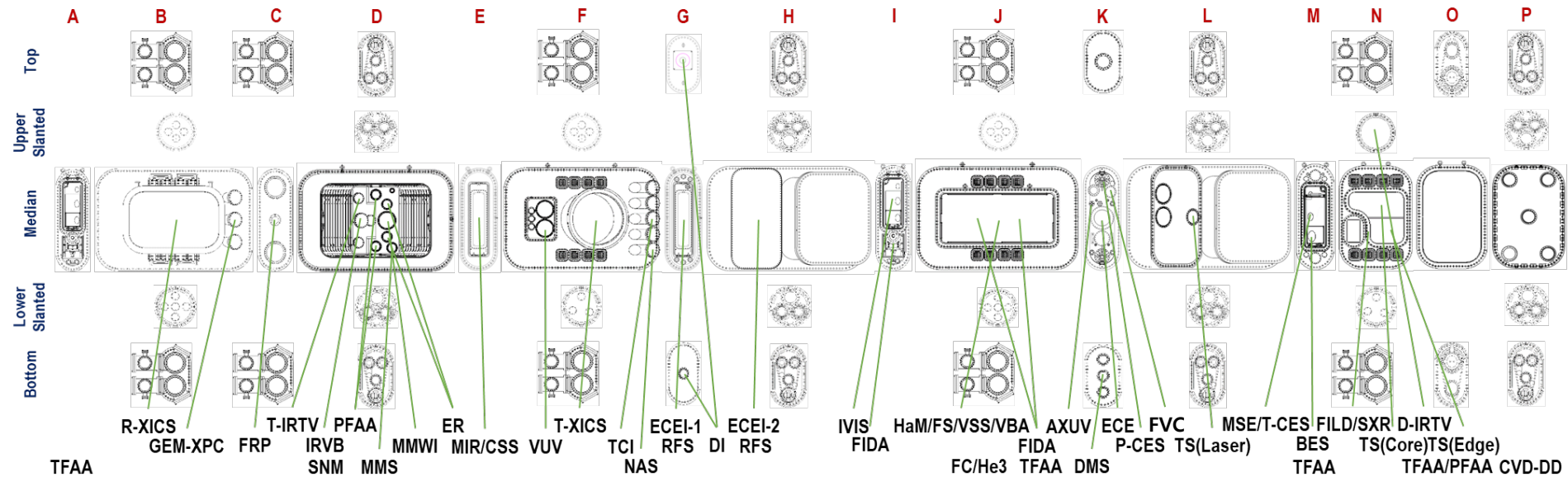
KSTAR Diagnostics

General Information



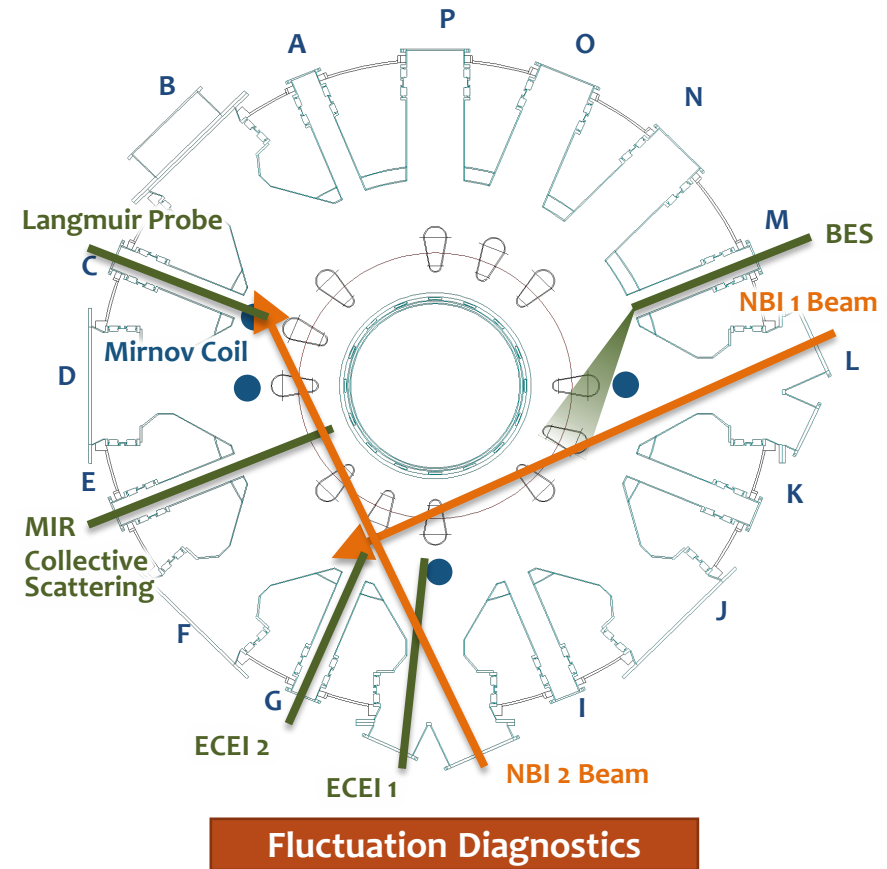
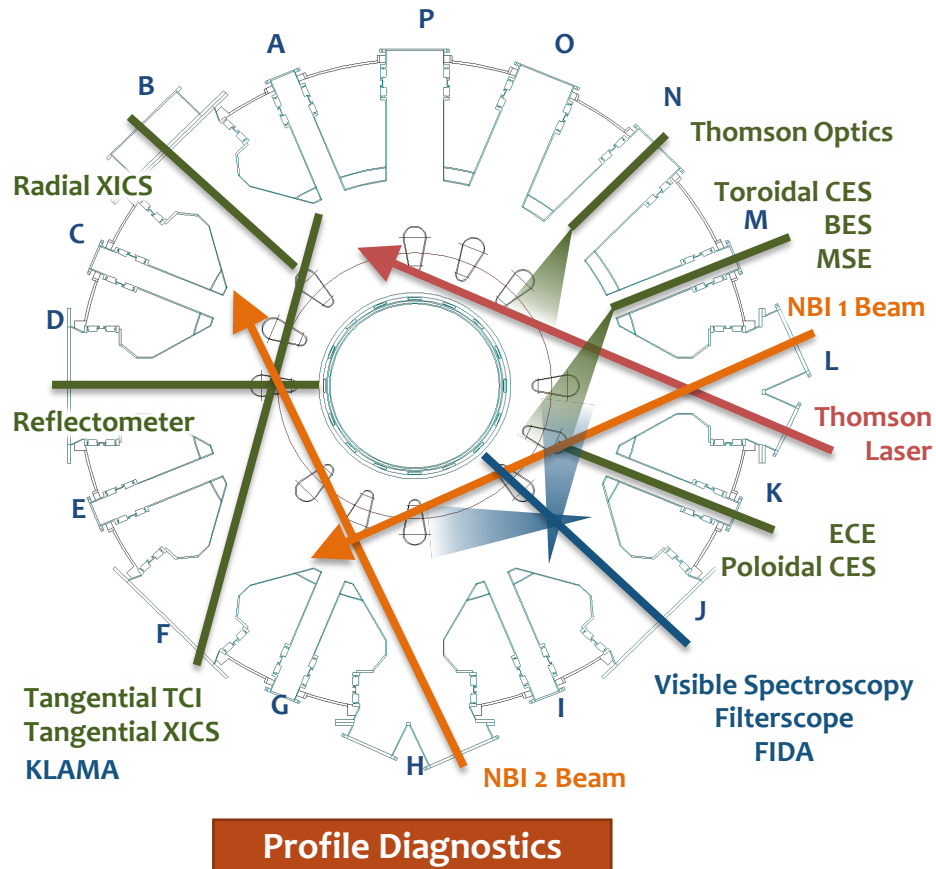
Diagnostics Port Allocation

General Information



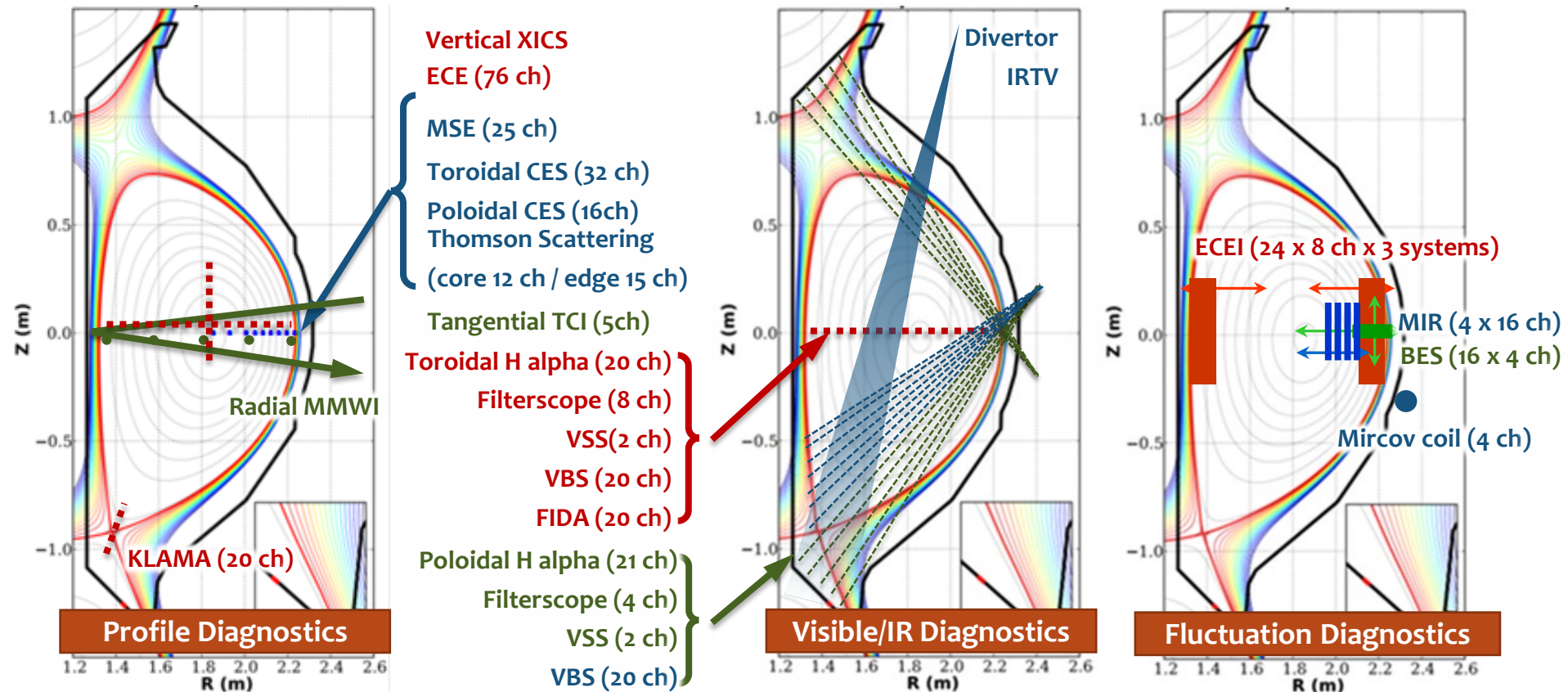
Diagnostics Layouts : Toroidal

General Information



Diagnostics Layouts : Poloidal

General Information



Diagnostics Device Status WebPage

General Information

- Daily Status of KSTAR Diagnostics can be checked on the following KSTAR website.

<https://kstar.kfe.re.kr/diagnostics?mid=a12000000000>

* sign up required. contact H.M. Wi (hanmin@kfe.re.kr)

KSTAR
Logout My Page

공지·알림 Proposal ShotPlan **Diagnostic Device Status** Device Maintenance Schedule Experiment Operator Experimental Schedule Shot briefing & Summary Download Old Proposal Old ShotPlan & Summary

Diagnostic Device Status

Diagnostic Device Status

Status by device
OK
LIMITED
STOP
N.A
Save

DL OK	FL / LV OK	LM / SL OK	RC / VCM OK	HCM OK	MP OK	MC OK	FRPA STOP	FPA STOP
TS STOP	ECE OK	TCI STOP	THI STOP	DI STOP	ER STOP			
CES OK	MSE STOP	BES STOP	XICS STOP	ECEI STOP	MIR STOP	CSS STOP	RF STOP	
H-alpha STOP	VSS STOP	FS STOP	VBS STOP	VUV STOP	FILD LIMITED	FIDA OK		
IVIS OK	FVC STOP	Sur IR STOP	Div IR STOP	IRVB STOP	AXUV STOP	FAA STOP	FC / He3 STOP	SND STOP
								HXR STOP

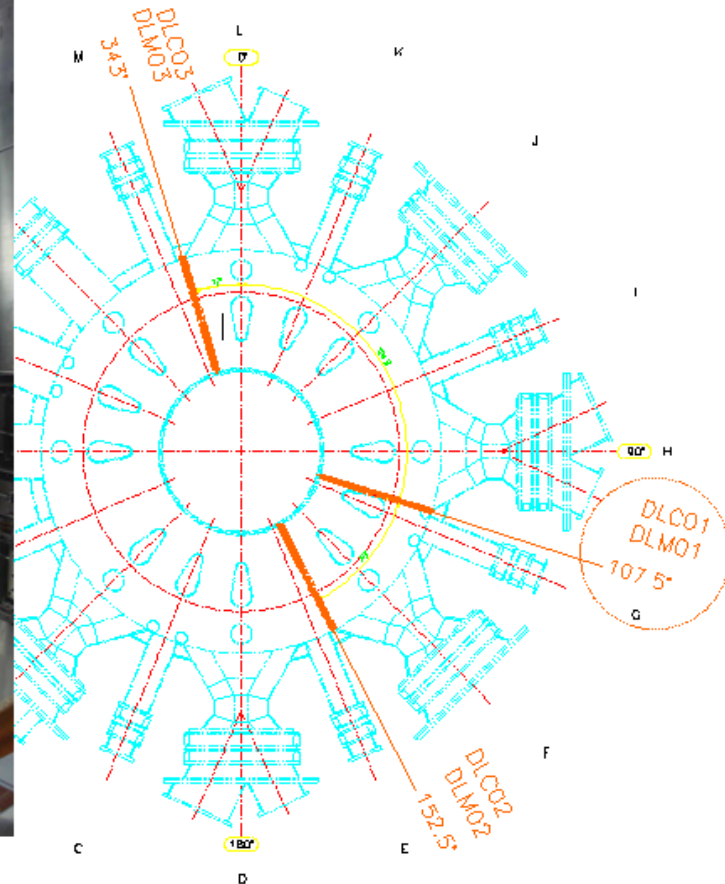
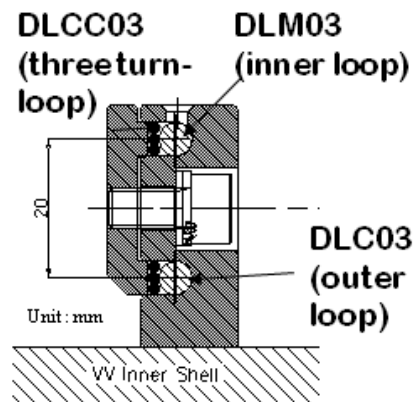
Magnetic Diagnostics - DL

General Information

Full Name	Diamagnetic Loop
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	diamagnetic flux (Φ_d), poloidal beta (β_p), stored energy (W_{tot})
Port assignment	loop (E, G, M), feedthrough (B, F, J, N)
Channel information	3 loops
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

Magnetic Diagnostics - DL

Channels Layout



Magnetic Diagnostics - DL

MDS+ Node Information

Name	Sampling	Unit	Description
\DLMo1,\DLMo2,\DLMo3	20 kHz	V	raw data
\DLCo1,\DLCCo2,\DLCCo3	20 kHz	V	raw data
\DMF_DLMo3	20 kHz	Wb	diamagnetic flux
\BETAP_DLMo3	20 kHz		poloidal beta
\WTOT_DLMo3	20 kHz	kJ	stored energy
\BETAP_kappa			poloidal beta via RTEFIT
\WTOT_kappa			stored energy via RTEFIT

Notes

- \BETAP_kappa and \WTOT_kappa are evaluated by using both DL measurements and kappa from the EFITRT1 code.

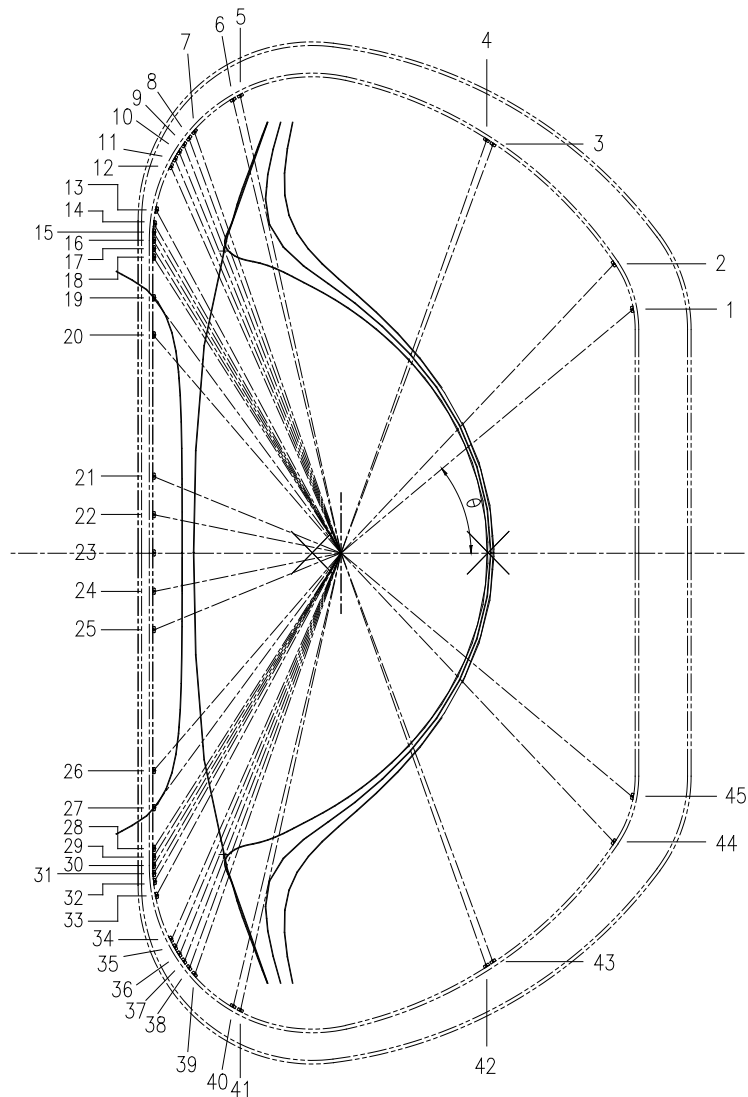
Magnetic Diagnostics – FL / LV

General Information

Full Name	Flux loop / Loop voltage
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	poloidal flux (Φ_{pol}), loop voltage (V_l)
Port assignment	feedthrough (B, F, J, N)
Channel information	45 loops
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

Magnetic Diagnostics - FL / LV

Channels Layout



Magnetic Diagnostics - FL / LV

MDS+ Node Information

Name	Sampling	Unit	Description
\FL01 ... \FL45	20 kHz	Wb	poloidal flux
\LV01, \LV12, \LV23, \LV34, \LV45	20 kHz	V	loop voltage

Notes

-

Magnetic Diagnostics – LM / SL

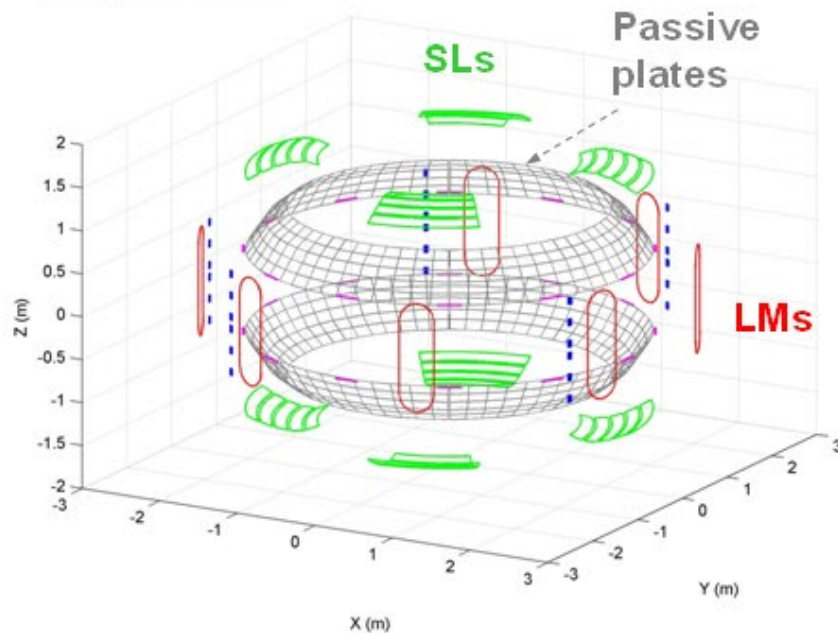
General Information

Full Name	Locked Mode coils / Saddle loop
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	LM : local radial magnetic field (Φ_r) SL : local vertical magnetic field (Φ_v)
Port assignment	feedthrough (B, F, J, N)
Channel information	LM : 4 coils SL : 40 loops
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

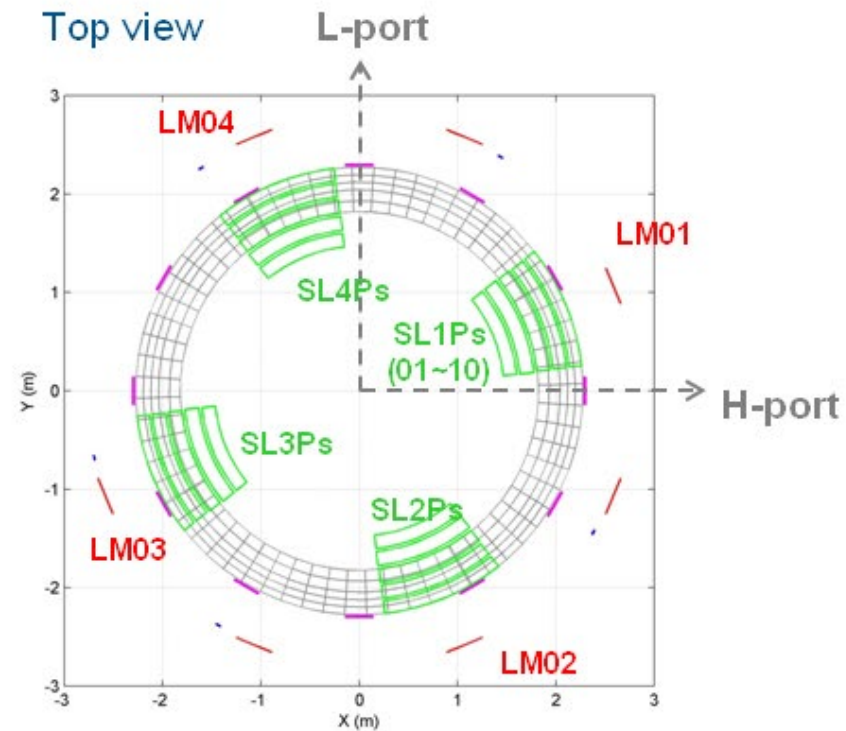
Magnetic Diagnostics - LM / SL

Channels Layout

Diagonal view



Top view



Magnetic Diagnostics - LM / SL

MDS+ Node Information

Name	Sampling	Unit	Description
\LMo1 ... \LMo4	20 kHz	Wb	Local radial magnetic flux at midplane.
\SL1Po1 ... \SL1P10	20 kHz	Wb	Local vertical magnetic flux at the toroidal angle of 67.5 Degr.
\SL2Po1 ... \SL2P10	20 kHz	Wb	Local vertical magnetic flux at the toroidal angle of 157.5 Degr.
\SL3Po1 ... \SL3P10	20 kHz	Wb	Local vertical magnetic flux at the toroidal angle of 247.5 Degr.
\SL4Po1 ... \SL4P10	20 kHz	Wb	Local vertical magnetic flux at the toroidal angle of 337.5 Degr.

Notes

- Local radial magnetic field can be obtained as $B_r = \frac{1}{r} \cdot \frac{d\Phi}{d\theta}$ where ALM is effective area of LM (ALM= 0.4603 m²).

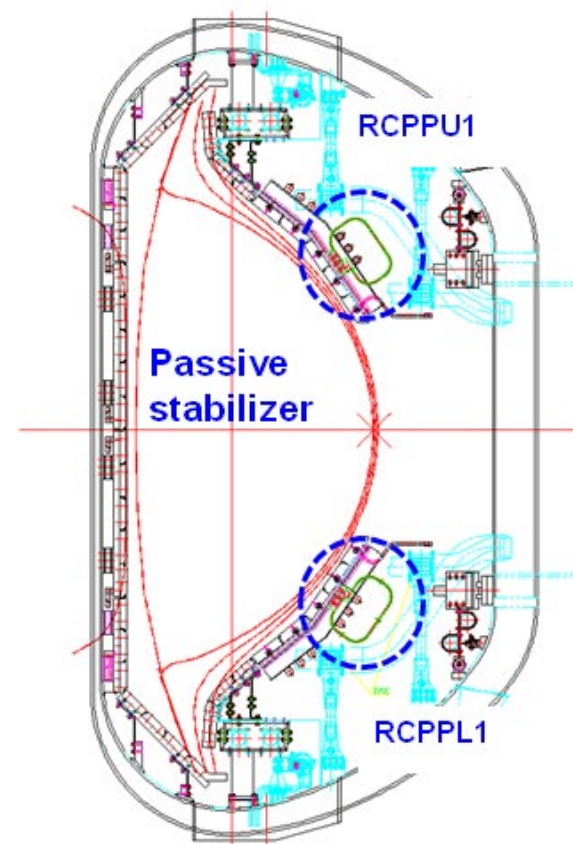
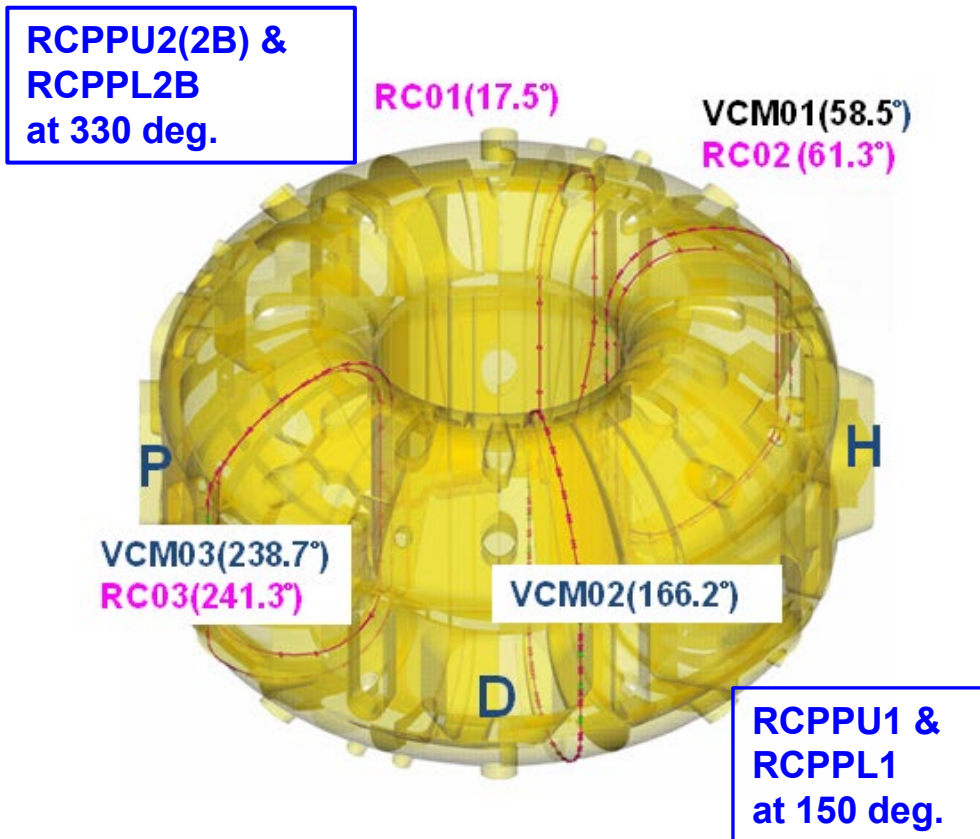
Magnetic Diagnostics – RC / VCM

General Information

Full Name	Rogowski Coil, Vessel Current Monitor
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	RC : plasma current (I_p), eddy current at passive stabilizer (I_{eddy}) VCM : total current (I_{tot})
Port assignment	feedthrough (B, F, J, N, Cb)
Channel information	RC : 3 coils, VCM : 3 loops
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

Magnetic Diagnostics - RC / VCM

Channels Layout



Poloidal cross-section at the location of the E-port

Magnetic Diagnostics - RC / VCM

MDS+ Node Information

Name	Sampling	Unit	Description
\RC03	20 kHz	A	Plasma current
\VCM03	20 kHz	A	Total toroidal current
\RCPPU1, \RCPPU2:FOO, \RCPPU2B:FOO	20 kHz	A	Toroidal eddy current at backplate for upper passive stabilizer
\RCPPL1, \RCPPL2B:FOO	20 kHz	A	Toroidal eddy current at backplate for lower passive stabilizer
\RC02,\RC01	20kHz	A	Plasma current
\VCM02, \VCM01	20kHz	A	Total toroidal current

Notes

- \VCM03 means total current, plasma current plus current induced at vacuum vessel during a plasma discharge. The vessel current I_v can be obtained as $I_v = I_{tot} - I_p$
- Three RCs and VCMs were installed at different toroidal locations (see figure). Now All of RC and VCM can provide data at the MDSplus. \RCPPU2,\RCPPU2B,\RCPPL2B are available as raw data

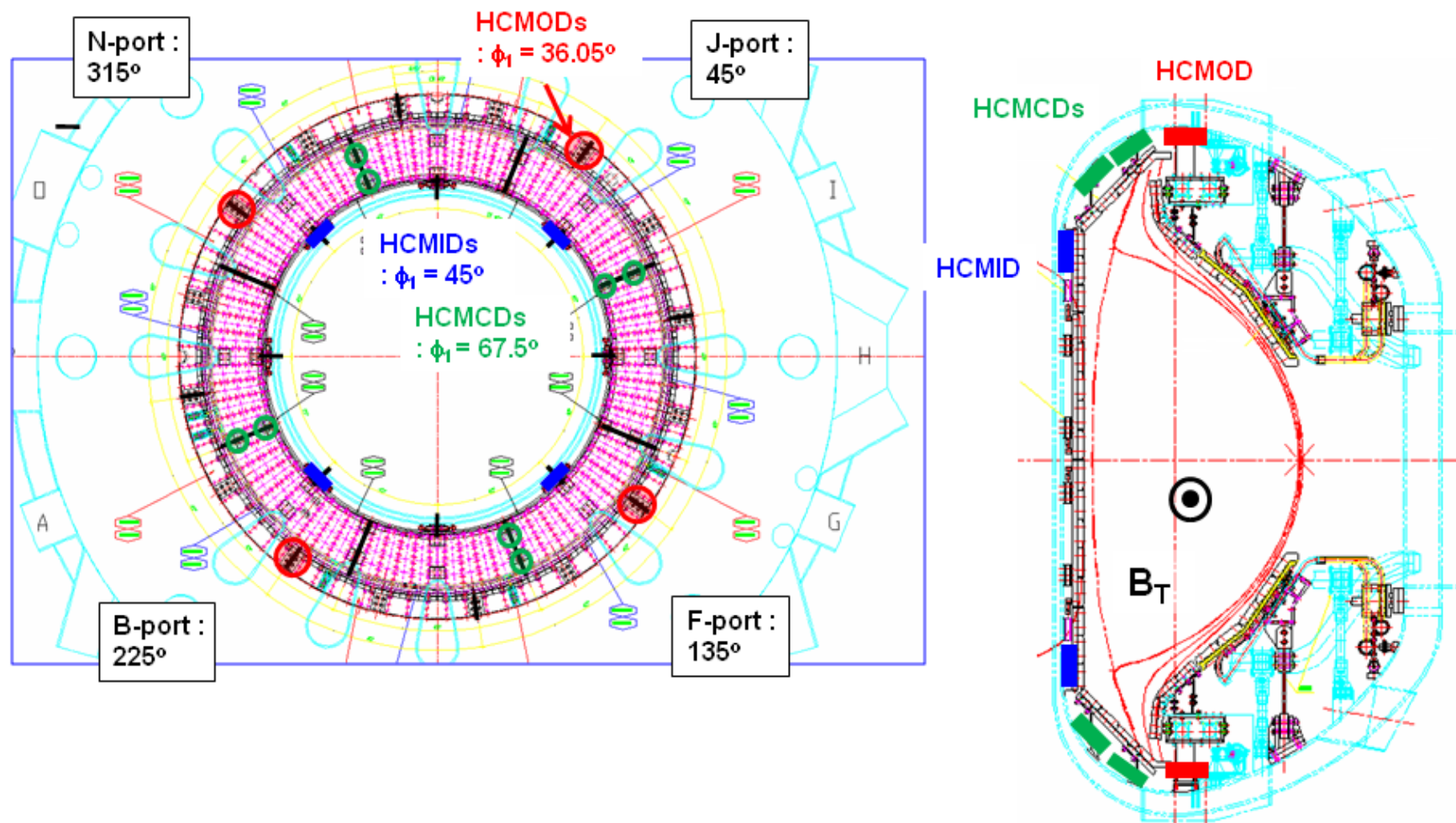
Magnetic Diagnostics – HCM

General Information

Full Name	Halo current monitor
Contact	J. G. Bak (jgbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	halo current (I_h)
Port assignment	feedthrough (B, F, J, N)
Channel information	48 coils
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

Magnetic Diagnostics - HCM

Channels Layout



Magnetic Diagnostics - HCM

MDS+ Node Information

Name	Sampling	Unit	Description
\HCMILo1 ~ \HCMILo8 (UPPER) \HCMILo9 ~ \HCMIL16 (LOWER)	20 kHz	A	Halo currents through the supporting structures of back-plates for inboard limiters
\HCMIDo1 ~ \HCMIDo4 (UPPER) \HCMIDo5 ~ \HCMIDo8 (LOWER)	20 kHz	A	Halo currents through the supporting structures of back-plates for inboard divertors
\HCMCDo1 ~ \HCMCDo8 (UPPER) \HCMCDo9 ~ \HCMCD16 (LOWER)	20 kHz	A	Halo currents through the supporting structures of back-plates for central divertors
\HCMODo1 ~ \HCMODo4 (UPPER) \HCMODo5 ~ \HCMODo8 (LOWER)	20 kHz	A	Halo currents through the supporting structures of back-plates for outboard divertors

Notes

- Please an e-mail to the contact person if you want to see the HCM data.

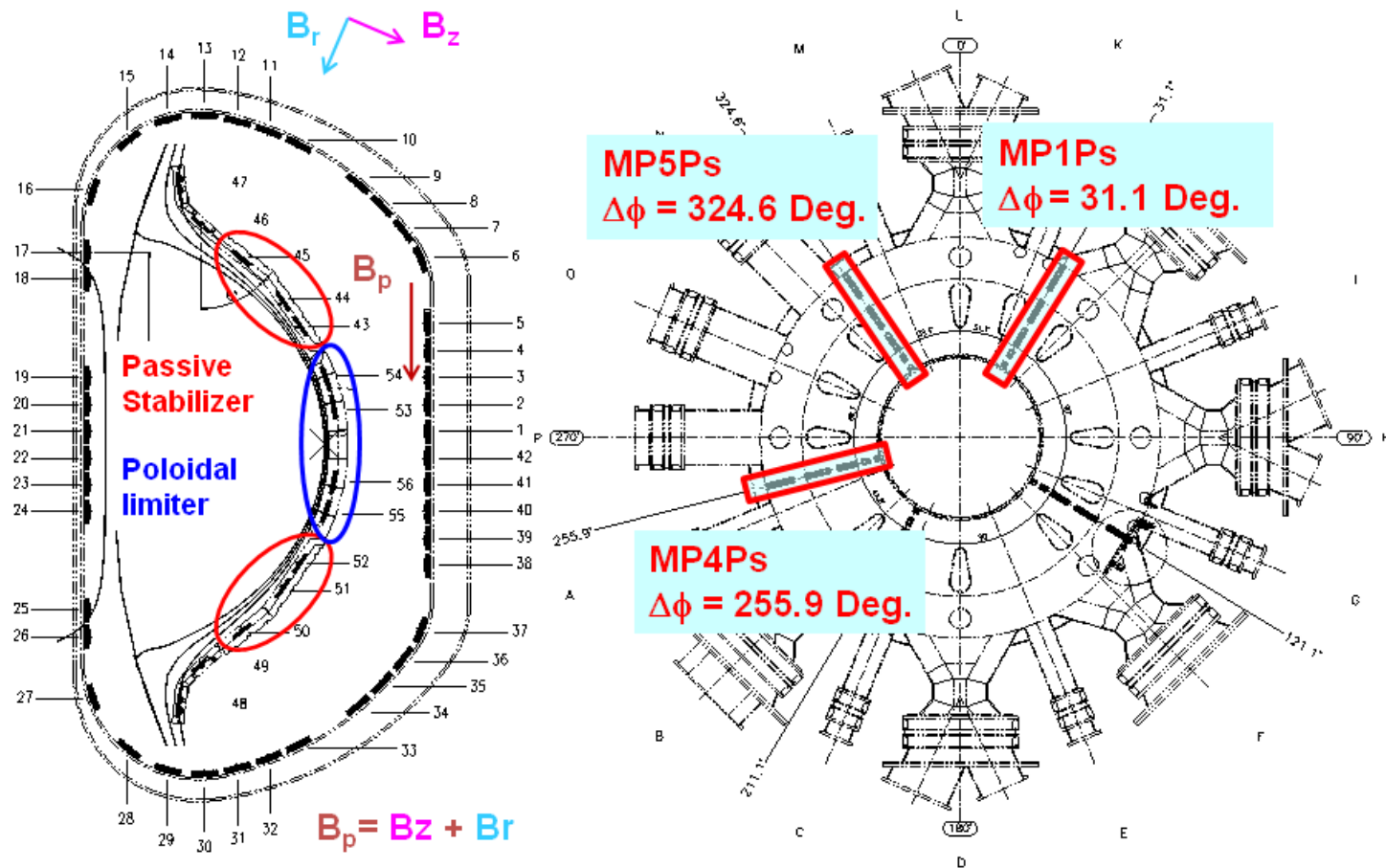
Magnetic Diagnostics – MP

General Information

Full Name	Magnetic field probe
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	tangential and normal components of local poloidal field (B_z , B_r)
Port assignment	feedthrough (B, F, J, N)
Channel information	3 arrays
Time resolution	20 kHz
Spatial Resolution	-
Availability	routinely available

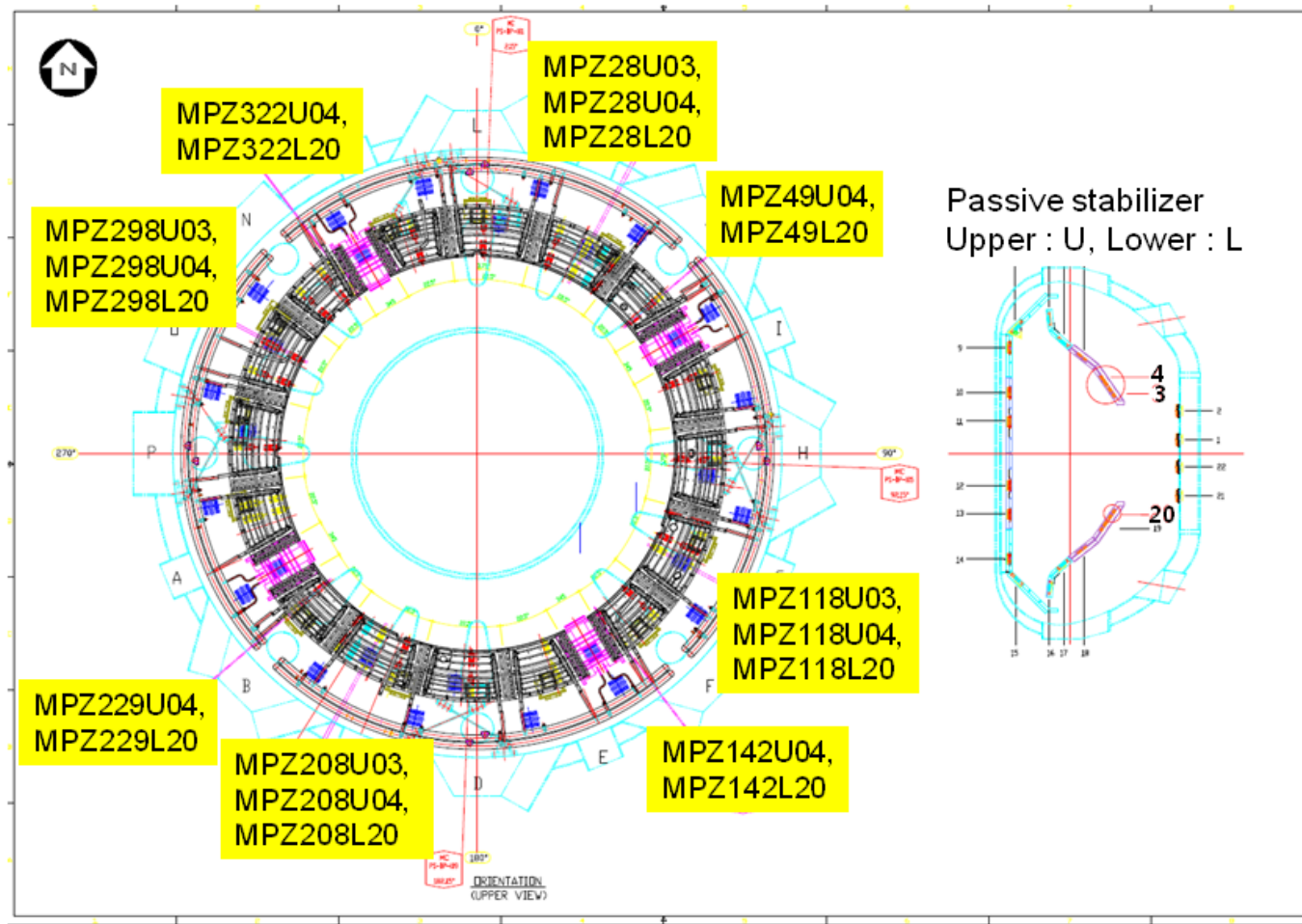
Magnetic Diagnostics - MP

Channels Layout



Magnetic Diagnostics - MP

Channels Layout



Magnetic Diagnostics - MP

MDS+ Node Information

Name	Sampling	Unit	Description
\MP4P01Z ... \MP4P42Z	20 kHz	T	Tangential component of local poloidal field at the toroidal angle of 255.9 deg. The MPs were mounted at the wall of vacuum vessel
\MP4P01R ... \MP4P42R	20 kHz	T	Normal component of local poloidal field at the toroidal angle of 255.9 deg. The MPs were mounted at the wall of vacuum vessel
\PLMP01Z ... \PLMP04Z	20 kHz	T	Tangential component of local poloidal field at the toroidal angle of 324 deg. The MPs were mounted at the poloidal limiter
\MP4P43Z ... \MP4P45Z \MP4P50Z ... \MP4P52Z	20 kHz	T	Tangential component of local poloidal field at the toroidal angle of 255.9 deg. The MPs were mounted at the passive stabilizer.
\MP1P01Z ... \MP1P42Z	20 kHz	T	Tangential component of local poloidal field at the toroidal angle of 31.1 deg. The MPs were mounted at the wall of vacuum vessel
\MP1P01R ... \MP1P42R	20 kHz	T	Normal component of local poloidal field at the toroidal angle of 31.1 deg. The MPs were mounted at the wall of vacuum vessel

Magnetic Diagnostics - MP

MDS+ Node Information

Name	Sampling	Unit	Description
\MPZ028U03 \MPZ028U04 \MPZ028L20	20 kHz	T	at the toroidal angle of 28 deg. (passive stabilizer)
\MPZ049U04 \MPZ049L20	20 kHz	T	at the toroidal angle of 49 deg.
\MPZ118U03 \MPZ118U04 \MPZ118L20	20 kHz	T	at the toroidal angle of 118 deg. (passive stabilizer)
\MPZ142U04 \MPZ142L20	20 kHz	T	at the toroidal angle of 142 deg. (passive stabilizer)
\MPZ208U03 \MPZ208U04 \MPZ208L20	20 kHz	T	at the toroidal angle of 208 deg. (passive stabilizer)

Magnetic Diagnostics - MP

MDS+ Node Information

Name	Sampling	Unit	Description
\MPZ229U04 \MPZ229L20	20 kHz	T	at the toroidal angle of 229 deg. (passive stabilizer)
\MPZ298U03 \MPZ298U04 \MPZ298L20	20 kHz	T	at the toroidal angle of 298 deg. (passive stabilizer)
\MPZ322U04 \MPZ322L20	20 kHz	T	at the toroidal angle of 322 deg. (passive stabilizer)

Notes

- Several data from \MP5P01Z to \MP5P42Z provide integrated tangential components of local poloidal fields at the toroidal angle of 324.6 Degr.
- Several data from \MP5P06R to \MP5P16R and from \MP5P27R to \MP5P37R provide integrated normal components of local poloidal fields at the toroidal angle of 324.6 Degr.

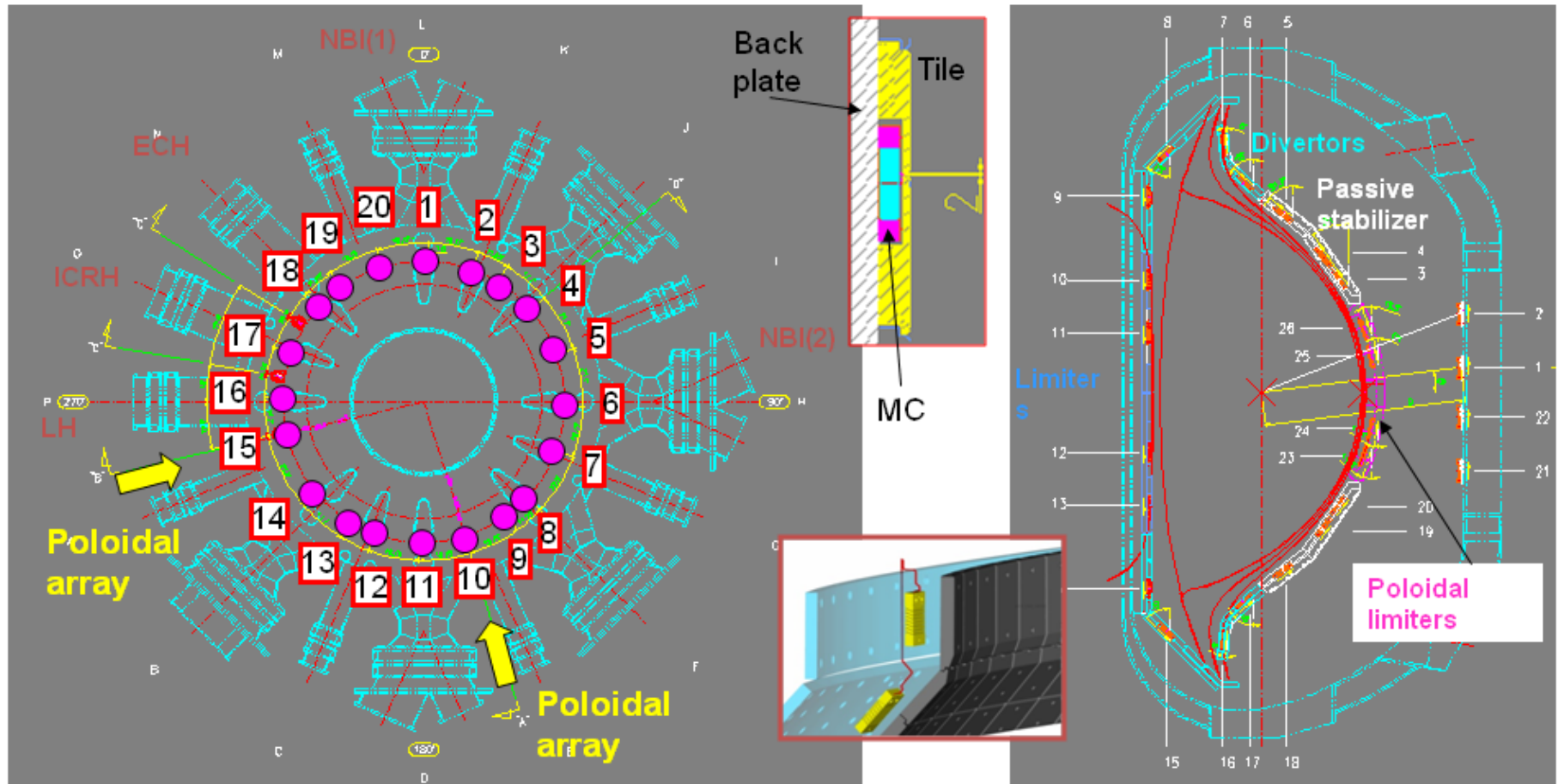
Magnetic Diagnostics – MC

General Information

Full Name	Mirnov Coil
Contact	J. G. Bak (igbak@kfe.re.kr) , H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	time derivative of tangential component of poloidal magnetic field, magnetic fluctuation (dB_z/dt)
Port assignment	feedthrough (B, F, J, N)
Channel information	toroidal array (20 ch), poloidal array (22 ch)
Time resolution	2MHz (default)
Spatial Resolution	-
Availability	routinely available

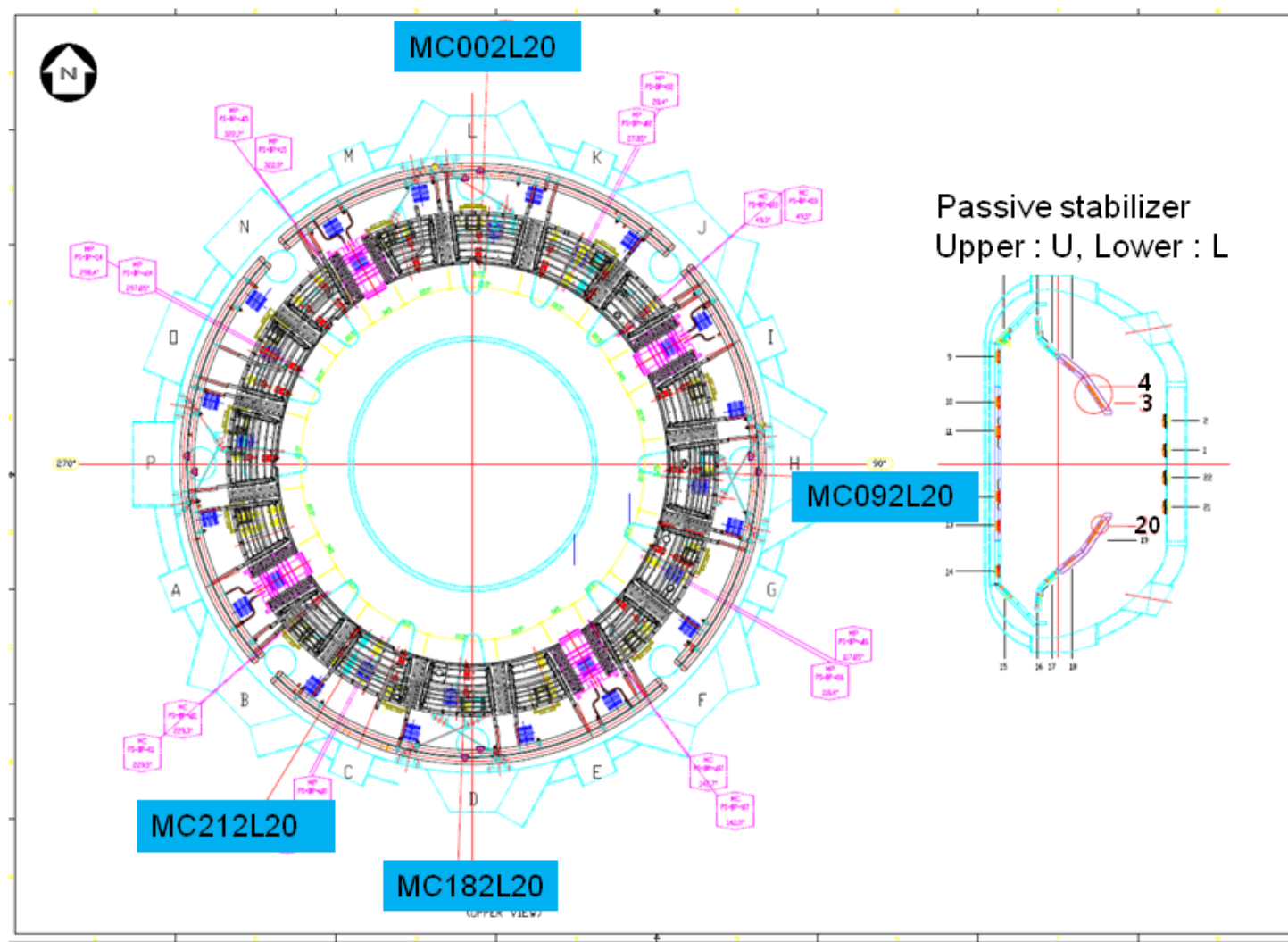
Magnetic Diagnostics - MC

Channels Layout



Magnetic Diagnostics - MC

Channels Layout



Magnetic Diagnostics - MC

MDS+ Node Information

Name	Sampling	Unit	Description
\MC1T02 ~ \MC1T08 \MC1T10 ~ \MC1T16 \MC1T19	2 MHz (default)	T/s	Magnetic fluctuation (time derivative of tangential component of poloidal magnetic field) for evaluation of toroidal mode number of the MHD (toroidal array)
\MC1P01 ~ \MC1P06 \MC1P08, \MC1P10~\MC1P13, \MC1P15 ~ \MC1P17, MC1P20~\MC1P22	2 MHz (default)	T/s	Magnetic fluctuation for evaluation of poloidal mode number of the MHD (poloidal array)
\MC2P10, \MC2P11	2 MHz (default)	T/s	Magnetic fluctuation for evaluation of poloidal mode number of the MHD
\PCMCTL01 \PCMCTL02 \PCMCTL05 \PCMCTL09	100 kHz	T/s	Magnetic fluctuation for NTM control (toroidal angle : 2, 92,182,212 degs)

Notes

- In order to get “\MC1T10”, please use “ \MC1P03”
- Signal whose polarity should be corrected by just multiplying “-1”
: MC1T12~16,19 (toroidal array), MC1P01~04,06,10,11,15,17, MC2P11 (poloidal array)

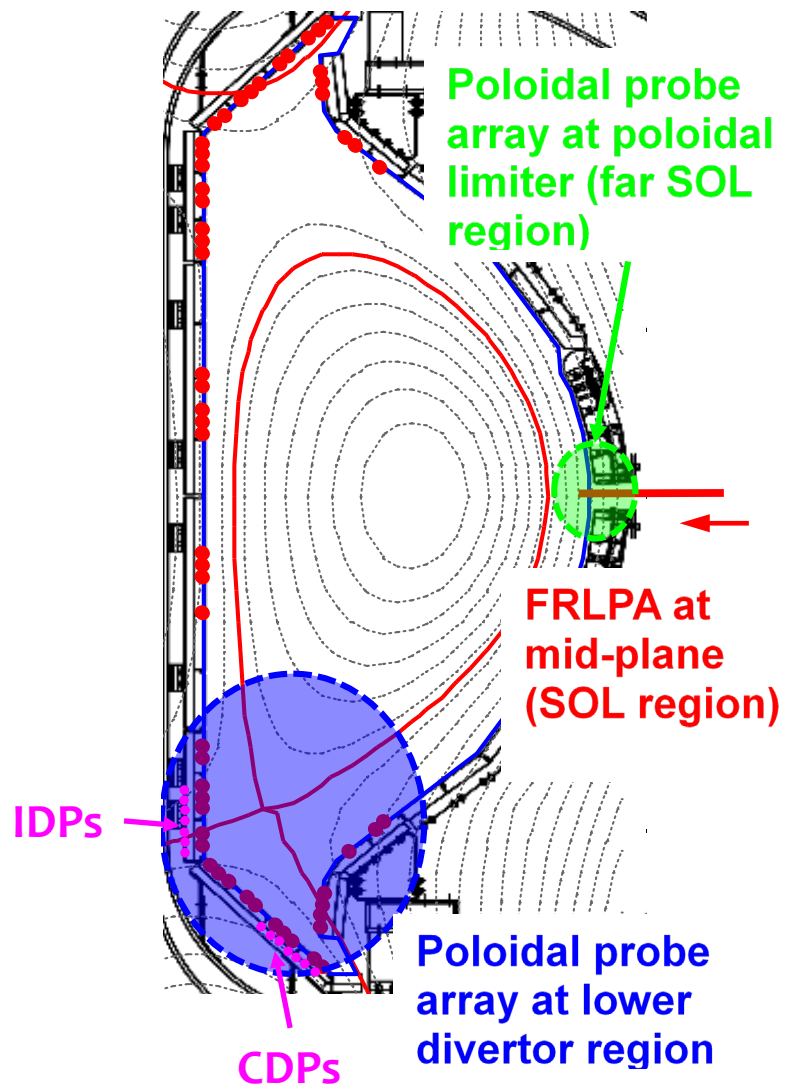
Electric Probe Diagnostics - FRPA / ELPA

General Information

Full Name	Fast reciprocating Langmuir Probe Assembly Edge Langmuir Probe Array
Contact	J. G. Bak (jgbak@kfe.re.kr), H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	FRPA : ion saturation current (I_{is}), plasma & floating potential (V_p , V_f) at SOL ELPA : ion saturation current (I_{is}) at divertor region
Port assignment	C (FRPA / ELPA), N (IDP / CDP)
Channel information	FRPA : 5 ch ELPA : 57 ch
Time resolution	100 kHz (default), 1 MHz (maximum on request)
Spatial Resolution	-
Availability	on request

Electric Probe Diagnostics - FRPA / ELPA

Channels Layout



Note :

- Only EP40- Ep63 (in the lower divertor region at the Bay C) are available
- In addition, IDP01- IDP13, and CDP01-CDP13 are available in the lower divertor region at the Bay N.
- FRPA data for only certain shots before the campaign of 2019 are available.
- ELPA data (EP#s) before the campaign of 2023 are provided.
- Please ask the contact person for further information of the FRPA and ELPA data

Electric Probe Diagnostics - FRPA / ELPA

MDS+ Node Information

Name	Sampling	Unit	Description
\FRPA01:FOO : \FRPA05:FOO	100 kHz	V	Raw data of ion saturation currents, floating and plasma potentials in the scrape-off-layer(SOL) region
\EP07:FOO : \EP63:FOO	100 kHz	V	Raw data of ion saturation currents at the divertor region

Notes

- FRPA : Data analysis on the raw data should be required for the evaluation of radial profiles of plasma density, electron temperature and Mach number.
- ELPA : Data analysis on the raw data should be required for the evaluation of radial profiles of heat/particle flux

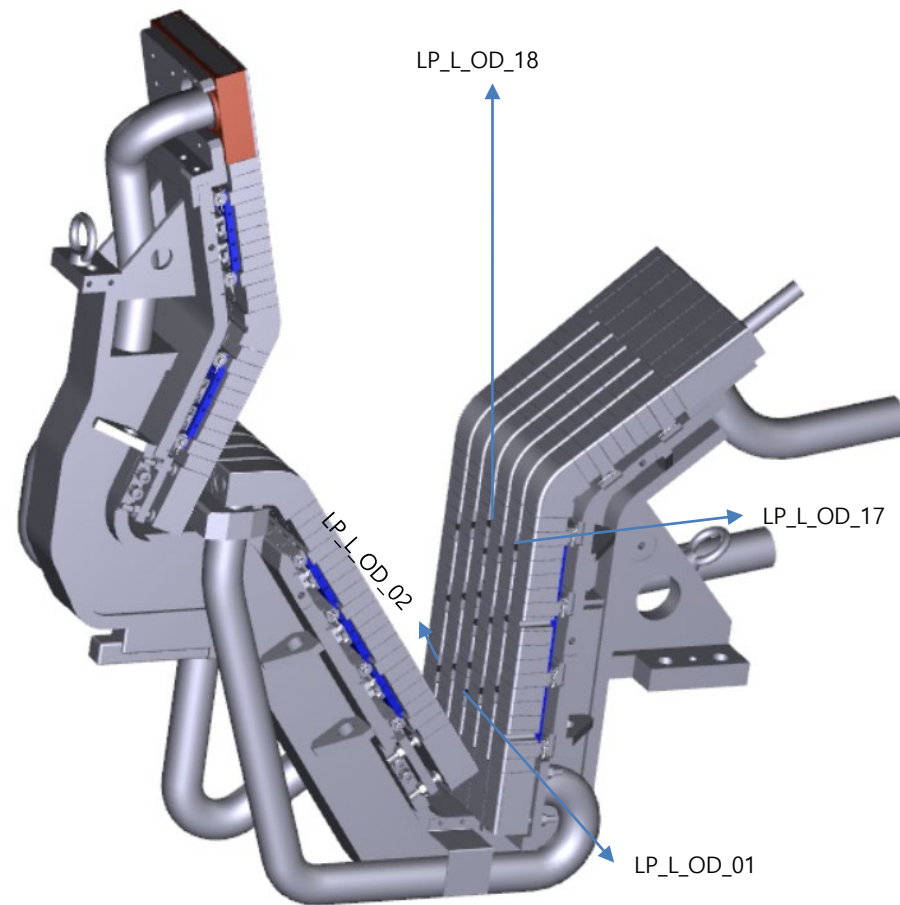
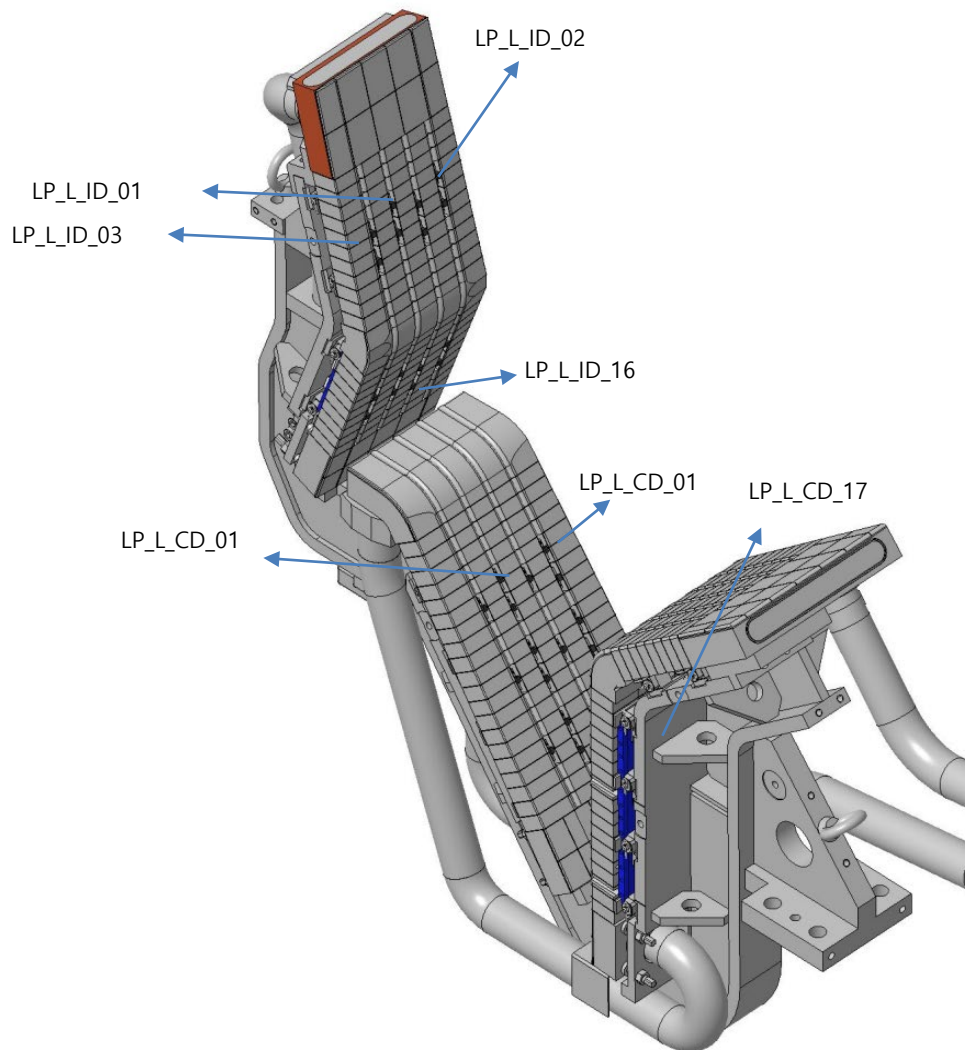
Electric Probe Diagnostics – TCD LP

General Information

Full Name	Tungsten cassette divertor Langmuir Probe Inner Divertor Probe / Central Divertor Probe/ Outer Divertor Probe
Contact	Eunnam Bang(bang14@kfe.re.kr), J. G. Bak (jgbak@kfe.re.kr), H. S. Kim (kimhs@kfe.re.kr)
Measuring properties	Ion saturation current (I_{is}), plasma & floating potential (V_p , V_f) at SOL
Port assignment	D, L
Channel information	D port: 54ch, L port : 54ea (ID: 16ch, CD: 18ch, OD: 18ch)
Time resolution	200 kHz (default)
Spatial Resolution	25~28mm
Availability	routinely available

Electric Probe Diagnostics – TCD LP

Channels Layout



Electric Probe Diagnostics – TCD LP

MDS+ Node Information

Name	Sampling	Unit	Description
\LP_D_ID_01 ~ \LP_D_ID_16:foo \LP_D_CD_01 ~ \LP_D_CD_18:foo \LP_D_OD_01 ~ \LP_D_OD_18:foo * 채널 제외	200 kHz (default)	V	<u>Raw data of ion saturation currents floating (main) and plasma potentials in the scrape-off-layer(Isat) region (changeable)</u> Distance between probe tips: 2.5 ~2.8cm 2025.11.27일 이후 10배 증폭된 값
\LP_L_ID_01 ~ \LP_L_ID_16:foo \LP_L_CD_01 ~ \LP_L_CD_18:foo \LP_L_OD_01 ~ \LP_L_OD_18:foo * 채널 제외	200 kHz (default)	V	<u>Raw data of ion saturation currents floating and plasma potentials in the scrape-off-layer(SOL, main) region (changeable)</u> Distance between probe: 2.5 ~ 2.8cm
*\LP_D_CD_08,10,12,14,16,18:foo \LP_D_OD_08,09,11:foo => EP51~59		V	LP-> PCS Isat
*\LP_L_CD_07,10,11,13,15 \LP_L_OD_08,11,12,13, \LP_D_OD12 => LP_NE,TE01~10	200 kHz (default)	V	LP voltage sweeping(1kHz) Ne, Te on strike point region
\EP07~24	200 kHz (default)	V	Isat (dom type tip on upper divertor EP) ~2022 used EP reconnected.

- 2025.11.26 #39321 : Isat 1/10 scale down
- 2025.11.26 #38909 ne, Te data uploaded in MDSplus

Electric Probe Diagnostics – TCD LP

MDS+ Node Information

Name	Sampling	Unit	Description
\LP_D_ID_01 ~ \LP_D_ID_16:foo \LP_D_CD_01 ~ \LP_D_CD_18:foo \LP_D_OD_01 ~ \LP_D_OD_18:foo	200 kHz (default)	V	<u>Raw data of ion saturation currents floating (main)</u> and plasma potentials in the scrape-off-layer(Isat) region (changeable) Distance between probe tips: 2.5 ~ 2.8cm
\LP_L_ID_01 ~ \LP_L_ID_16:foo \LP_L_CD_01 ~ \LP_L_CD_18:foo \LP_L_OD_01 ~ \LP_L_OD_18:foo	200 kHz (default)	V	<u>Raw data of ion saturation currents floating and</u> <u>plasma potentials in the scrape-off-layer(SOL, main)</u> region (changeable) Distance between probe: 2.5 ~ 2.8cm

- There is a possibility of changing the channel for V sweeping as needed.
- 2025.11.26 #39321 : Isat 1/10 scale down
- 2025.11.26 #38909 ne, Te data uploaded in MDSplus

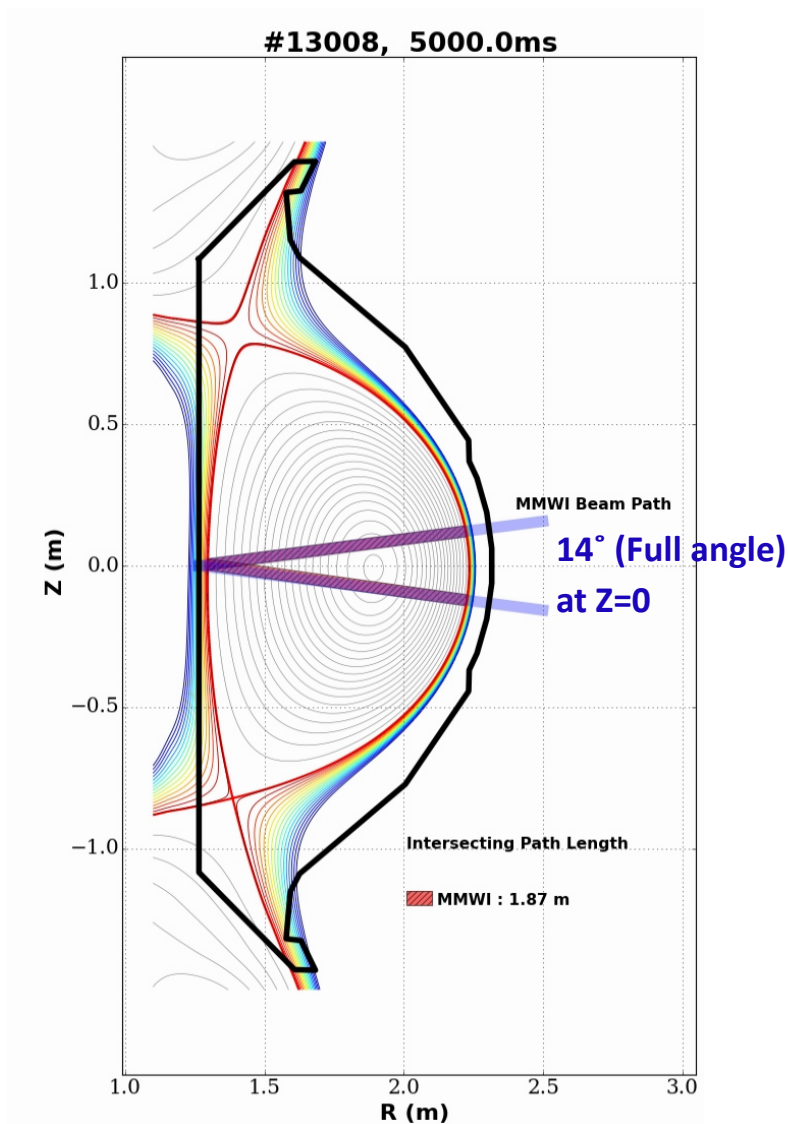
Interferometer – THI

General Information

Full Name	Tera-Hertz Interferometer (THI, former MMWI)
Contact	Y. U. Nam (yunam@kfe.re.kr), Dong-jae Lee (djlee124@kfe.re.kr)
Measuring properties	line-integrated electron density (n_{el})
Port assignment	D -> H (planned)
Channel information	Horizontal 1 ch on midplane ($Z=0.0$ m with 14° full angle)
Time resolution	100 kHz
Spatial Resolution	-
Availability	Data available only for MMWI

Interferometer - THI

Channels Layout



Interferometer - THI

MDS+ Node Information

Name	Sampling	Unit	Description
\ne_intero1	100 kHz	10^{19} m^{-2}	Double-path line-integrated electron density at Z=0

Notes

- The main EM-wave source has been changed from millimeter-wave to tera-hertz wave.
- But the point name is kept same.
- Line-density is integrated for double-path. So dividing by 2 will be necessary for the single-path line-integration.
- For line-average density calculation, dividing by 1.9 will be more appropriate in case of the typical diverted plasmas.
- Be careful for dividing by 2 or 1.9 whichever, especially when the vertical position of the plasma column is biased usually known as vertical displacement.
- During disruption, nel may not follow the abrupt density drop so final density may not fall down to 0.

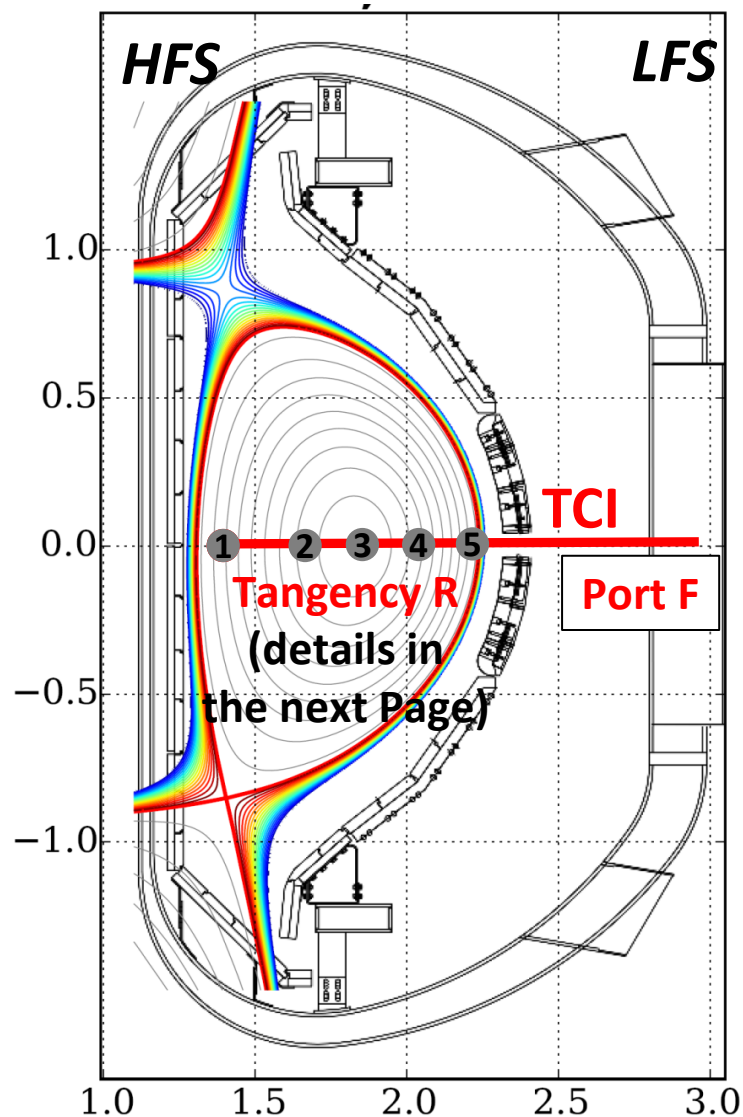
Interferometer - TCI

General Information

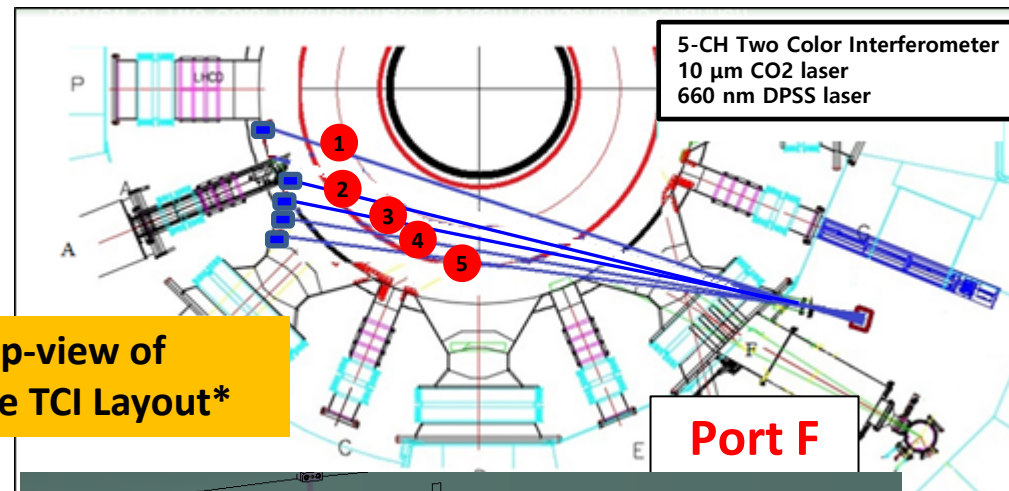
Full Name	Two-Color Interferometer
Contact	K.C. Lee(kcllee@kfe.re.kr), J. W. Juhn(jwjuhn@kfe.re.kr) , J. H. Kim (jhwakim@kfe.re.kr), Y. S. Kim (ablesun@kfe.re.kr)
Measuring properties	Tangentially line-averaged electron densities (n_{el}) in different tangency radii (up to 5 chords)
Port assignment	Tangential Fm
Channel information	2017 : 1 2018 : 1, 2, 5 2019 : 1, 2, 3, 4 2020 : 1, 2, 3, 4, 5
Time resolution (Sampling Rate)	>100 kHz (typ. 313 or 625 kHz of raw data are available in 2019. If needs these raw data, please consult jwjuhn@kfe.re.kr)
Spatial Resolution	N/A
Availability	2018. 8. 27~

Interferometer - TCI

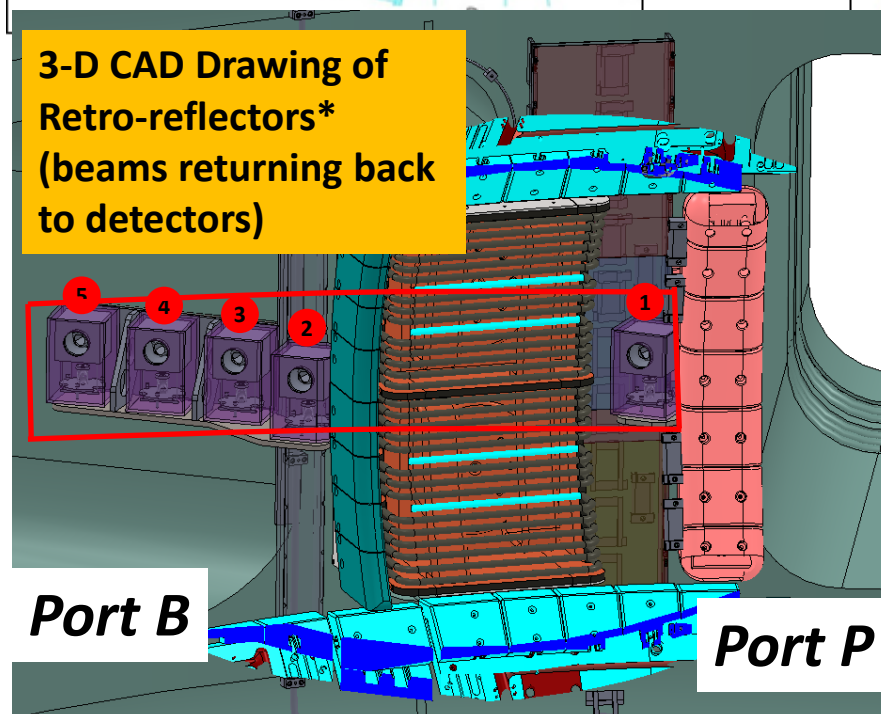
Channels Layout



Top-view of the TCI Layout*



3-D CAD Drawing of Retro-reflectors* (beams returning back to detectors)



Interferometer - TCI

MDS+ Node Information

Name	Sampling Rate	Unit	Tangency R (m)	Path Length (m)	Description
\ne_tcio1	>100 kHz (typ. 313 or 625 kHz of raw data are available since 2019)	10^{19} m^{-3}	1.34	7.23	All are LINE-averaged density. See the notes below. Path lengths here count double-path for round-trip beam path.
\ne_tcio2			1.78	5.51	
\ne_tcio3			1.91	4.76	#03, 04 channels are added in 2019. CH #04 started after #22958
\ne_tcio4			2.04	3.80	
\ne_tcio5			2.16	2.52	#5 channel is available since 2020.

Notes

- In principle, TCI is also double-path line-integrated measurement like MMWI and FIR. However, the path-length is already divided for each cord in the MDS+ data based on the typical numbers given above. Do not confuse other interferometer data that are line-integrated. If line-integrated densities are necessary, use the path lengths given above for each cord.
- In general, tangentially line-averaged data of CH #01 are accurate within at most 10% of discrepancy from the radial interferometer MMWI depending on the position of the plasmas.
- CH#04 started after #22958 while CH#05 started from yr 2020.
- Slow drift were suppressed after #22970. A few of drift cases still exist after #22970 depending on the status of the optical reference channel (temporarily stored using the node of CH #05. Since 2020 optical reference data is stored on #00).
- Tangency R remains the same since 2019
- Tangency R was calculated with the typical R of LCFS as 2.25 m. For the special cases, the path length should be changed w.r.t. the actual R of LCFS.
- Slow drift, fringe jumps and phase jumps may exist for your shots.
- Therefore, before use TCI data , **consult June-Woo Juhn (jwjuhn@kfe.re.kr)**

Interferometer - TCI

EPICS CA & PV

Name	Default	Usage
TCI_DAQ_TIME_AFTER_DISRUPT	3 (30 for ECWC*)	TCI operation time after IP min fault time set on TCI Host PC #1 and #2.
TCI2_DAQ_TIME_AFTER_DISRUPT	3 (30 for ECWC*)	

*Manually adjusted for ECWC

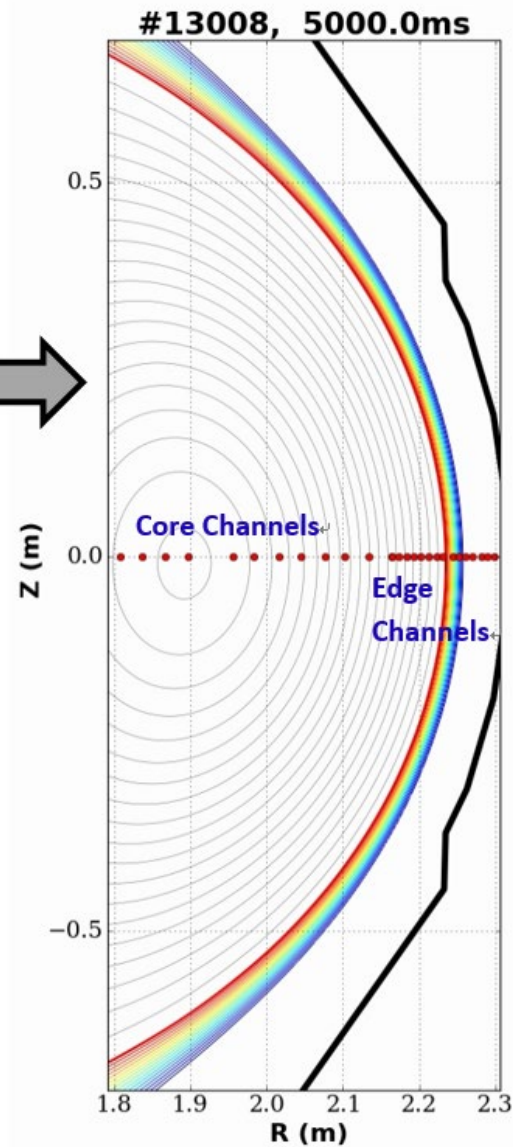
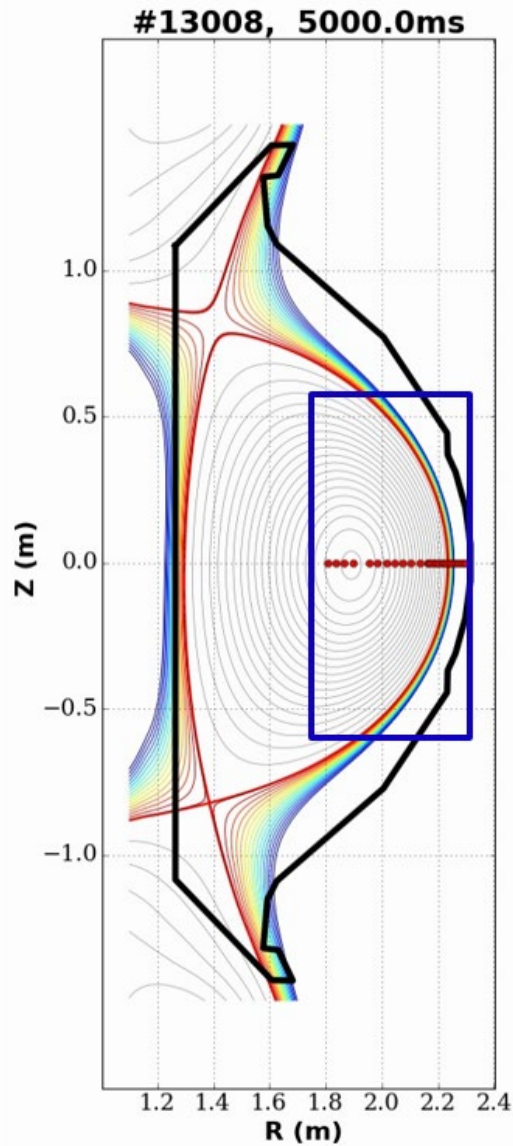
Thomson Scattering

General Information

Full Name	Thomson Scattering
Contact	J. H. Lee (jhlee@kfe.re.kr), H. J. Kim (jjinkim1146@kfe.re.kr), G. H. Park(ghpark@kfe.re.kr), J. H. Kim (jhwakim@kfe.re.kr), Y. G. Kim (ykim@kfe.re.kr)
Measuring properties	electron temperature (T_e), electron density (n_e)
Port assignment	laser input (L), collection optics (N)
Channel information	core (14 points), edge (15points)
Time resolution	20Hz, 50 Hz(main) : depends on the laser condition
Spatial Resolution	~ <20 mm (core), < 5 mm (edge)
Availability	Available by request

Thomson Scattering

Channels Layout



Thomson Scattering

MDS+ Node Information

Name	Sampling	Unit	Description
\TS_CORE1:CORE1_TE ... \TS_CORE14:CORE14_TE	50 Hz	eV	Te at core with 14 channels
\TS_EDGE1:EDGE1_TE ... \TS_EDGE15:EDGE15_TE	50 Hz	eV	Te at edge with 15 channels
\TS_CORE1:CORE1_NE ... \TS_CORE14:CORE14_NE	50 Hz	m ⁻³	ne at core with 14 channels
\TS_EDGE1:EDGE1_NE ... \TS_EDGE15:EDGE15_NE	50 Hz	m ⁻³	ne at edge with 15 channels
\TS_CORE1:CORE1_POS ... \TS_CORE14:CORE14_POS	50 Hz	m ⁻³	radial channel position at core : see next page
\TS_EDGE1:EDGE1_POS ... \TS_EDGE15:EDGE15_POS	50 Hz	m ⁻³	radial channel position at edge : see next page

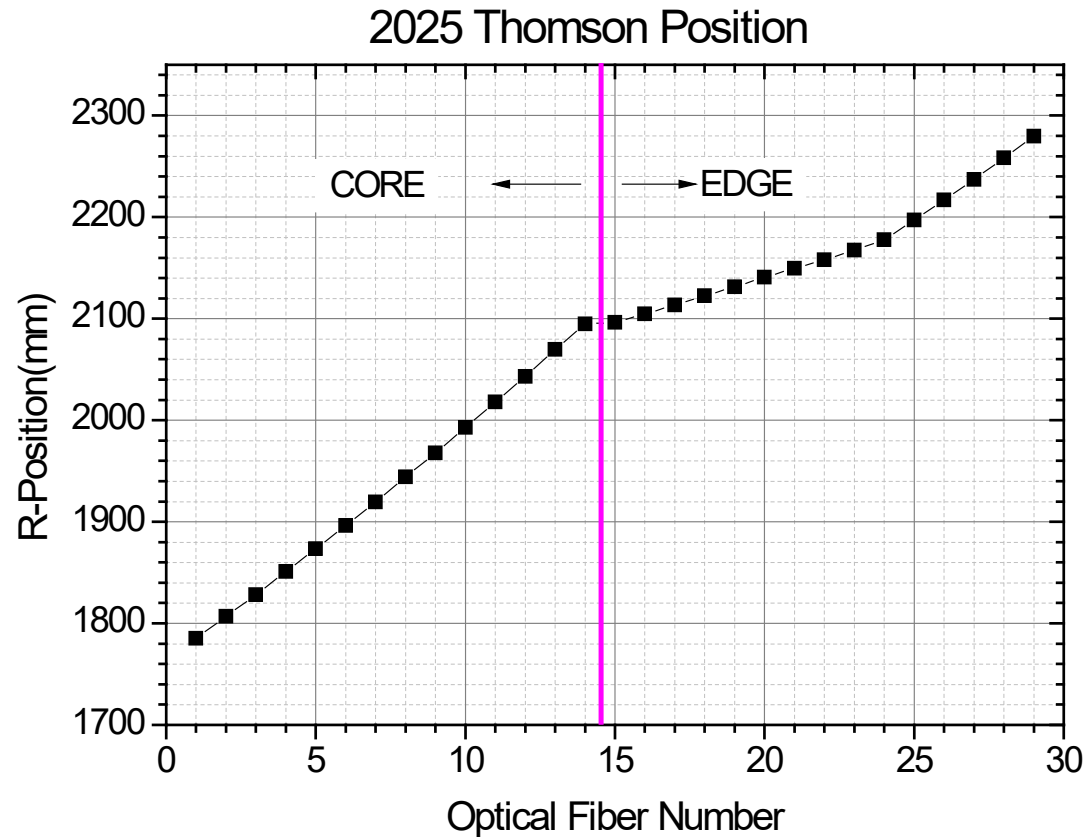
Notes

-

Thomson scattering

Position information

1	1785.20091
2	1806.74887
3	1828.24475
4	1850.83171
5	1873.30334
6	1896.23067
7	1919.59738
8	1944.03634
9	1967.58604
10	1992.84772
11	2017.82838
12	2043.17369
13	2069.5695
14	2094.90512
15	2096.3218
16	2104.84179
17	2113.39556
18	2122.6998
19	2131.32267
20	2140.70081
21	2149.39104
22	2158.11299
23	2167.59713
24	2177.85142
25	2197.00482
26	2217.04247
27	2237.22651
28	2258.30854
29	2279.53952



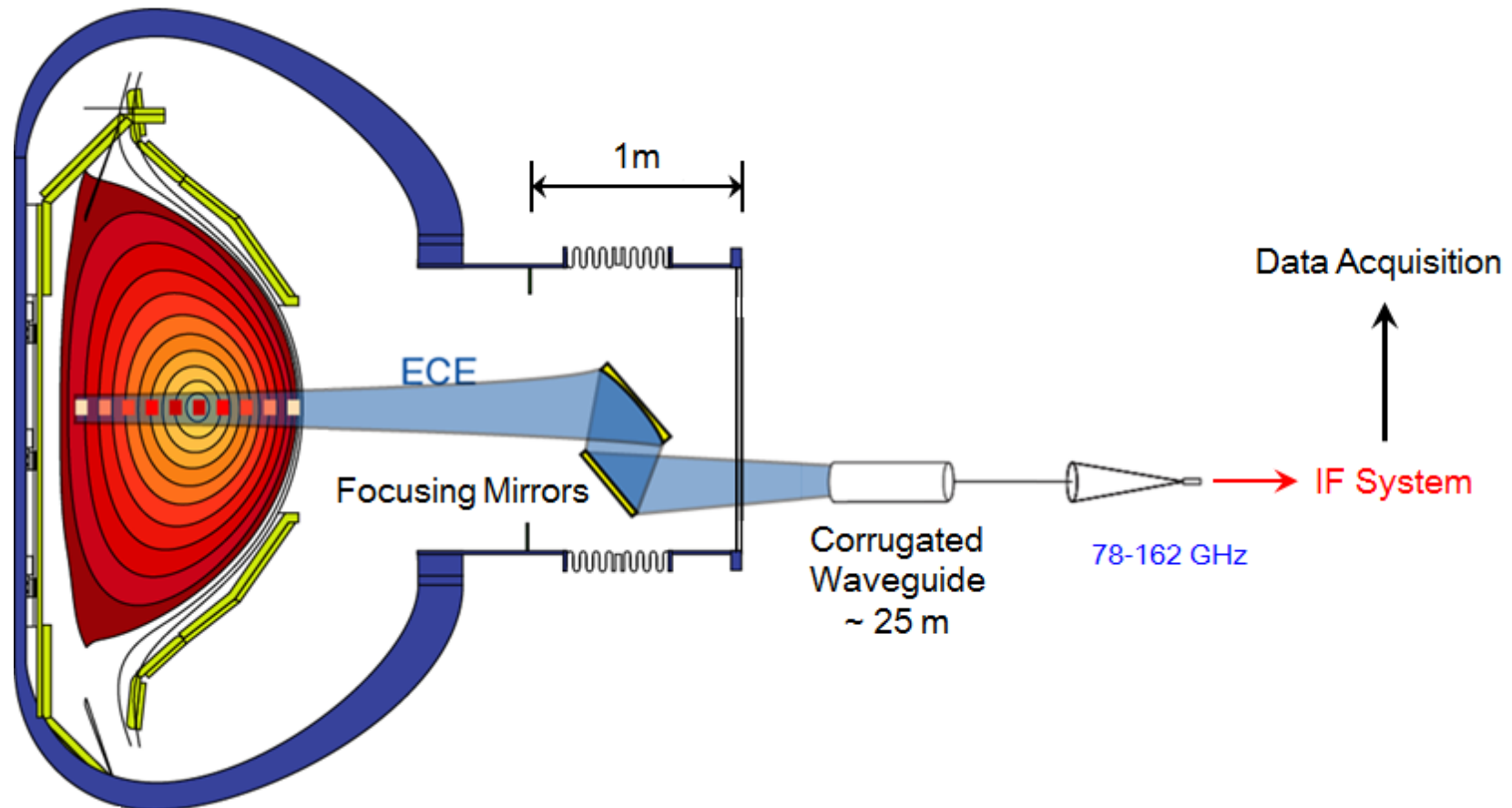
ECE Radiometer

General Information

Full Name	Electron Cyclotron Emission Radiometer
Contact	K. D. Lee (kdlee@kfe.re.kr)
Measuring properties	electron temperature (T_e)
Port assignment	K
Channel information	76 ch (horizontal, midplane)
Measuring position	full radial profile (depending on B_T)
Time resolution	500 kHz (default), 2 MHz (maximum)
Spatial Resolution	~ 1 cm
Availability	routinely available

ECE Radiometer

Channels Layout



ECE Radiometer

MDS+ Node Information

Name		Sampling	Unit	Description
\ECE01 ... \ECE76		500 kHz	eV	electron temperature (T_e)
Subnodes	:FOO	500 kHz	V	raw data
	:FREQ		GHz	ECE frequency
	:RPOS2ND		m	radial position for ECE 2 nd harmonics
	:RPOS3RD		m	radial position for ECE 3 rd harmonics
\ECE_I_TF			kA	TF current value for calculating radial position
\ECE_R2ND			m	76ch radial positions for ECE 2 nd harmonics
\ECE_CENT				central ECE channel
\ECE_CPOS			m	central ECE radial position

Notes

- Since 2016, 76 channels have been available and full radial profiles are available from $B_T=1.8T$ to $2.5T$.
- ECE27 has no calibration factor due to the 140 GHz notch filter and also ECE72 has no calibration factor due to the 105 GHz notch filter.
- ECE01, 02, 03, 28, 48, 49, 52, and ECE59 have no calibration factor due to the poor detector sensitivity.
- Information of radial positions based on the 2nd harmonic channels is listed on the next two pages. The harmonic overlap channels (yellow color numbers) are not recommended to be used.

ECE Radiometer

Position Information

CH	f [GHz]	ECE position (latest version, 2020.07.07)															
		B=1.5 T	B=1.6 T	B=1.7 T	B=1.8 T	B=1.9 T	B=2.0 T	B=2.1 T	B=2.2 T	B=2.3 T	B=2.4 T	B=2.5 T	B=2.6 T	B=2.7 T	B=2.8 T	B=2.9 T	B=3.0 T
Core		ECE54	ECE59	ECE63	ECE67	ECE73	ECE03	ECE08	ECE13	ECE18	ECE24	ECE26	ECE32	ECE37	ECE42	ECE48	(ECE48)
ECE01	110	1.369	1.461	1.552	1.643	1.734	1.826	1.917	2.008	2.100	2.191	2.282					
ECE02	111	1.357	1.447	1.538	1.628	1.719	1.809	1.900	1.990	2.081	2.171	2.262					
ECE03	112	1.345	1.435	1.524	1.614	1.704	1.793	1.883	1.972	2.062	2.152	2.241					
ECE04	113	1.333	1.422	1.511	1.600	1.688	1.777	1.866	1.955	2.044	2.133	2.222	2.310				
ECE05	114	1.321	1.409	1.497	1.586	1.674	1.762	1.850	1.938	2.026	2.114	2.202	2.290				
ECE06	115	1.310	1.397	1.484	1.572	1.659	1.746	1.834	1.921	2.008	2.096	2.183	2.270				
ECE07	116	1.299	1.385	1.472	1.558	1.645	1.731	1.818	1.904	1.991	2.078	2.164	2.251				
ECE08	117	1.287	1.373	1.459	1.545	1.631	1.717	1.802	1.888	1.974	2.060	2.146	2.231				
ECE09	118	1.276	1.362	1.447	1.532	1.617	1.702	1.787	1.872	1.957	2.042	2.127	2.213	2.298			
ECE10	119	1.266	1.350	1.435	1.519	1.603	1.688	1.772	1.856	1.941	2.025	2.110	2.194	2.278			
ECE11	120		1.339	1.423	1.506	1.590	1.674	1.757	1.841	1.925	2.008	2.092	2.176	2.259			
ECE12	121		1.328	1.411	1.494	1.577	1.660	1.743	1.826	1.909	1.992	2.075	2.158	2.241			
ECE13	123		1.306	1.388	1.470	1.551	1.633	1.714	1.796	1.878	1.959	2.041	2.123	2.204	2.286		
ECE14	124		1.296	1.377	1.458	1.539	1.620	1.701	1.782	1.863	1.944	2.025	2.106	2.187	2.267		
ECE15	125		1.285	1.366	1.446	1.526	1.607	1.687	1.767	1.848	1.928	2.008	2.089	2.169	2.249		
ECE16	126		1.275	1.355	1.435	1.514	1.594	1.674	1.753	1.833	1.913	1.992	2.072	2.152	2.231	2.311	
ECE17	127		1.265	1.344	1.423	1.502	1.581	1.660	1.740	1.819	1.898	1.977	2.056	2.135	2.214	2.293	
ECE18	128			1.334	1.412	1.491	1.569	1.647	1.726	1.804	1.883	1.961	2.040	2.118	2.197	2.275	
ECE19	129			1.323	1.401	1.479	1.557	1.635	1.713	1.790	1.868	1.946	2.024	2.102	2.18	2.257	
ECE20	130			1.313	1.390	1.468	1.545	1.622	1.699	1.777	1.854	1.931	2.008	2.086	2.163	2.24	
ECE21	131			1.303	1.380	1.456	1.533	1.610	1.686	1.763	1.840	1.916	1.993	2.070	2.146	2.223	2.300
ECE22	132			1.293	1.369	1.445	1.521	1.598	1.674	1.750	1.826	1.902	1.978	2.054	2.13	2.206	2.282
ECE23	133			1.284	1.359	1.435	1.510	1.586	1.661	1.737	1.812	1.888	1.963	2.039	2.114	2.19	2.265
ECE24	134			1.274	1.349	1.424	1.499	1.574	1.649	1.724	1.799	1.873	1.948	2.023	2.098	2.173	2.248
ECE25	138				1.310	1.383	1.455	1.528	1.601	1.674	1.746	1.819	1.892	1.965	2.037	2.11	2.183
ECE26	139				1.300	1.373	1.445	1.517	1.589	1.662	1.734	1.806	1.878	1.951	2.023	2.095	2.167
ECE27	140				1.291	1.363	1.435	1.506	1.578	1.650	1.721	1.793	1.865	1.937	2.008	2.08	2.152
ECE28	141				1.282	1.353	1.424	1.496	1.567	1.638	1.709	1.780	1.852	1.923	1.994	2.065	2.137
ECE29	142				1.273	1.344	1.414	1.485	1.556	1.626	1.697	1.768	1.839	1.909	1.98	2.051	2.121
ECE30	143				1.264	1.334	1.404	1.475	1.545	1.615	1.685	1.756	1.826	1.896	1.966	2.036	2.107
ECE31	144					1.325	1.395	1.464	1.534	1.604	1.674	1.743	1.813	1.883	1.953	2.022	2.092
ECE32	145					1.316	1.385	1.454	1.524	1.593	1.662	1.731	1.801	1.870	1.939	2.008	2.078
ECE33	146					1.307	1.376	1.444	1.513	1.582	1.651	1.719	1.788	1.857	1.926	1.995	2.063
ECE34	147					1.298	1.366	1.435	1.503	1.571	1.639	1.708	1.776	1.844	1.913	1.981	2.049
ECE35	148					1.289	1.357	1.425	1.493	1.561	1.628	1.696	1.764	1.832	1.9	1.968	2.035
ECE36	149					1.280	1.348	1.415	1.483	1.550	1.617	1.685	1.752	1.820	1.887	1.954	2.022
ECE37	151					1.264	1.330	1.397	1.463	1.530	1.596	1.663	1.729	1.796	1.862	1.929	1.995
ECE38	152						1.321	1.387	1.453	1.519	1.586	1.652	1.718	1.784	1.85	1.916	1.982

ECE Radiometer

Position Information

		ECE position (latest version, 2020.07.07)															
CH	f [GHz]	B=1.5 T	B=1.6 T	B=1.7 T	B=1.8 T	B=1.9 T	B=2.0 T	B=2.1 T	B=2.2 T	B=2.3 T	B=2.4 T	B=2.5 T	B=2.6 T	B=2.7 T	B=2.8 T	B=2.9 T	B=3.0 T
Core		ECE54	ECE59	ECE63	ECE67	ECE73	ECE03	ECE08	ECE13	ECE18	ECE24	ECE26	ECE32	ECE37	ECE42	ECE48	(ECE48)
ECE39	153						1.313	1.378	1.444	1.510	1.575	1.641	1.706	1.772	1.838	1.903	1.969
ECE40	154						1.304	1.369	1.435	1.500	1.565	1.630	1.695	1.761	1.826	1.891	1.956
ECE41	155						1.296	1.360	1.425	1.490	1.555	1.620	1.684	1.749	1.814	1.879	1.944
ECE42	156						1.287	1.352	1.416	1.481	1.545	1.609	1.674	1.738	1.802	1.867	1.931
ECE43	157						1.279	1.343	1.407	1.471	1.535	1.599	1.663	1.727	1.791	1.855	1.919
ECE44	158						1.271	1.335	1.398	1.462	1.525	1.589	1.652	1.716	1.78	1.843	1.907
ECE45	159						1.263	1.326	1.389	1.453	1.516	1.579	1.642	1.705	1.768	1.832	1.895
ECE46	160							1.318	1.381	1.444	1.506	1.569	1.632	1.695	1.757	1.82	1.883
ECE47	161							1.310	1.372	1.435	1.497	1.559	1.622	1.684	1.746	1.809	1.871
ECE48	162							1.302	1.364	1.426	1.488	1.550	1.612	1.674	1.736	1.798	1.860
ECE49	78	1.931	2.060	2.189													
ECE50	79	1.907	2.034	2.161	2.288												
ECE51	81	1.860	1.984	2.108	2.231												
ECE52	82	1.837	1.959	2.082	2.204												
ECE53	83	1.815	1.936	2.057	2.178	2.299											
ECE54	84	1.793	1.913	2.032	2.152	2.271											
ECE55	85	1.772	1.890	2.008	2.126	2.245											
ECE56	86	1.751	1.868	1.985	2.102	2.219											
ECE57	87	1.731	1.847	1.962	2.078	2.193	2.308										
ECE58	88	1.712	1.826	1.940	2.054	2.168	2.282										
ECE59	89	1.692	1.805	1.918	2.031	2.144	2.257										
ECE60	90	1.674	1.785	1.897	2.008	2.120	2.231										
ECE61	91	1.655	1.766	1.876	1.986	2.097	2.207										
ECE62	92	1.637	1.746	1.856	1.965	2.074	2.183	2.292									
ECE63	96	1.569	1.674	1.778	1.883	1.987	2.092	2.197	2.301								
ECE64	97	1.553	1.656	1.760	1.863	1.967	2.070	2.174	2.278								
ECE65	98	1.537	1.639	1.742	1.844	1.947	2.049	2.152	2.254								
ECE66	99	1.521	1.623	1.724	1.826	1.927	2.029	2.130	2.231								
ECE67	100	1.506	1.607	1.707	1.808	1.908	2.008	2.109	2.209	2.310							
ECE68	101	1.491	1.591	1.690	1.790	1.889	1.988	2.088	2.187	2.287							
ECE69	102	1.477	1.575	1.674	1.772	1.871	1.969	2.067	2.166	2.264							
ECE70	103	1.462	1.560	1.657	1.755	1.852	1.950	2.047	2.145	2.242							
ECE71	104	1.448	1.545	1.641	1.738	1.835	1.931	2.028	2.124	2.221							
ECE72	105	1.435	1.530	1.626	1.721	1.817	1.913	2.008	2.104	2.200	2.295						
ECE73	106	1.421	1.516	1.610	1.705	1.800	1.895	1.989	2.084	2.179	2.274						
ECE74	107	1.408	1.502	1.595	1.689	1.783	1.877	1.971	2.065	2.159	2.252						
ECE75	109	1.382	1.474	1.566	1.658	1.750	1.843	1.935	2.027	2.119	2.211	2.303					
ECE76	110	1.369	1.461	1.552	1.643	1.734	1.826	1.917	2.008	2.100	2.191	2.282					

R [m]

2nd Harmonic

Harmonic overlap

Core

ECE##

Notch

Bad

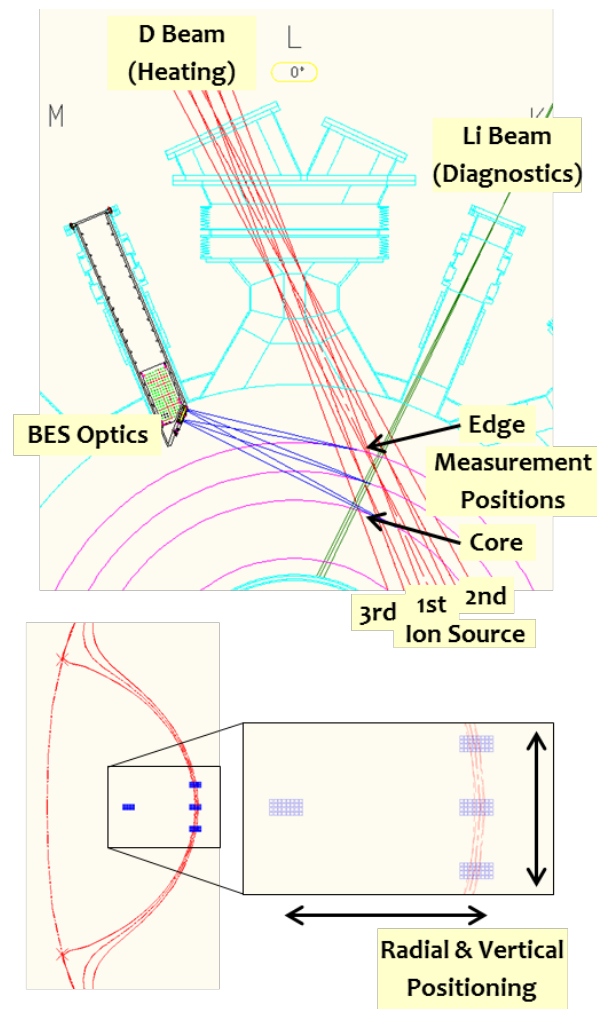
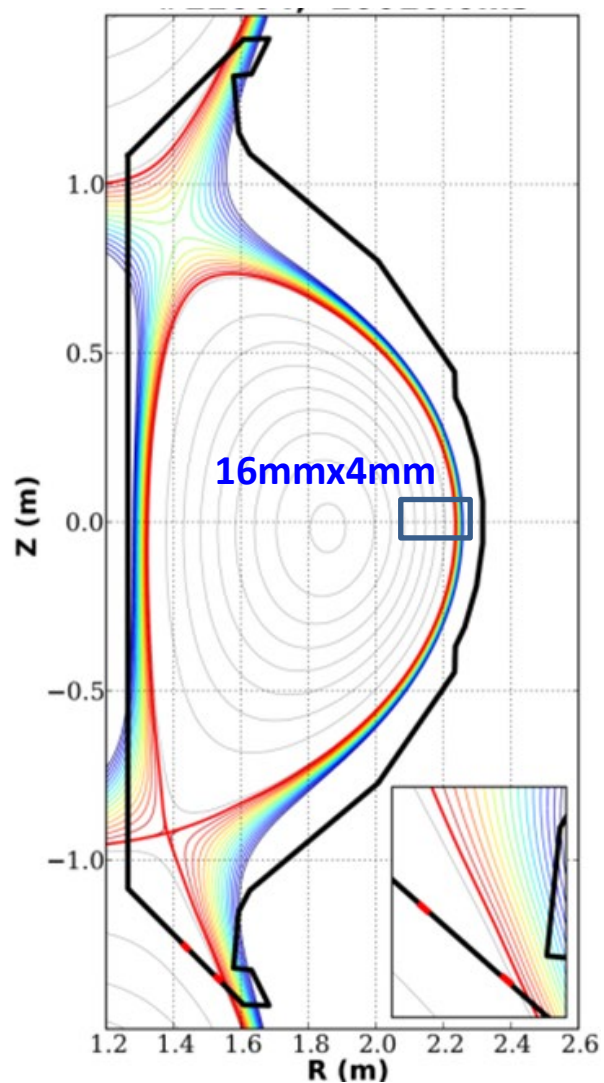
R [m]

2nd Harmonic
Harmonic overlap
Core

ECE##

Notch
Bad

Full Name	Beam Emission Spectroscopy
Contact	Jaewook Kim (ijwkim@kfe.re.kr) , Y. U. Nam (yunam@kfe.re.kr)
Measuring properties	electron density (n_e , relative)
Port assignment	M (collection optics), K (measuring position)
Channel information	16 x 4 channels (16 x 4 cm)
Measuring position	adjustable (radial position, array angle, default at edge midplane)
Time resolution	2MHz
Spatial Resolution	1 cm
Availability	NBI needed

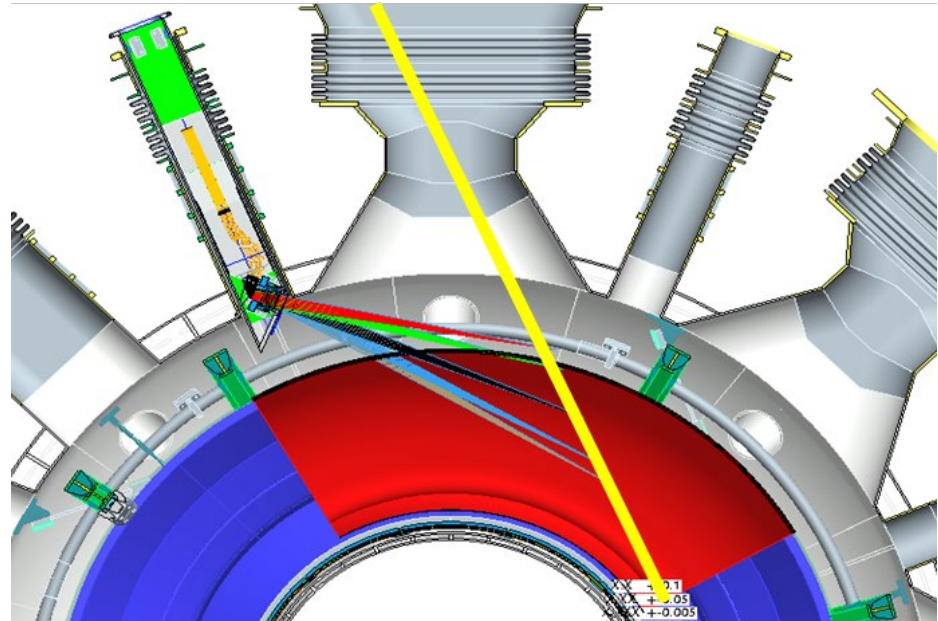
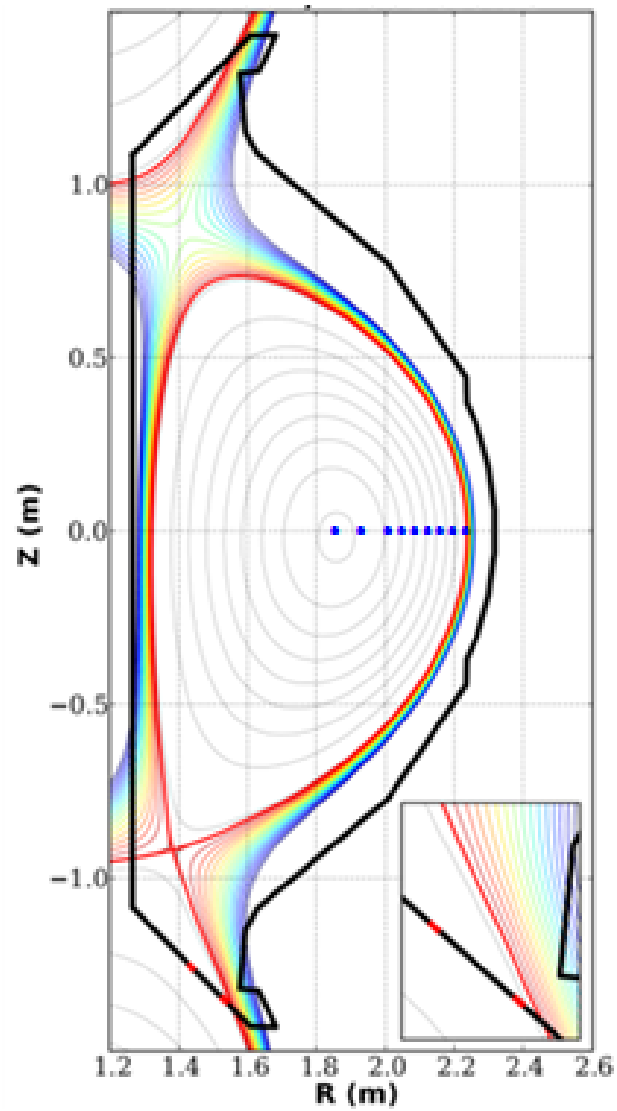


Name		Sampling	Unit	Description
\BES_0101 : \BES_0416		500kHz	V	\BES+[vertical channel number:01-04] + [radial channel number:01-16] = (:FOO x :CAL)
Subnodes	:FOO	500kHz	V	500kHz down-sampled data (photodiode intensity)
	:CAL			calibration factor (for relative cal. btw channels)
	:RPOS		mm	radial channel position
	:VPOS		mm	vertical channel position

Notes

- Measurements position can be varied shot-by-shot.
- When the calibration factor is not available, please refer data at \BES_0101:FOO
- If 2MHz raw data is required, please consult with contact persons.
- Measured voltage is mainly proportional to the local electron density with background subtraction. Effect of the electron temperature is under 5% in typical condition. The background signal level can be obtained during NBI turn-off.
- Measured signal fluctuation is proportional to the local electron density fluctuation in case of edge measurements with heating beam. For absolute electron density profile, a reconstruction analysis is required.

Full Name	Charge Exchange Spectroscopy
Contact	W. H. Ko (whko@kfe.re.kr), J. K. Lee (jkleee@kfe.re.kr)
Measuring properties	ion temperature (T_i), toroidal/poloidal rotation velocity (V_t , V_p)
Port assignment	M (toroidal optics), K (poloidal optics), K (measuring position)
Channel information	32 ch (toroidal), 16 ch (poloidal)
Time resolution	100 Hz (10 ms)
Spatial Resolution	5 mm (edge), 20-30 mm (core)
Availability	NBI modulation needed (1 or 2 Hz with 99% duty)



Name		Sampling	Unit	Description
\CES_Tl01 ... \CES_Tl32		100 Hz	eV	ion temperature from toroidal CES (in M port)
\CES_VT01 ... \CES_VT32		100 Hz	km/s	toroidal rotation from toroidal CES (in M port)
\CES_RT01 ... \CES_RT32			mm	major radius of toroidal CES (in M port)
\CES_TP01 ... \CES_TP16		100 Hz	eV	ion temperature from poloidal CES (in K port)
\CES_VP01 ... \CES_VP16		100 Hz	km/s	poloidal rotation from poloidal CES (in K port)
\CES_RP01 ... \CES_RP16			mm	major radius of poloidal CES (in M port)
Subnodes	:FOO			raw data
	:ERR_BAR			error bar

Notes

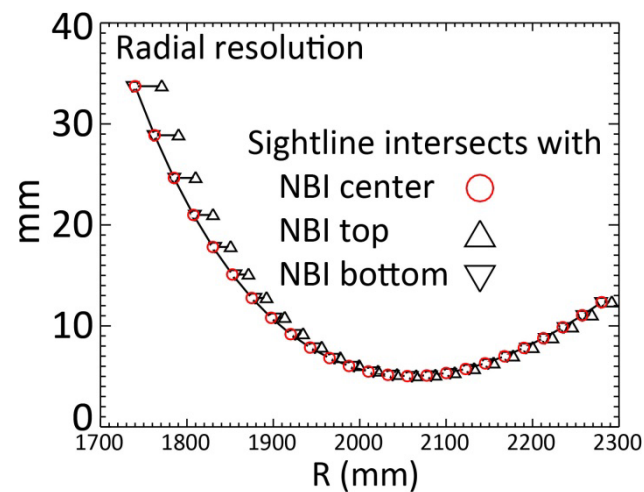
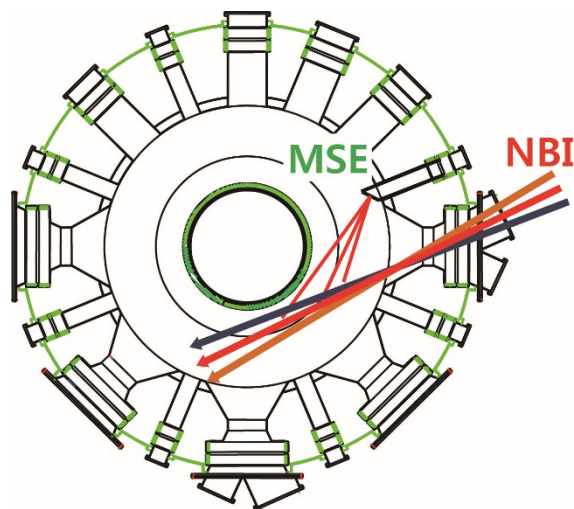
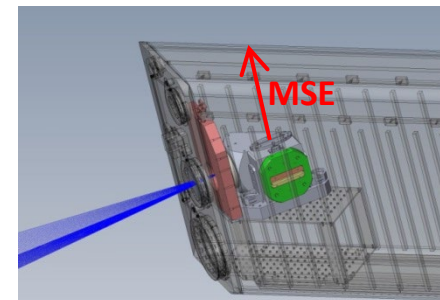
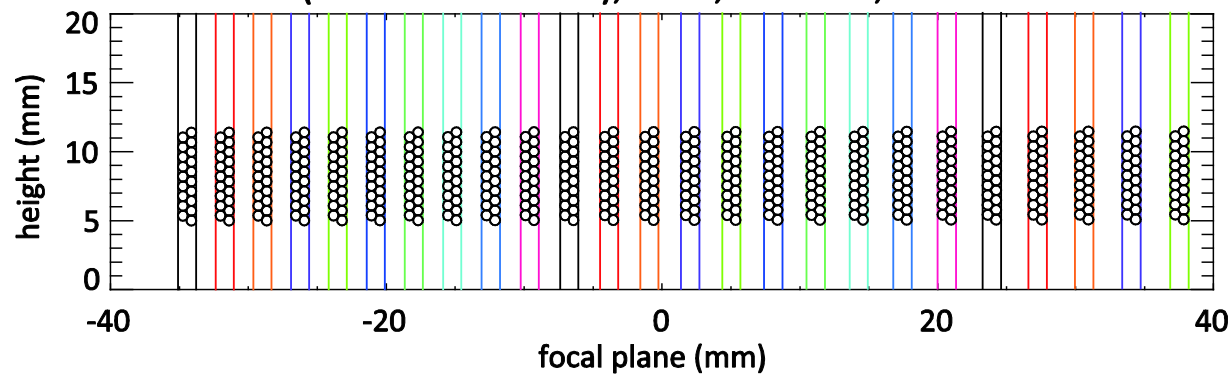
- **It is necessary to be checked by the person in charge of CES before using it because Ti and Vt data is automatically calculated and uploaded in MDSplus.**
- If you need T_i and V_t or V_p , you should have NBI modulation with a few Hz because CES need the background signal obtained during NBI turn-off.
- There are toroidal and poloidal rotation velocities which have unit km/s. If Rotation divided by major radius or effective radius, the value is angular velocity and the unit is [kw/s].

Toroidal CES node

					R [mm]						
T _i [eV]	T _i error bar [eV]	V _T [km/s]	V _T error bar [km/s]	R [mm]	2011	2012	2013 2014	2015	#13522~ #27382	2021	2022~
CES_TI01	CES_TI01:err_bar	CES_VT01	CES_VT01:err_bar	CES_RT01	1795	1800	1795	1801	1801	1800	1794
CES_TI02	CES_TI02:err_bar	CES_VT02	CES_VT02:err_bar	CES_RT02	1800	1850	1850	1822	1822	1823	1819
CES_TI03	CES_TI03:err_bar	CES_VT03	CES_VT03:err_bar	CES_RT03	1850	1900	1900	1843	1843	1853	1849
CES_TI04	CES_TI04:err_bar	CES_VT04	CES_VT04:err_bar	CES_RT04	1900	1950	1950	1874	1874	1884	1879
CES_TI05	CES_TI05:err_bar	CES_VT05	CES_VT05:err_bar	CES_RT05	1950	2000	2000	1895	1895	1914	1910
CES_TI06	CES_TI06:err_bar	CES_VT06	CES_VT06:err_bar	CES_RT06	2000	2050	2050	1945	1945	1945	1939
CES_TI07	CES_TI07:err_bar	CES_VT07	CES_VT07:err_bar	CES_RT07	2050	2100	2100	1995	1995	1985	1979
CES_TI08	CES_TI08:err_bar	CES_VT08	CES_VT08:err_bar	CES_RT08	2100	2150	2150	2016	2016	2013	2009
CES_TI09	CES_TI09:err_bar	CES_VT09	CES_VT09:err_bar	CES_RT09	2140	2170	2170	2047	2047	2043	2038
CES_TI10	CES_TI10:err_bar	CES_VT10	CES_VT10:err_bar	CES_RT10	2160	2180	2180	2078	2078	2073	2069
CES_TI11	CES_TI11:err_bar	CES_VT11	CES_VT11:err_bar	CES_RT11	2170	2190	2190	2099	2099	2094	2089
CES_TI12	CES_TI12:err_bar	CES_VT12	CES_VT12:err_bar	CES_RT12	2180	2200	2200	2125	2125	2122	2119
CES_TI13	CES_TI13:err_bar	CES_VT13	CES_VT13:err_bar	CES_RT13	2190	2205	2205	2150	2150	2147	2144
CES_TI14	CES_TI14:err_bar	CES_VT14	CES_VT14:err_bar	CES_RT14	2200	2210	2210	2171	2171	2171	2168
CES_TI15	CES_TI15:err_bar	CES_VT15	CES_VT15:err_bar	CES_RT15	2205	2215	2215	2192	2187	2187	2183
CES_TI16	CES_TI16:err_bar	CES_VT16	CES_VT16:err_bar	CES_RT16	2210	2220	2220	2203	2192	2192	2187
CES_TI17	CES_TI17:err_bar	CES_VT17	CES_VT17:err_bar	CES_RT17	2215	2225	2225	2213	2198	2198	2192
CES_TI18	CES_TI18:err_bar	CES_VT18	CES_VT18:err_bar	CES_RT18	2220	2230	2230	2223	2203	2203	2198
CES_TI19	CES_TI19:err_bar	CES_VT19	CES_VT19:err_bar	CES_RT19	2225	2235	2235	2228	2208	2208	2203
CES_TI20	CES_TI20:err_bar	CES_VT20	CES_VT20:err_bar	CES_RT20	2230	2240	2240	2233	2213	2213	2208
CES_TI21	CES_TI21:err_bar	CES_VT21	CES_VT21:err_bar	CES_RT21	2235	2245	2245	2238	2218	2218	2213
CES_TI22	CES_TI22:err_bar	CES_VT22	CES_VT22:err_bar	CES_RT22	2240	2250	2250	2243	2223	2223	2218
CES_TI23	CES_TI23:err_bar	CES_VT23	CES_VT23:err_bar	CES_RT23	2245	2255	2255	2248	2228	2228	2223
CES_TI24	CES_TI24:err_bar	CES_VT24	CES_VT24:err_bar	CES_RT24	2250	2260	2260	2253	2233	2233	2228
CES_TI25	CES_TI25:err_bar	CES_VT25	CES_VT25:err_bar	CES_RT25	2255	2265	2265	2259	2238	2238	2233
CES_TI26	CES_TI26:err_bar	CES_VT26	CES_VT26:err_bar	CES_RT26	2265	2270	2270	2264	2243	2243	2238
CES_TI27	CES_TI27:err_bar	CES_VT27	CES_VT27:err_bar	CES_RT27	2275	2275	2275	2269	2248	2248	2243
CES_TI28	CES_TI28:err_bar	CES_VT28	CES_VT28:err_bar	CES_RT28	2280	2280	2280	2273	2253	2253	2248
CES_TI29	CES_TI29:err_bar	CES_VT29	CES_VT29:err_bar	CES_RT29	2285	2285	2285	2280	2259	2259	2253
CES_TI30	CES_TI30:err_bar	CES_VT30	CES_VT30:err_bar	CES_RT30	2290	2290	2290	2286	2264	2264	2259
CES_TI31	CES_TI31:err_bar	CES_VT31	CES_VT31:err_bar	CES_RT31	2295	2295	2295	2291	2269	2269	2264
CES_TI32	CES_TI32:err_bar	CES_VT32	CES_VT32:err_bar	CES_RT32	2300	2300	2300	2296	2273	2273	2269

Full Name	Motional Stark effect diagnostic system
Contact	J. Ko (jinseok@kfe.re.kr) , J. Ko (juyoung05@kfe.re.kr), J. Chung (jinil@kfe.re.kr)
Measuring properties	Magnetic pitch angle (J and q profiles when combined with EFIT)
Port assignment	M port – middle window
Channel information	25 spatial channels
Time resolution	100 Hz
Spatial Resolution	1 – 3 cm
Availability	Available by request

710 μm (core+clad+buffer), 25ch, 22.5mm, 1740-2280



Name		Data Type	Unit	Description
\spectro::top.mse.pmse.pmse_efit001				
Subnodes	:aa1gam ... :aa7gam	# of channels		'a' coefficients
	:rrrgam			
	:zzzgam			
	:pmse_ver	scalar		mse analysis version number
	:pmse_ic	string		invessel calibration flag name
	:pmse_fc	string		faraday calibration flag name
	:pmse_ge	string		geometry flag name
	:pmse_bc	string		bandpass filter calibration flag name
	:pmse_avg_dt	scalar		integration time

Name		Data Type	Unit	Description
\spectro::top.mse.pmse.pmse_efit001.tgamma				
Subnodes	:tgamma01 : :tgamma25	# of time points		'tgamma' for efit mse snap file for channel 1 - 25
\spectro::top.mse.pmse.pmse_efit001.sgamma				
Subnodes	:sgamma01 : :sgamma25	# of time points		'sgamma' for efit mse snap file for channel 1 - 25

Notes

- Recommended NB1 energy combinations:
 - When NB1A is used as MSE signal source, $\text{NB1A energy} \geq \text{NB1B energy} + 10 \text{ keV}$
 - When NB1B is used as MSE signal source, $\text{NB1B energy} \geq \text{NB1A energy}$
 - Using NB1C as MSE signal source is rare and not recommended.
- Although we need to experimentally confirm, the injection of NB12 is expected to have no effect on MSE measurements.

Current density (j)

Node name	Node name	Notes
\pmse_jv01	\pmse_je01	j value & error at $r/a = 0$ [MA/m ²]
\pmse_jv02	\pmse_je02	...
...	...	...
\pmse_jv20	\pmse_je20	j value & error at $r/a = 0.98$ [MA/m ²] (Constrained to zero)

Safety factor (q)

Node name	Node name	Notes
\pmse_qv01	\pmse_qe01	q value & error at $r/a = 0$ (a.k.a q ₀)
\pmse_qv02	\pmse_qe02	...
...	...	...
\pmse_qv20	\pmse_qe20	q value & error at $r/a = 0.98$ (Constrained to q ₉₅ from rtEFIT)

Normalized minor radius = $(R_{\text{major}} - R_{\text{magx}}) / (R_{\text{edge}} - R_{\text{magx}})$

where R_{magx} is the major radius at the magnetic axis inferred from the MSE pitch angle profile (zero-crossing).

Node name	Notes
\pmse_av01	1 st normalized minor radius ($r/a = 0$)
\pmse_av02	2 nd normalized minor radius
...	...
\pmse_av20	Last normalized minor radius ($r/a = 0.98$)

Others (q₀, R_{magx})

Node name	Notes
\pmse_magxv	Magnetic axis inferred from MSE pitch angle [m]
\pmse_magxe	Its error [m]
\pmse_qov	q ₀
\pmse_qoe	Its error

Ref for direct q & j

C. C. Petty et al, 'Analysis of current drive using MSE polarimetry without equilibrium reconstruction', Nucl. Fusion 42 (2002) 1124-1133

Name	Value	Usage
CCS_SHOT_NUMBER	Shot number	MSE filter setup for Bt and NB11 energy before shot.
PASS_SHOT_COUNTDOWN	Time left before shot start [sec]	
CCS_ITF_MEASURED	Current in TF coil [A]	Final settings at PASS_SHOT_COUNTDOWN = 20.
NB1_P1G1_RTN_VOLT	NB1A energy [keV]	
NB1_P2G1_RTN_VCMD	NB1B energy [keV]	
NB1_P3G1_RTN_VCMD	NB1C energy [keV]	

Visible Spectroscopy – H alpha

General Information

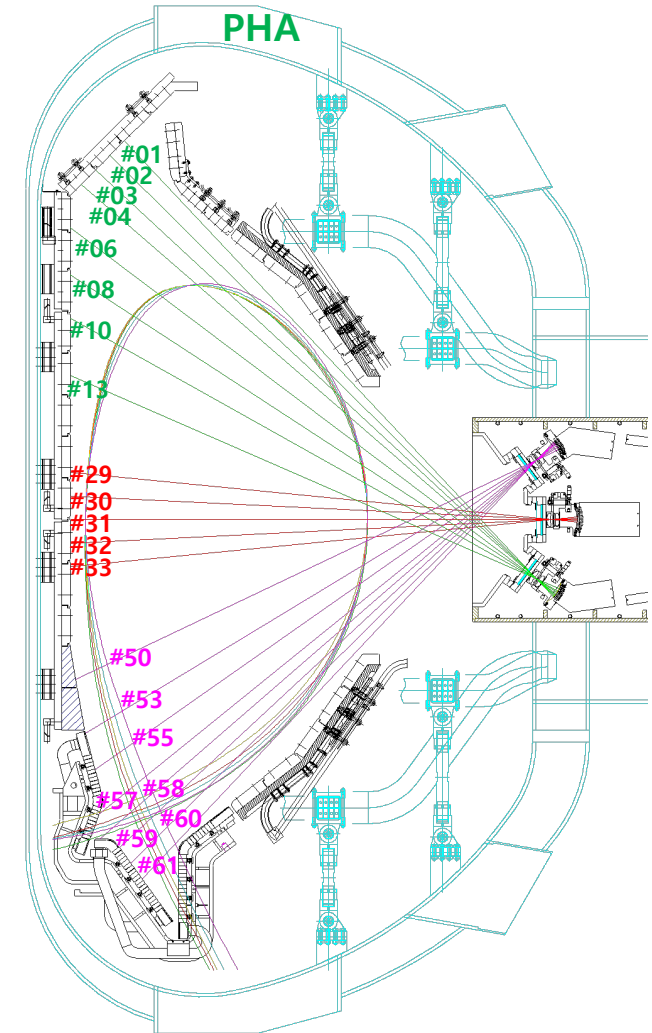
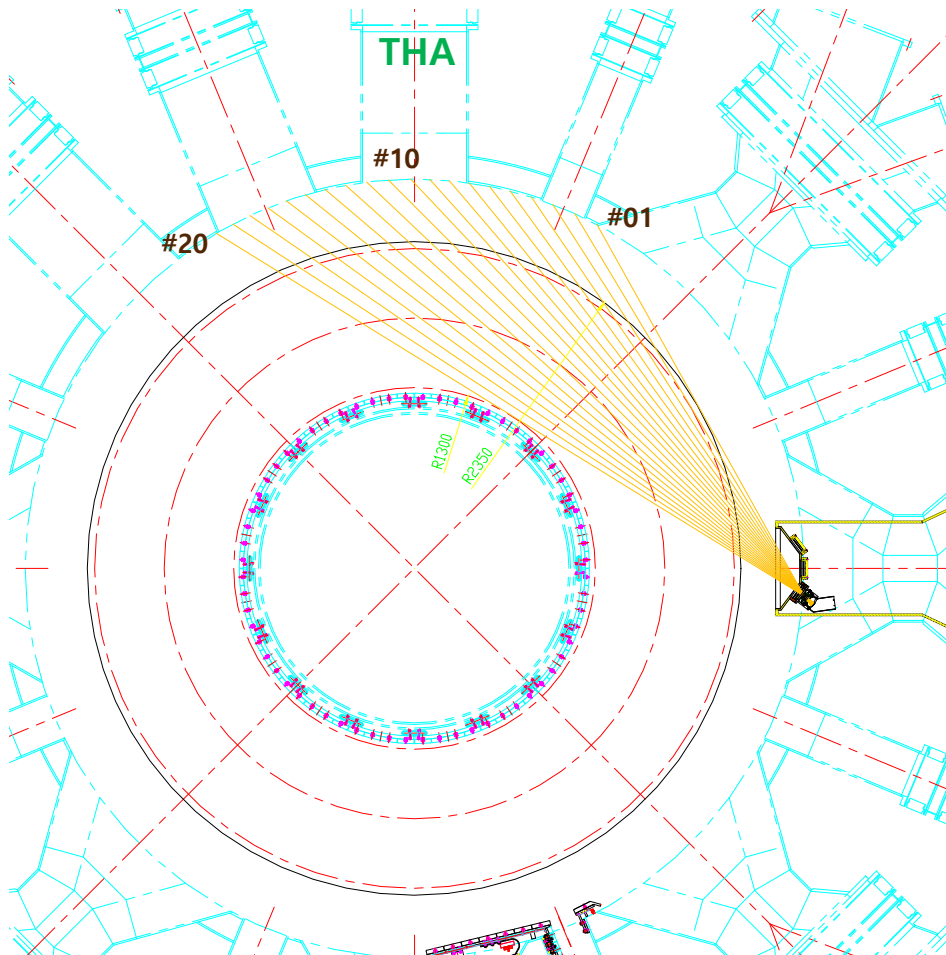
Full Name	H α monitoring system
Contact	J. Jang (jjh4368@kfe.re.kr), D.C. Seo (dcseo@kfe.re.kr)
Measuring properties	H alpha emission intensity
Port assignment	J
Channel information	Horizontal 20 ch on midplane, Poloidal 21 ch
Time resolution	20 kHz (default), ~ 500kHz (maximum on request)
Spatial Resolution	-
Availability	routinely available

Notes

- In 2024, 61 poloidal channels were upgraded, 21 channels are in use.

Visible Spectroscopy - H alpha

Channels Layout



Visible Spectroscopy - H alpha

MDS+ Node Information

Name	Sampling	Unit	Description
\TOR_HA01 : \TOR_HA20	20 kHz	photon flux (ph/m ² /n m/sr/s)	Toroidal channels for H alpha emission photon flux
\POL_HA01 : \POL_HA61			Poloidal channels for H alpha emission photon flux
Subnodes :FOO		V	raw data

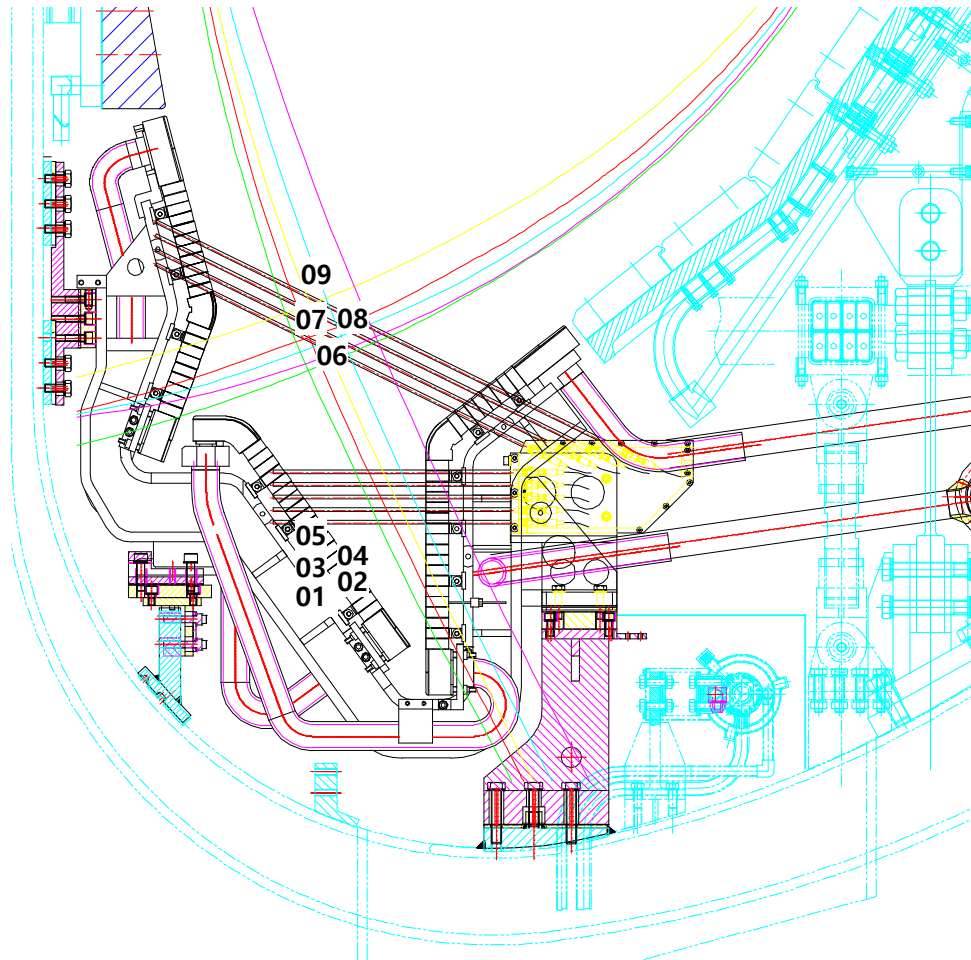
Visible Spectroscopy – Divertor H alpha

General Information

Full Name	Divertor H α monitoring system
Contact	D.C. Seo (dcseo@kfe.re.kr), J. Jang (jjh4368@kfe.re.kr)
Measuring properties	H alpha emission intensity in divertor
Port assignment	K, G
Channel information	divertor 9 ch / Port
Time resolution	20 kHz (default), ~ 500kHz (maximum on request)
Spatial Resolution	-
Availability	routinely available

Visible Spectroscopy – Divertor H alpha

Channels Layout



Visible Spectroscopy – Divertor H alpha

MDS+ Node Information

Name		Sampling	Unit	Description
₩DIV_KHA01 : ₩DIV_KHA09		20 kHz	photon flux (ph/m ² / nm/sr/s)	Divertor channels for H alpha emission photon flux
₩DIV_GHA01 : ₩DIV_GHA09		20 kHz	photon flux (ph/m ² / nm/sr/s)	Divertor channels for H alpha emission photon flux
Subnodes	:FOO		V	raw data

Visible Spectroscopy - FS

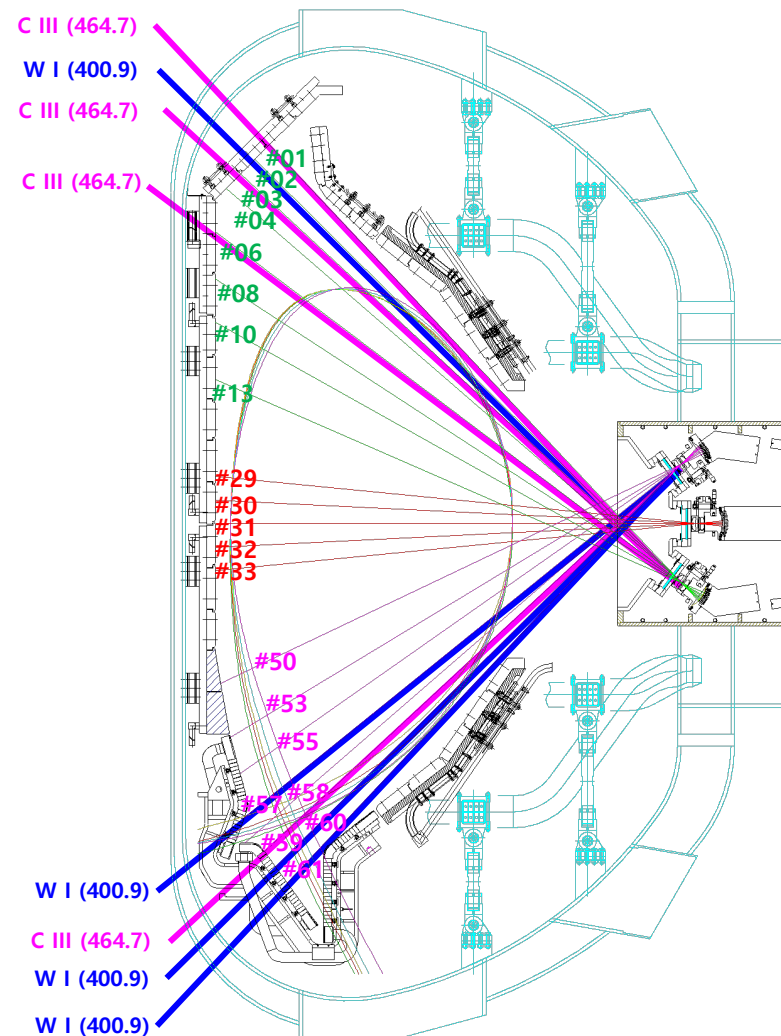
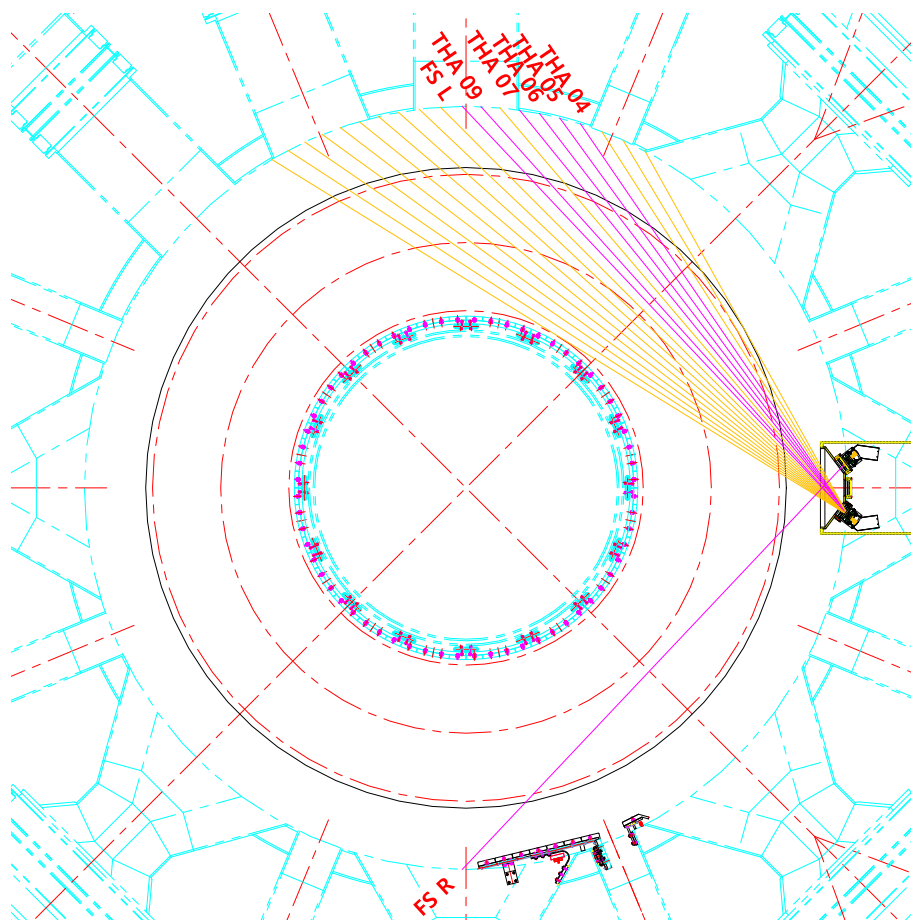
General Information

Full Name	Filterscope
Contact	D.C. Seo (dcseo@kfe.re.kr), J. Jang (jjh4368@kfe.re.kr)
Measuring properties	Impurities intensity
Port assignment	J
Channel information	8 ch in poloidal direction
Time resolution	20 kHz (default)
Spatial Resolution	-
Availability	routinely available

Visible Spectroscopy - FS

Channels Layout

Toroidal No Operation



Visible Spectroscopy - FS

MDS+ Node Information

Name		Sampling	Unit	Description
\POL_VB41		20 kHz	photon flux (ph/m²/nm/sr/s)	C(III), poloidal direction (PHA_1)
\POL_VB42				W(I), poloidal direction (PHA_2)
\POL_VB43				C(III), poloidal direction (PHA_3)
\POL_VB44				C(III), poloidal direction (PHA_6)
\POL_VB45				W(I), poloidal direction (PHA_57)
\POL_VB46				C(III), poloidal direction (PHA_59)
\POL_VB47				W(I), poloidal direction (PHA_60)
\POL_VB48				W(I), poloidal direction (PHA_61)
Subnodes	:FOO		V	raw data

* Visible Bremsstrahlung

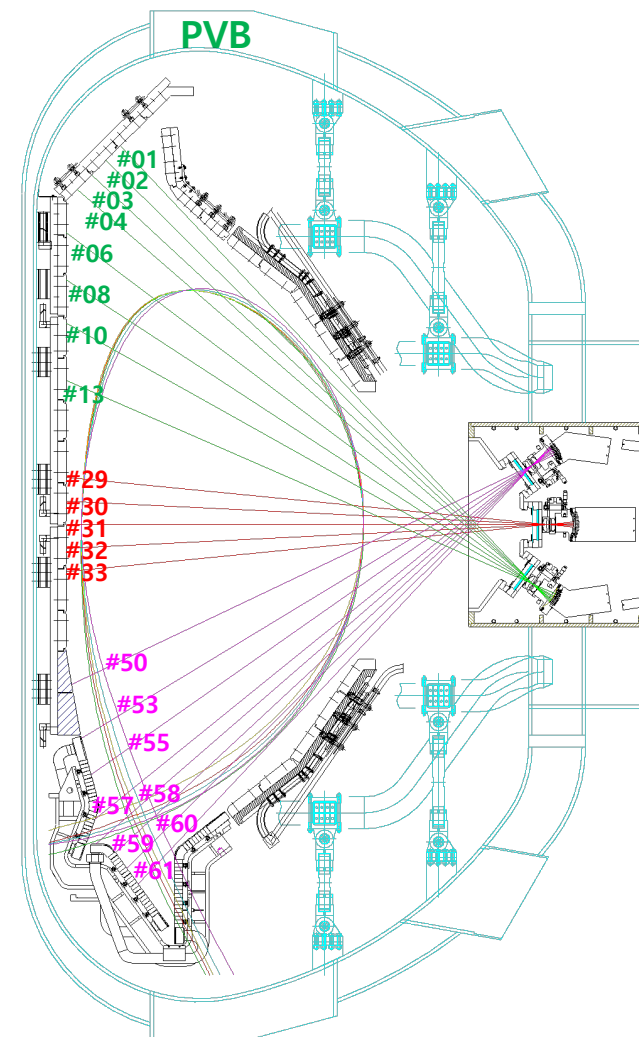
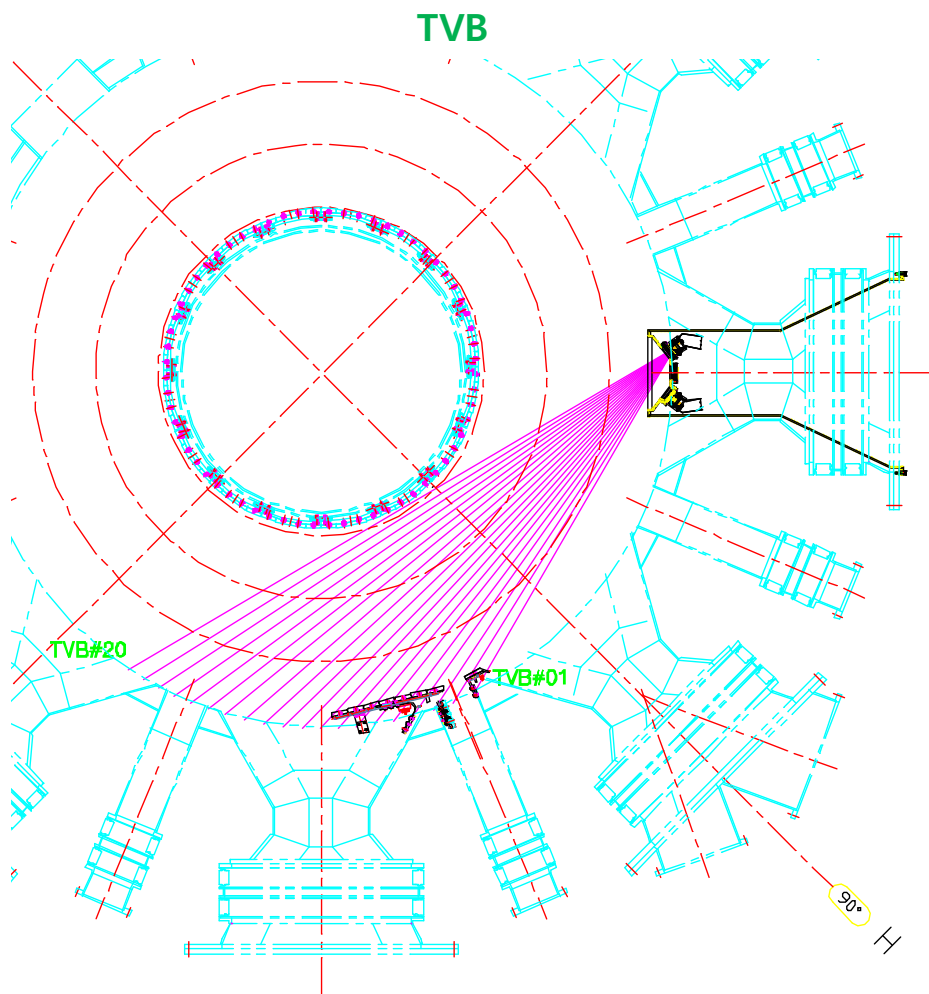
Visible Spectroscopy - VBS

General Information

Full Name	Visible Bremsstrahlung System
Contact	J. Jang (jjh4368@kfe.re.kr), D.C. Seo (dcseo@kfe.re.kr)
Measuring properties	effective Z number (Z_{eff} , impurities)
Port assignment	J
Channel information	Horizontal 20 ch on midplane, poloidal 21 ch
Time resolution	20 kHz (default), ~ 500kHz (maximum on request)
Spatial Resolution	-
Availability	routinely available

Visible Spectroscopy - VBS

Channels Layout



Visible Spectroscopy - VBS

MDS+ Node Information

Name		Sampling	Unit	Description
\TOR_VB01 : \TOR_VB20		20 kHz	V	toroidal channels
\POL_VB01 : \POL_VB61		20 kHz	V	poloidal channels
Subnodes	:FOO		V	raw data

Notes

- In 2020, toroidal channel #07 and #08 have been changed to He I (467nm) and Ne I (640 nm) filters, which are used for SPI experiments.
 - In 2021, toroidal channel #07 and #08 returned to VBS
 - In 2024, 10 toroidal channels were added.
 - In 2024, 61 poloidal channels were upgraded, 20 channels are in use.
 - Please consult Juhyeok Jang (jjh4368@kfe.re.kr) for Z_{eff} calculation.
 - In 2024 campaign, the followings were for CIII and WI monitoring:
 - CIII (464.7 nm) → -\POL_VB59:foo
 - WI (400.9 nm) → -\POL_VB60:foo
- Please consult Juhyeok Jang (jjh4368@kfe.re.kr)

Visible Spectroscopy - VSS (I)

General Information

Full Name	Visible Survey Spectrometer (AVANTES)
Contact	D.C. Seo (dcseo@kfe.re.kr), J. Jang (jjh4368@kfe.re.kr)
Measuring properties	A variety of impurities species in visible range
Port assignment	J
Channel information	Horizontal 1 ch on midplane (same LoS with THA#13)
Time resolution	~50Hz
Spatial Resolution	-
Availability	Routinely available

Notes

- Not available on MDS+, please consult to contact person.
- This system is operated using Toroidal H-alpha #13 channel. Please check VSS(IV) section for the channels layout.

Visible Spectroscopy - VSS (II)

General Information

Full Name	Visible Survey Spectrometer (Princeton Instruments, PI)
Contact	J. Jang (jjh4368@kfe.re.kr), D.C. Seo (dcseo@kfe.re.kr)
Measuring properties	A variety of impurities species in visible range
Port assignment	J
Channel information	poloidal 1 ch (same LoS with PVB#56, PHA#56)
Time resolution	100Hz
Spatial Resolution	-
Availability	Not available

Notes

- Not available on MDS+, please consult to contact person.
- This system is operated using Poloidal VB #56 channel. Please check VSS(IV) section for the channels layout.

Visible Spectroscopy - VSS (III)

General Information

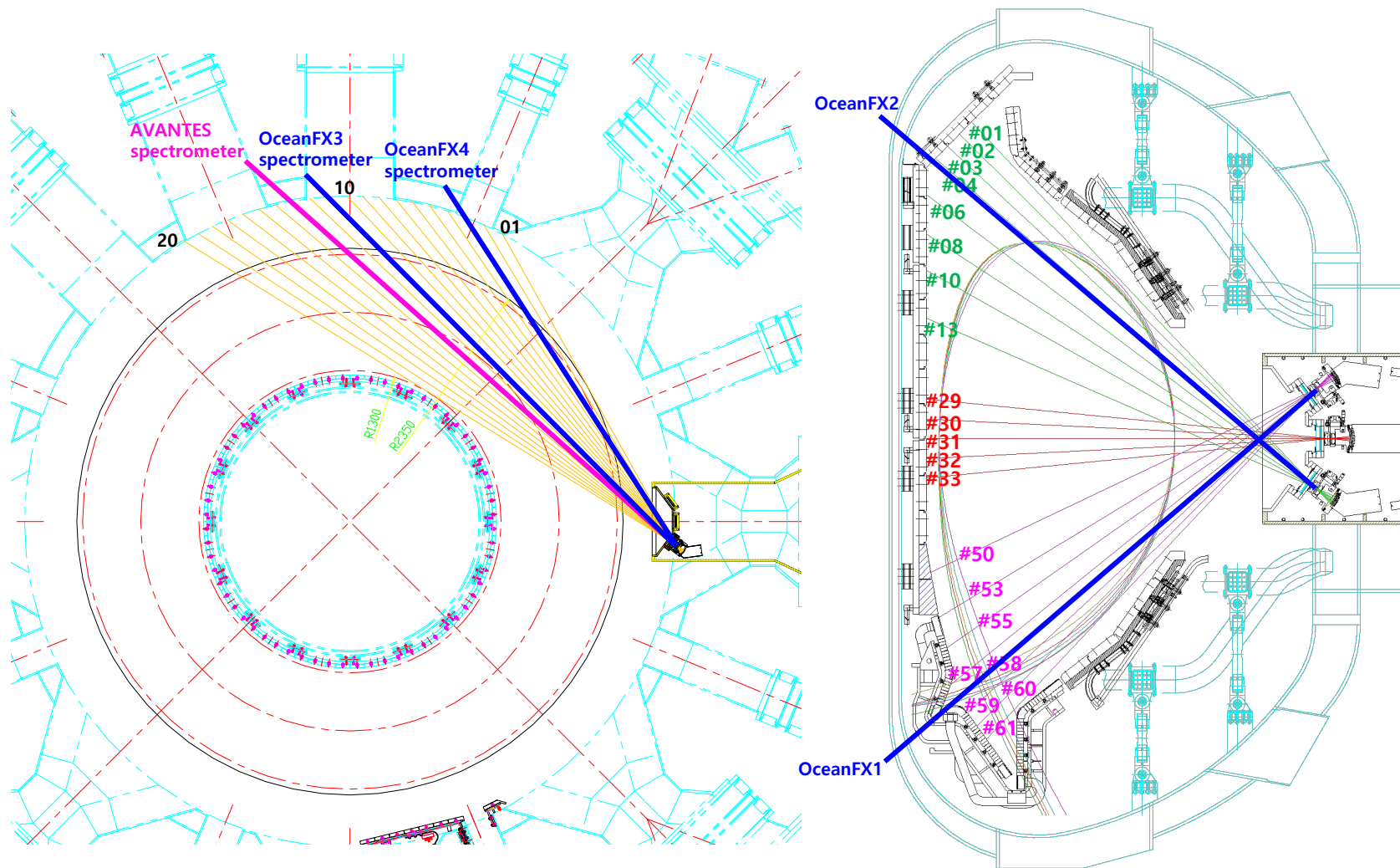
Full Name	Visible Survey Spectrometer (OceanFX)
Contact	D.C. Seo (dcseo@kfe.re.kr), J. Jang (jjh4368@kfe.re.kr)
Measuring properties	A variety of impurities species in visible range
Port assignment	J
Channel information	Horizontal 2 ch on midplane, poloidal 2 ch
Time resolution	500Hz
Spatial Resolution	-
Availability	Routinely available

Notes

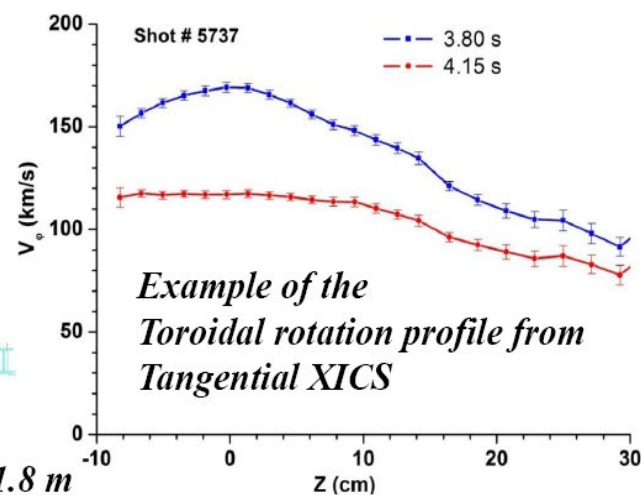
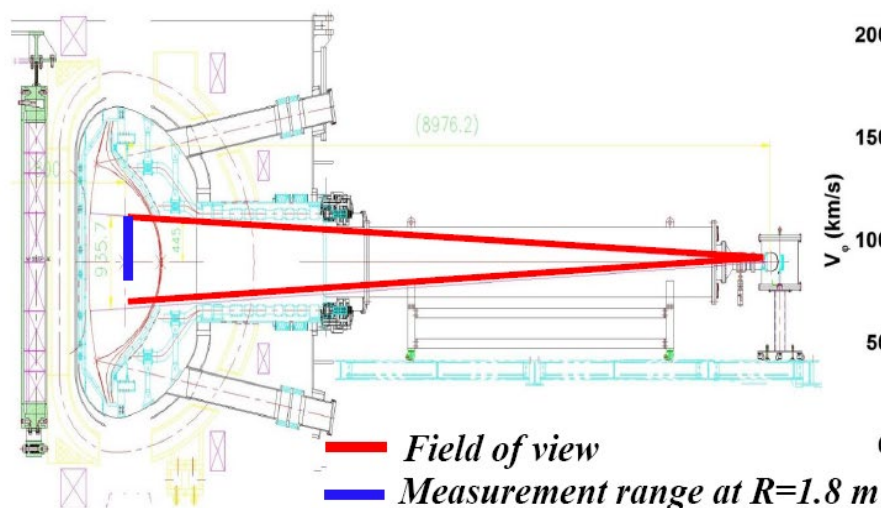
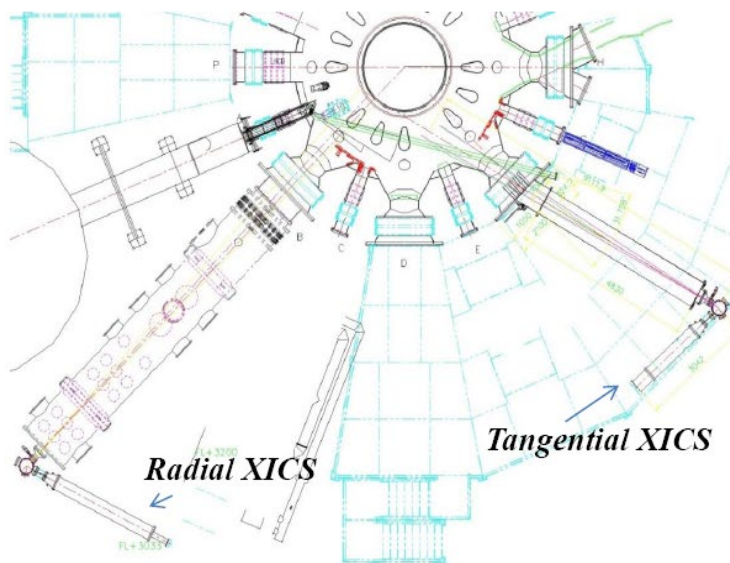
- Not available on MDS+, please consult to contact person.
- This system is operated using toroidal H-alpha #3, #11 and poloidal H-alpha #4, #57 channels. Please check VSS(IV) section for the channels layout.

Visible Spectroscopy - VSS (IV)

Channels Layout



Full Name	X-ray Imaging Crystal Spectrometer
Contact	S. G. Lee (sglee@kfe.re.kr)
Measuring properties	toroidal rotation velocity (V_ϕ), ion temperature(T_i), electron temperature (T_e)
Port assignment	B (radial), F (tangential)
Channel information	Vertical profile in the core region at ($R=1.8\text{m}$, $-0.1\text{ m} < Z < +0.3\text{ m}$)
Time resolution	2 – 10 ms depending on the photon count rate
Spatial Resolution	< 2 cm depending on the photon count rate
Availability	Argon puffing required



Name	Sampling	Unit	Description
\TXCS_VR053		km/s	Toroidal rotation at R = 1.8 m, Z = 0 m
\TXCS_Tl053		keV	Ion temperature at R = 1.8 m, Z = 0 m
\TXCS_TE053		keV	Electron temperature at R = 1.8 m, Z = 0 m

Notes

- The radial and tangential XICS systems have been installed as shown above layout. The main diagnostic is the tangential XICS system since it can provide the toroidal rotation, ion and electron temperatures simultaneously. Whereas the radial XICS system can provide the poloidal rotation, ion and electron temperatures simultaneously.
- Although XICS system can provide core profile measurement, one core location at R=1.8 m and Z=0 m from the tangential XICS system is routinely uploaded in the MDS+ server due to the detector limitation. The core profile measurements from both the radial and tangential XICS systems will be uploaded when the proper detector systems are ready.
- The measurement wavelength range of Ar XVII (helium-like) is from the resonance line w, $1s^2\ 1s0-1s2p\ 1P_1$ at 3.9494 Å, to the forbidden line z, $1s^2\ 1s0-1s2p\ 3S_1$, at 3.9944 Å.
- A proper argon gas puffing (or multiple puffing for long-pulse discharges) is required for the XICS operation.

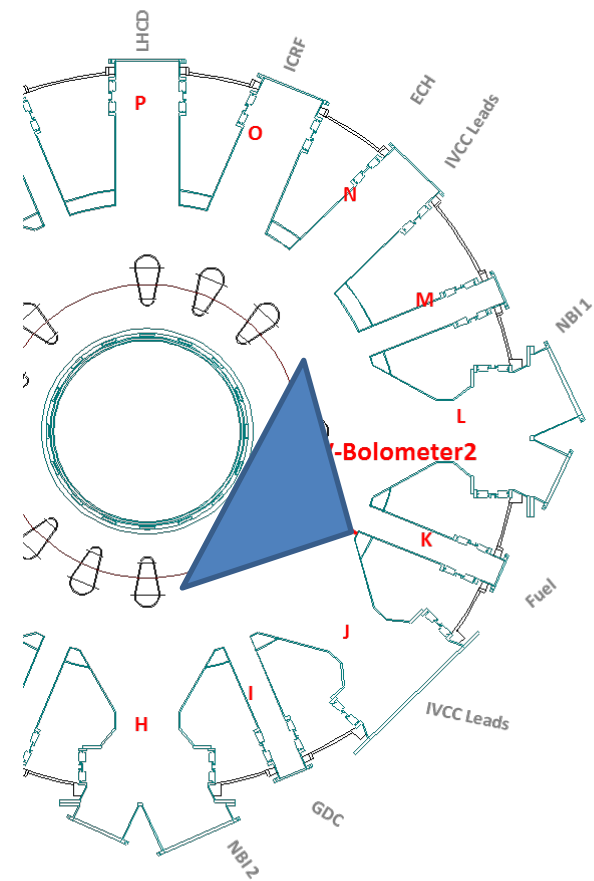
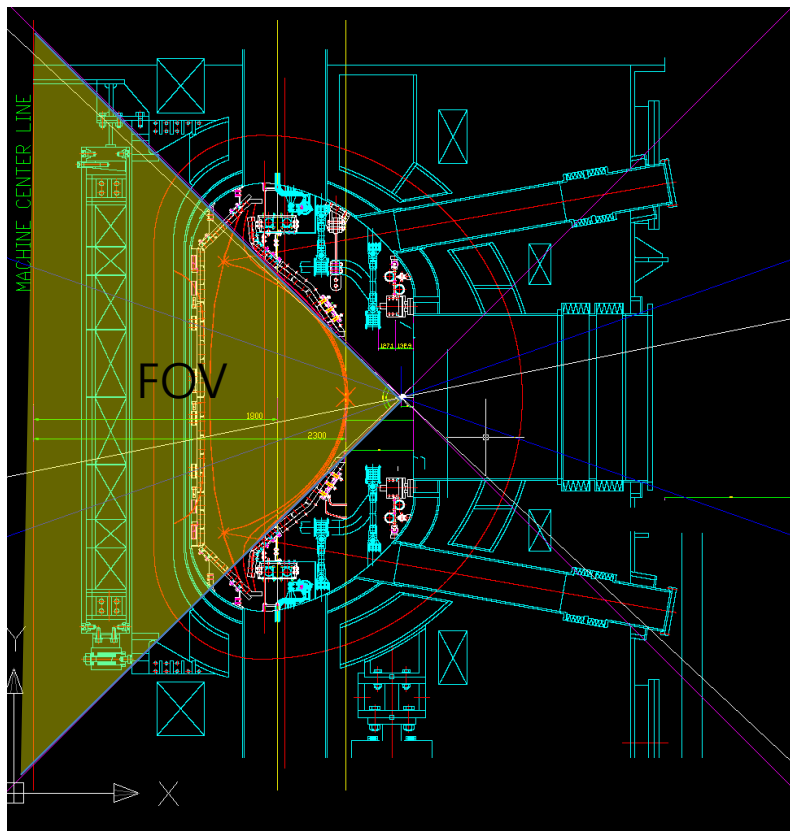
AXUV bolometer

General Information

Full Name	Absolute eXtreme Ultra-Violet photodiode bolometer
Contact	S. Oh (stoh@kfe.re.kr)
Measuring properties	Radiation Power (P_{rad})
Port assignment	J-K (feedthrough K)
Channel information	1 single channel
Time resolution	100.0 μs
Spatial Resolution	
Availability	routinely available

AXUV bolometer

Channels Layout



Notes

- Data are sensor signals [V] representing relative radiation power values
- The absolute radiation power cannot be derived due to the sensitivity changes of the AXUV sensors during campaign.

After-Shot Data

Name	Sampling	Unit	Description
\ax3_bolo02:FOO		V	MDS+ node name Photodiode signal [V] for the total radiation power, at the toroidal location between J-Port and K-Port.

- basic sampling rate is 10 KHz. (max. 1 MHz)

Real-Time Data

RFM map :

category	HEX address	0	4	8	C	no. of bytes
IRVB-> PCS	0x3110100	id	AXUV_BOLO01 (float, V)	AXUV_BOLO02 (float, V)	AXUV_BOLO03 (float, V)	16

↑
Main sensor

- basic sampling rate is 1 KHz. (max. 2 kHz)

Filtered AXUV Array (FAA)

General Information

Full Name	Filtered AXUV Array
Contact	Jayhyun Kim (jayhyunkim@kfe.re.kr), Jae-Wook Kim (jjwkim@kfe.re.kr)
Measuring properties	Plasma radiation (during disruption)
Port assignment	D, O
Channel information	Bare AXUV, Al filter, Multilayer filter, Blind channel
Time resolution	10 μ s
Spatial Resolution	6 cm (for PFAA)
Availability	routinely available

Filtered AXUV Array (FAA)

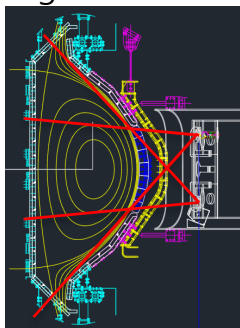
Channels Layout

Poloidal Filtered AXUV Array (PFAA)

- ✓ Radiation distribution on each poloidal plane (tomography)

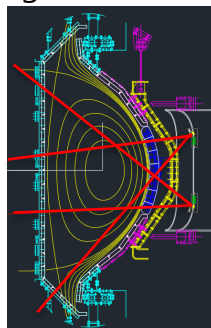
D-port

Poloidal viewing angle 54°



O-port

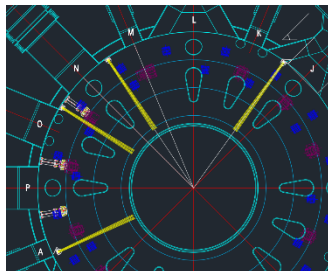
Poloidal viewing angle 41°



Toroidal Filtered AXUV Array (TFAA)

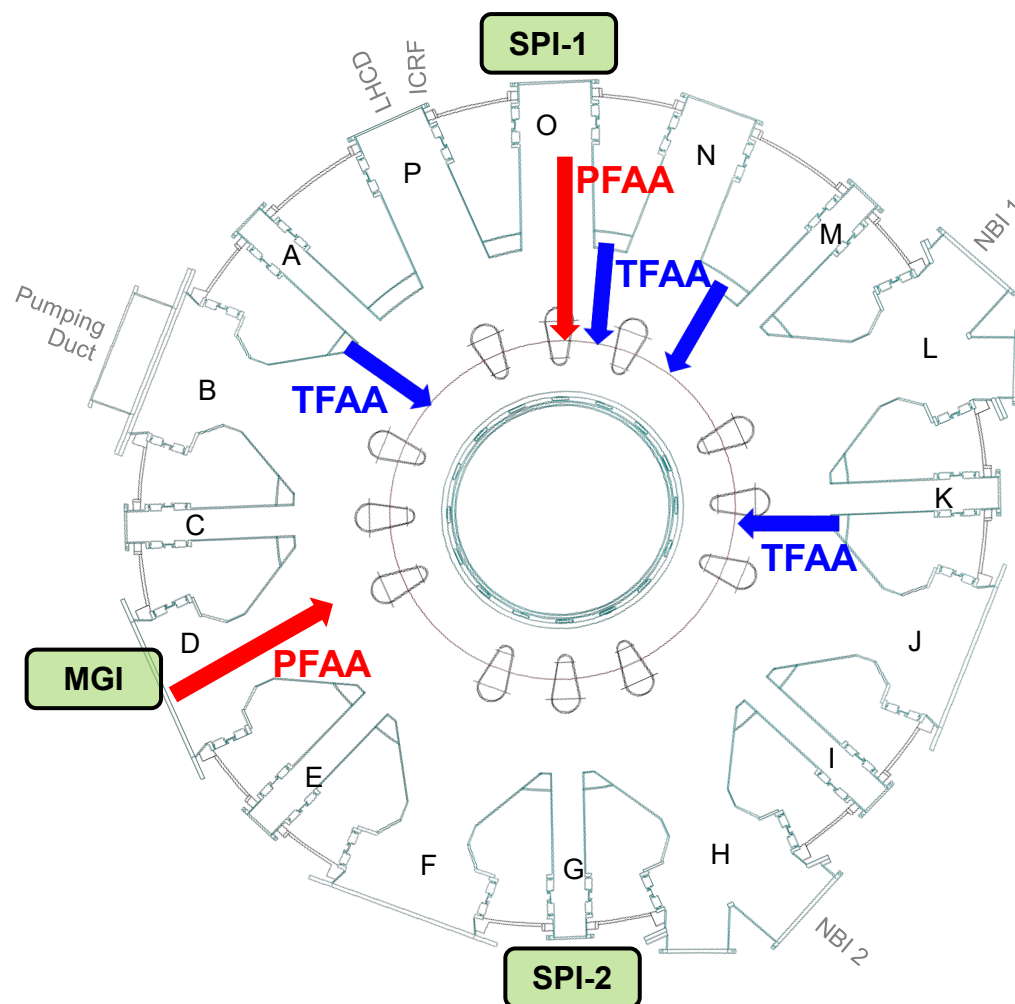
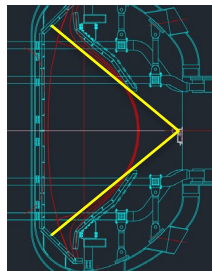
- ✓ Radiation power on different toroidal location

Top View



Side View

Poloidal viewing angle 79°



Filtered AXUV Array (FAA)

MDS+ Node Information

Name	Sampling	Unit	Description
\PFAA_DU001 ... \PFAA_DU060	100 kHz	V	Plasma radiation signal from upper PFAA at D port
\PFAA_DL001 ... \PFAA_DL060	100 kHz	V	Plasma radiation signal from lower PFAA at D port
\PFAA_OU001 ... \PFAA_OU060	100 kHz	V	Plasma radiation signal from upper PFAA at O port
\PFAA_OL001 ... \PFAA_OL060	100 kHz	V	Plasma radiation signal from lower PFAA at O port
\TFAA_001 ... \TFAA_016	100 kHz	V	Plasma radiation signal from TFAAs

Notes

- This node provides the raw voltage signal which is integrated through the line of sight. To assess exact radiated power, please contact Juhyeok Jang (jjh4368@kfe.re.kr)

IRTV - Tangential

General Information

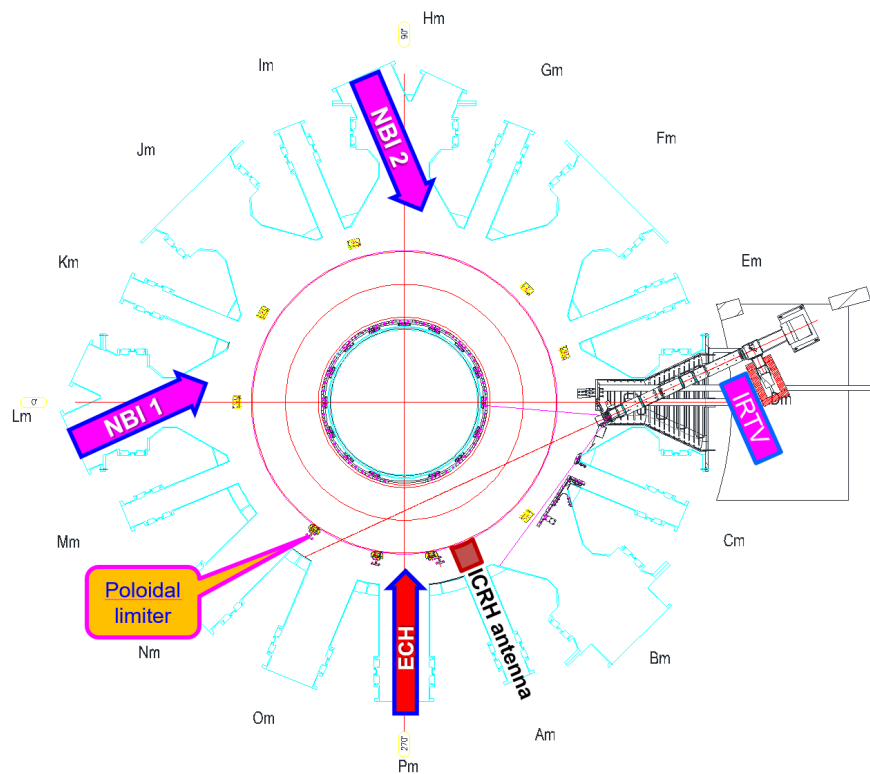
Full Name	Tangential Infrared TV
Contact	H. H. Lee (jdfm@kfe.re.kr)
Measuring properties	Surface temperature of first wall(T), heat flux(q_{tan})
Port assignment	D middle port
Channel information	2D image (640x512)
Time resolution	100 Hz ~ 36 kHz (depends on window size)
Spatial Resolution	3 ~ 8 mm/pixel
Availability	routinely available

Notes

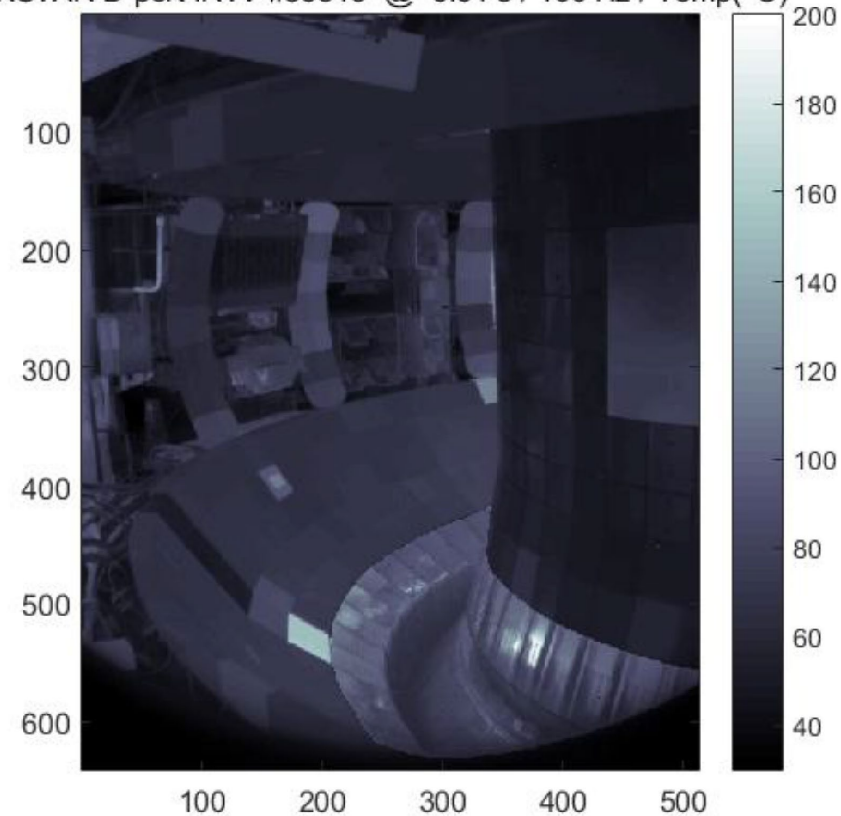
- MDS+ node not assigned yet but data can be provided by request.
- Video clip of the measurement result is available on KDIS (KSTAR Data Integration System) server

IRTV - Tangential

Channels Layout



KSTAR D-port IRTV #35819 @ -0.01 s / 100 Hz / Temp($^{\circ}$ C)



IRVB - Tangential

General Information

Full Name	Infra-Red Video Bolometer
Contact	S. Oh (stoh@kfe.re.kr)
Measuring properties	Radiation Profile
Port assignment	D middle port
Channel information	2D image (640x512, raw data)
Time resolution	100 Hz (integration time : 1 msec)
Availability	routinely available

Notes

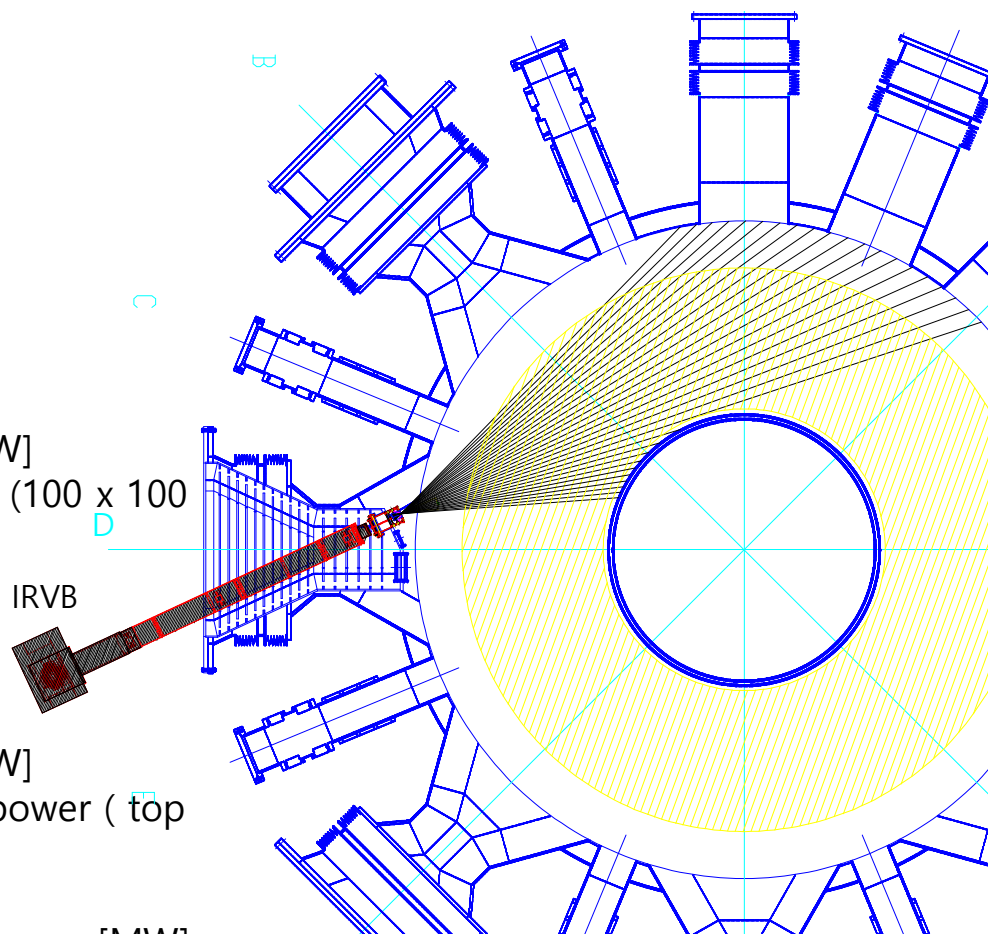
- RFM is available after Jan 8th 2024.
- DDS is available after 2024 campaign.

IRVB - Tangential

Channels Layout

Infra-red imaging video bolometer (IRVB)

- Main Bolometer System
 - >> absolute Value can be serviced.
- Data :
 - After-shot
 - Data rate : 100 Hz
 - Data contents:
 - total radiation power [MW]
 - 2D reconstructed Profiles (100 x 100 resolutions)
 - Streaming
 - Data rate: ~ 25 Hz
 - RFM :
 - ✓ total radiation power [MW]
 - ✓ Sampled Grid Radiation power (top 90% value)
 - DDS :
 - ✓ absolute total radiation power [MW]
 - ✓ 2D radiation profile (100 x 100 resolutions)



IRVB - Tangential

DATA Service

[After-shot]

- **MDS+**
 - Data type : total radiation power (No profile data)
 - Addr. : expName= "IMAGE", nodeName="\IRVB1_PRAD"
- **WebServer**
 - Data type : data plot, **2D profile data**, total radiation power
 - Visualization: movie file
 - Addr. : 172.17.112.125
- **KDIS**
 - Data type : plot data (movie file)

[Streaming (Real-Time)]

- **RFM**
 - Data : total radiation power, max power of sampled grid (No profile data)
 - Addr. : 0x3110000 ~
- **DDS**
 - Data: **2D profile data**, total radiation power
 - Addr. : topicName = "IRVB_RT_Recon" on network (172.17.112.XXX)

IRVB - Tangential

Service Layout

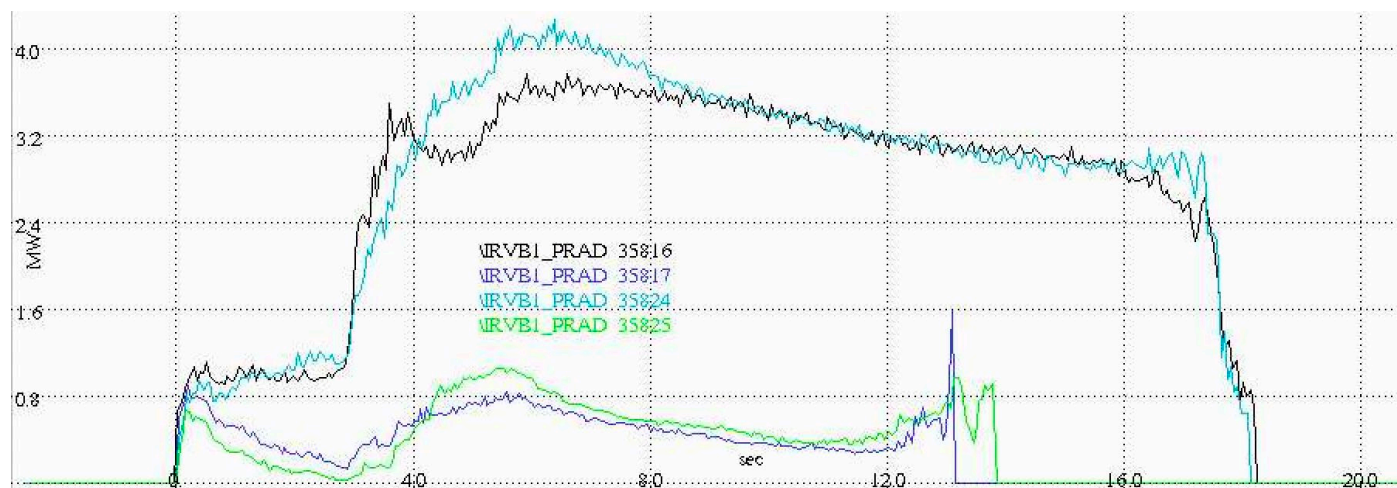
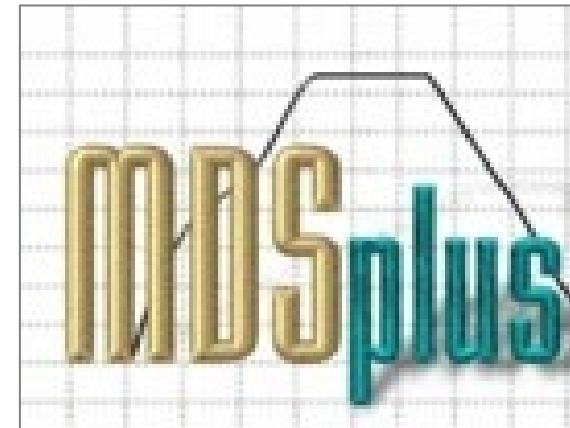
- After-shot data service

1. MDSplus

Data service:
total radiation power [MW]

Addr. :
expName= "IMAGE"
nodeName="\IRVB1_PRAD"
Description = "total radiation power [MW]"

Notice : plot using Jscope



IRVB - Tangential

Service Layout

• After-shot data service

2. Website

Data service:

- visualization (movie file)
- 2d profile [MW] (res.: 100 x 100)
- total radiation power [MW]

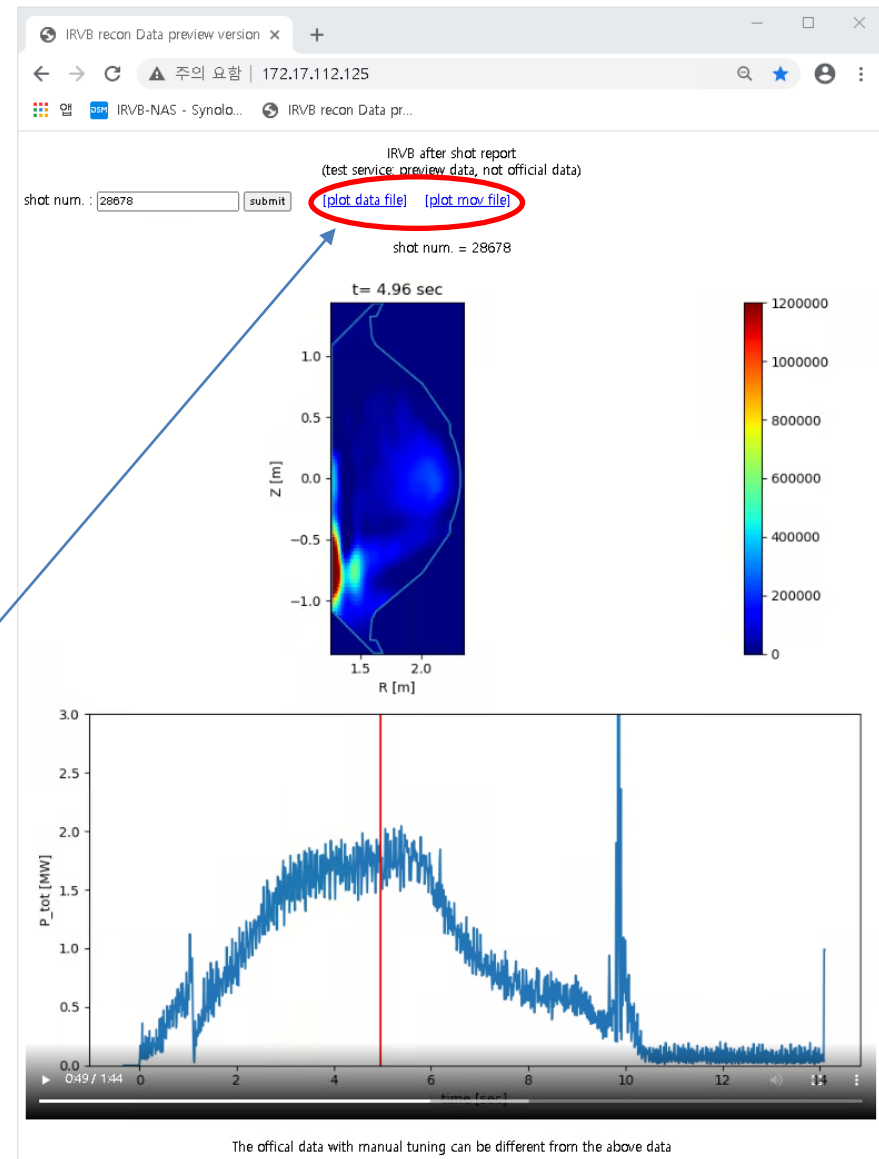
Addr. :

- <http://172.17.112.125>
- Accessible in KSTAR Exp. Network

Notice:

Data files can be downloaded from the website

If you have trouble in accessing the website, contact KSTAR network/computer team.



• Streaming (Real-Time) Data service

1. RFM

Data service:

- total radiation power [MW]
- Max power of each Grid (no full 2d profile)

Update Rate:

- ≤ 25 Hz

Addr. :

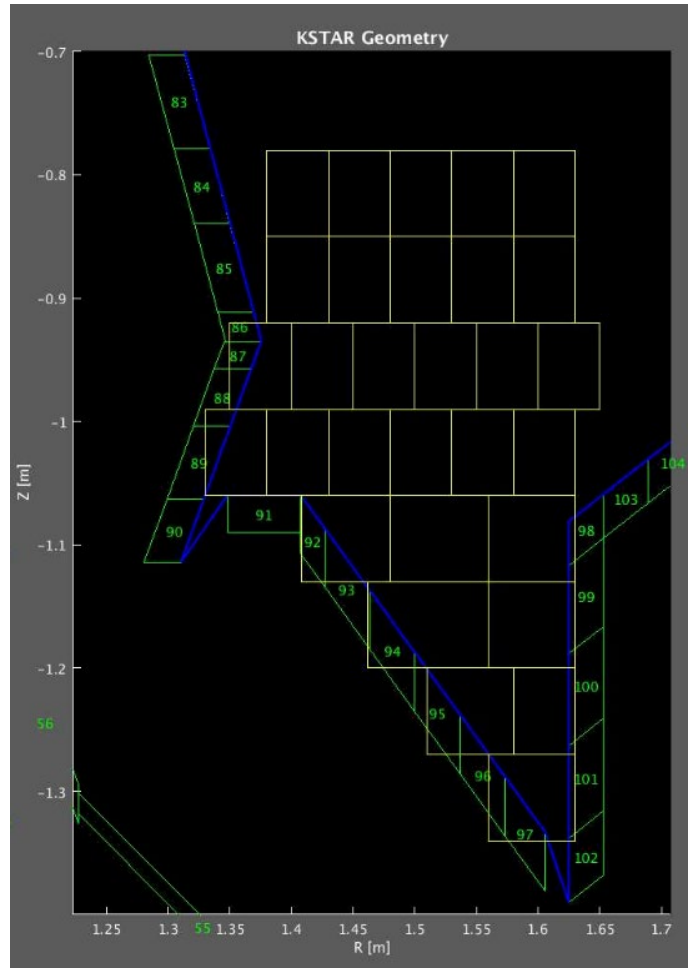
category	HEX address	0	4	8	C	no. of bytes
IRVB-> PCS	0x3110000	time_sec	timestamp_sec	total radiation_MW	fault_code	16
	0x3110010	p_MW[0]	p_MW[1]	p_MW[2]	p_MW[3]	32
	0x3110020	p_MW[4]	p_MW[5]	p_MW[6]	p_MW[7]	48
	0x3110030	p_MW[8]	p_MW[9]	p_MW[10]	p_MW[11]	64
	0x3110040	p_MW[12]	p_MW[13]	p_MW[14]	p_MW[15]	80
	0x3110050	p_MW[16]	p_MW[17]	p_MW[18]	p_MW[19]	96
	0x3110060	p_MW[20]	p_MW[21]	p_MW[22]	p_MW[23]	112
	0x3110070	p_MW[24]	p_MW[25]	p_MW[26]	p_MW[27]	128
	0x3110080	p_MW[28]	p_MW[29]	dummy	dummy	144
	0x3110090	dummy	dummy	dummy	dummy	160

Data type : all float

- Streaming (Real-Time) Data service

1. RFM

Grid coordinates:



	R0 [m]	Z0 [m]	dr	dz
p_MW[0]	1.408	-1.13	0.072	0.07
p_MW[1]	1.48	-1.13	0.08	0.07
p_MW[2]	1.56	-1.13	0.07	0.07
p_MW[3]	1.462	-1.2	0.098	0.07
p_MW[4]	1.56	-1.2	0.07	0.07
p_MW[5]	1.51	-1.27	0.07	0.07
p_MW[6]	1.58	-1.27	0.05	0.07
p_MW[7]	1.56	-1.34	0.07	0.07
p_MW[8]	1.33	-1.06	0.05	0.07
p_MW[9]	1.38	-1.06	0.05	0.07
p_MW[10]	1.43	-1.06	0.05	0.07
p_MW[11]	1.48	-1.06	0.05	0.07
p_MW[12]	1.53	-1.06	0.05	0.07
p_MW[13]	1.58	-1.06	0.05	0.07
p_MW[14]	1.35	-0.99	0.05	0.07
p_MW[15]	1.4	-0.99	0.05	0.07
p_MW[16]	1.45	-0.99	0.05	0.07
p_MW[17]	1.5	-0.99	0.05	0.07
p_MW[18]	1.55	-0.99	0.05	0.07
p_MW[19]	1.6	-0.99	0.05	0.07
p_MW[20]	1.38	-0.92	0.05	0.07
p_MW[21]	1.43	-0.92	0.05	0.07
p_MW[22]	1.48	-0.92	0.05	0.07
p_MW[23]	1.53	-0.92	0.05	0.07
p_MW[24]	1.58	-0.92	0.05	0.07
p_MW[25]	1.38	-0.85	0.05	0.07
p_MW[26]	1.43	-0.85	0.05	0.07
p_MW[27]	1.48	-0.85	0.05	0.07
p_MW[28]	1.53	-0.85	0.05	0.07
p_MW[29]	1.58	-0.85	0.05	0.07

IRVB - Tangential

Service Layout

• Streaming (Real-Time) Data service

2. DDS

Data service:

- visualization
- 2D profile [MW] (res.: 100 x 100)
- total radiation power [MW]

Update Rate:

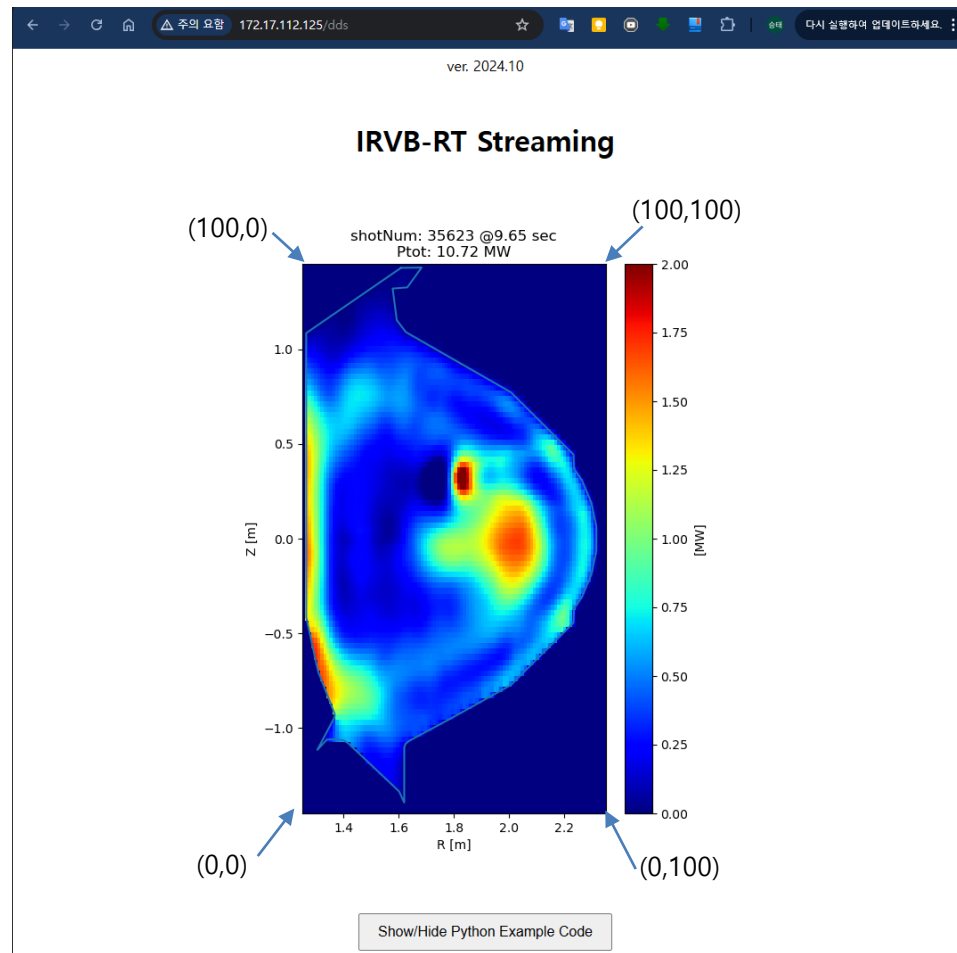
- ≤ 25 Hz

Addr. :

- DDS data:
Networks: 172.17.112.x
Topic name: "IRVB_RT_Recon"
- DDS visualization:
<http://172.17.112.125/dds>

Notice:

If you have trouble in accessing the web-site, contact KSTAR network/computer team.



- Streaming (Real-Time) Data service

2. DDS

Code Example:

```
global Nr, Nz
Nr=100
Nz=100

from dataclasses import dataclass
from cyclonedds.idl import IdlStruct
from cyclonedds.topic import Topic

from cyclonedds.domain import DomainParticipant
from cyclonedds.sub import Subscriber
from cyclonedds.sub import DataReader

from cyclonedds.util import duration
import numpy as np

@dataclass
class Message(IdlStruct):
    shotNum: int
    time: float
    recon1D_MW: array[float,10000]
    totRadPrad_MW: float

participant = DomainParticipant()
topic = Topic(participant, "IRVB_RT_Recon", Message)
subscriber = Subscriber(participant)
reader = DataReader(subscriber, topic)

for msg in reader.take_iter(timeout = duration(weeks=20)):
    print("I got shotNum:", msg.shotNum)
    print("I got time:", msg.time)
    print("I got recon1D_MW:", msg.recon1D_MW)
    print("I got totRadPrad_MW:", msg.totRadPrad_MW)

    recon2D_MW = np.reshape(msg.recon1D_MW,(Nr, Nz))
```

This code is a Python example for receiving IRVB-RT streaming data, specifically the entire IRVB 100x100 2D recon profile data, in real-time (streaming). CycloneDDS package must be installed, and the access is only possible in KSTAR experiment network.

Code is available on <http://172.17.112.125/dds>

IRTV - Divertor

General Information

Full Name	Divertor Infrared TV
Contact	H. H. Lee (jdfm@kfe.re.kr)
Measuring properties	divertor surface temperature (T_{div}), divertor heat flux (q_{div})
Port assignment	F middle port
Channel information	2D image (640 x 512)
Time resolution	150 Hz – 35 kHz (depending on the viewing size)
Spatial Resolution	~ 1.2 mm/pixel
Availability	To be updated

Notes

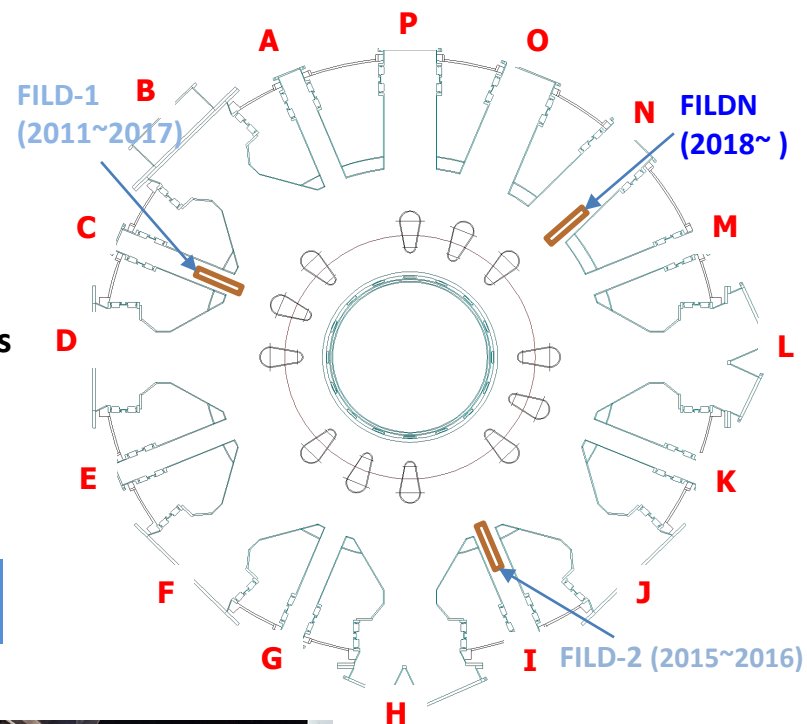
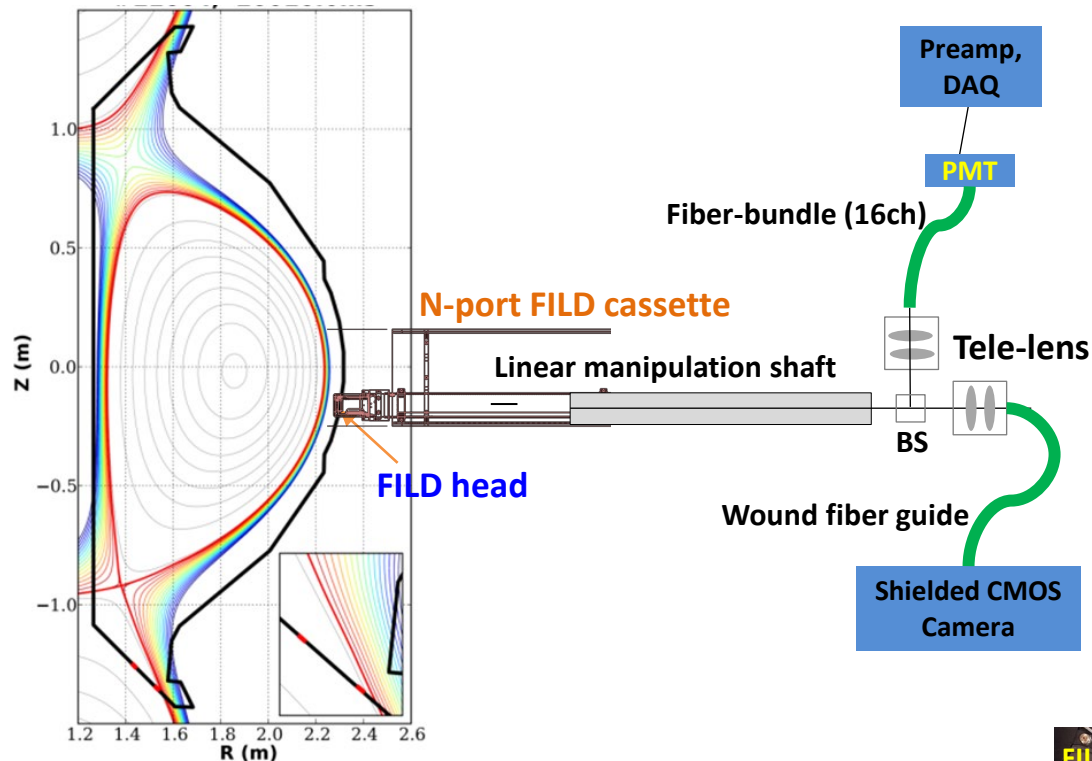
- MDS+ node not assigned yet but data can be provided by request.
- Divertor heat flux profile is estimated from the time history of the surface temperature. It takes some time to obtain the result. The divertor heat flux data is also provided by request.

Full Name	Fast-Ion Loss Detector
Contact	Junghee Kim (kimju@kfe.re.kr)
Measuring properties	- Local deposition power of the lost fast-ions - Phase-space of lost fast-ions: Detectable range is (40 – 500 keV) of energy and (5° – 87°) of pitch-angle.
Port assignment	C (FILD-1, 2011-2017), I (FILD-2, 2015 - 2016), N (FILDN, 2018 -)
Channel information	3 x 3 (virtual segments from 2D scintillator image → PMT or APD array)
Time resolution	60 - 5000 fps (CMOS), 2MS/s (PMT)
Resolution	2 – 5 keV in energy, 2.0° – 4.0° in pitch-angle
Availability	Only on request

* deduced from the Larmor radius and the pitch-angle mapped onto the FILD scintillator plane

* pitch-angle = $\arccos(v_{\parallel}/v)$

* Scintillator material: TG-Green or ZnS:Ag (2011, since 2019)



Scintillator-based fast-ion loss detector

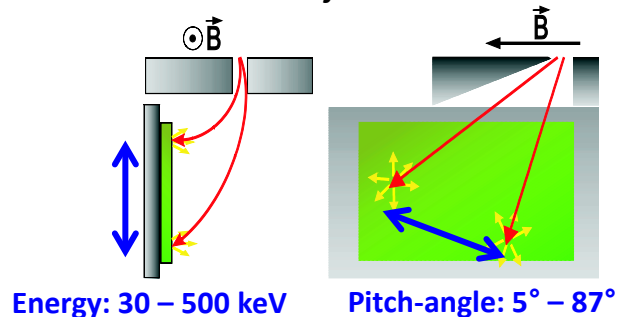
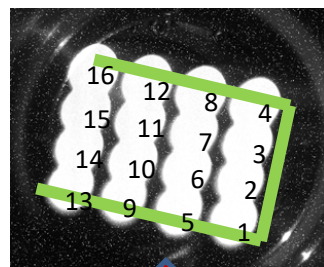
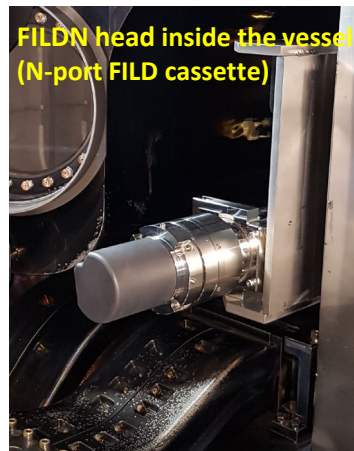


Figure courtesy of M. Garcia-Munoz



The region of observation on the scintillator, corresponding to the PMT channels

FILDN head inside the vessel (N-port FILD cassette)



Notes

- FILD-1 system has been removed from the C-port, and refurbished at the N-port (2018).
- APDCAM channels for the FILD-2 has not been uploaded to the MDSplus.

Name	System	Sampling	Unit	Description
\FIL1_INT:FOO or \FIL1_INT	FILD-1 CCD	125 - 250 fps (640 X 480)	a.u.	Time-dependent intensity record over whole scintillator plane (2011 - 2017)
\FILDN_INT	N-port FILD CCD	0.12 – 2 kfps (1024 X 1024)	a.u.	Time-dependent intensity record over whole scintillator plane, Photron Mini AX50: Max. 170 kfps /w 128x16 pixels (in 2018)
\FIL2_INT:FOO or \FIL2_INT	FILD-2 CCD	125 - 250 fps	a.u.	Time-dependent intensity record over whole scintillator plane
\FIL1_PMT01:FOO ... \FIL1_PMT16:FOO	FILD-1 PMT	0.1 - 2 MHz	V	Raw data of PMT signal for each segment are a of the scintillator plane (2011 - 2017) (2017 campaign: only\FIL1_PMT05:FOO is available.)
\FILDN_PMT01:FOO ... \FILDN_PMT16:FOO	N-port FILD PMT	0.1 - 2 MHz	V	Raw data of PMT signal for each segment are a of the scintillator plane (in 2018)
\FIL1_PMT01 ... \FIL1_PMT16	FILD-1 PMT	10 kHz	V	Down-sampling + smoothing (not available since 2015)

Notes

- Currently, uploading data to the MDS+ nodes is stopped temporarily. Post-processed data will be provided by the request.
- From 2018 campaign, N-port FILD system is operated. → \FILDN_INT, \FILDN_PMTXX:FOO are available instead of FILD-1 nodes.
- Energy and pitch-angle map is calculated separately on request, based on the magnetic equilibrium (3-D magnetic pitch at the FILD head) information.
- MDS node names of the FILD signals in 2011 or 2012 campaigns are different with the current one. Try '\FIL_INT' or '\FIL_INT:FOO' first for the 2011 & 2012 campaign data. If an error is encountered to acquire the MDS signal, please ask the contact persons

Full Name	Fast-Ion D α Spectroscopy
Contact	Junghee Kim (kimju@kfe.re.kr)
Measuring properties	• Fast-ion Dα intensity ($\propto$ fast-ion density) profile • Phase-space (Distribution of energy and pitch-angle) of confined fast-ions
Port assignment	J (tangential-view, 2019 -), J (oblique-view, 2025 -)
Channel information	2 arrays, 26 ch (FIDA01[bue-shift]: 18-19ch, FIDA02[red-shift]: 7-8ch)
Time resolution	10 msec
Resolution	Spatial: 2 – 5 cm
Availability	routinely available

* **Fast-ion phase-space** and **density profile** can be deduced by FIDASIM calculations combined with NUBEAM/TRANSP modelling on request.

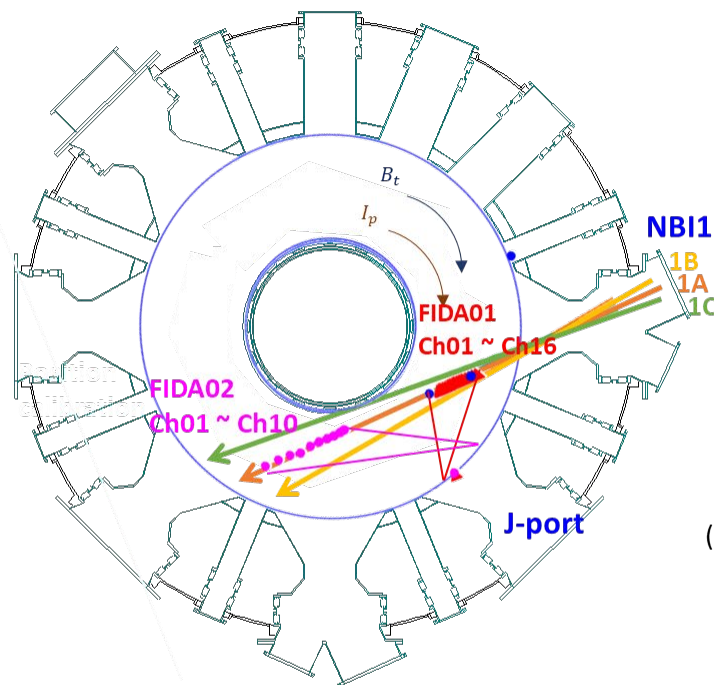
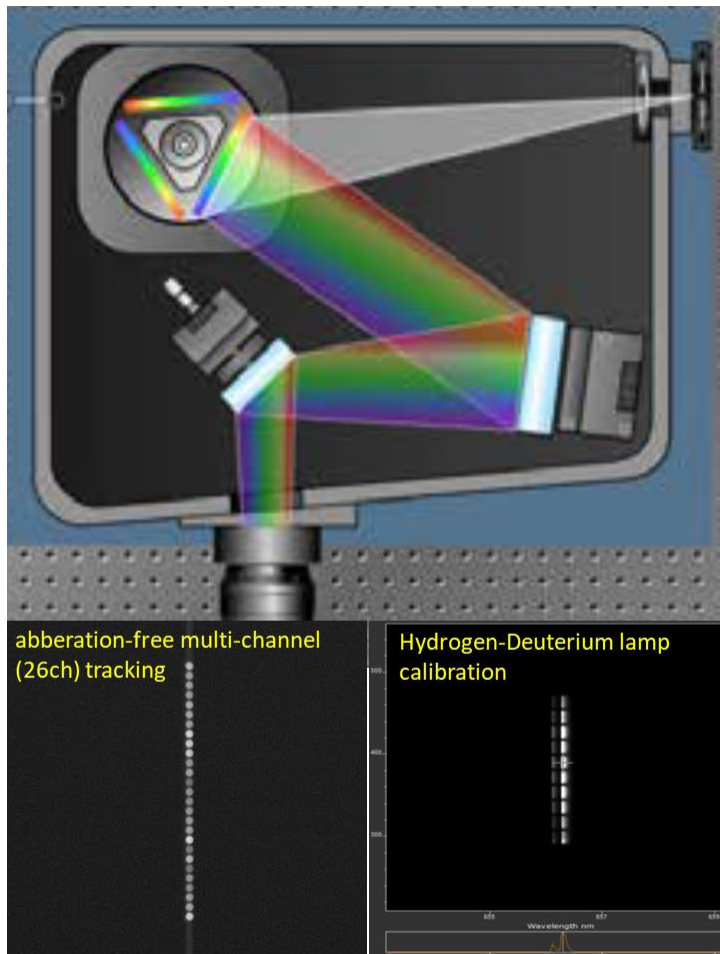
* In 2015 campaign, D spectra have been acquired by sharing MSE optics. In 2018, dedicated FIDA optics are installed at J-port (optical module #3 and #5).

* J-port Module #3: Blue-shifted FIDA (NB-1), J-port Module #5: Red-shifted FIDA (NB-1 or NB-2)

* Beam-modulation is essential. (minimum beam-off duration: 10msec (20 msec recommended))

* FIDA intensity profile data for the specific fast-ion energy band (e.g. 60 – 70 keV) will be acquired by OMFIT-FIDA module (HDF5 format).

Schmidt-Czerny-Turner Spectrograph (2021 ~)



FIDA array line-of-sights (two mid-plane tangential)
 FIDA01: blue-shifted, FIDA02: red-shifted

(2019) FIDA01

CH	R [cm]
1	224.71
2	220.28
3	217.00
4	213.11
5	209.52
6	206.69
7	203.90
8	201.03
9	198.79
10	196.32
11	193.75
12	191.28
13	189.32
14	186.93
15	184.93
16	182.92

(2019) FIDA02

CH	R [cm]
1	227.56
2	211.87
3	199.34
4	189.03
5	179.17
6	171.61
7	165.68
8	161.07
9	157.35
10	154.5

SXR array

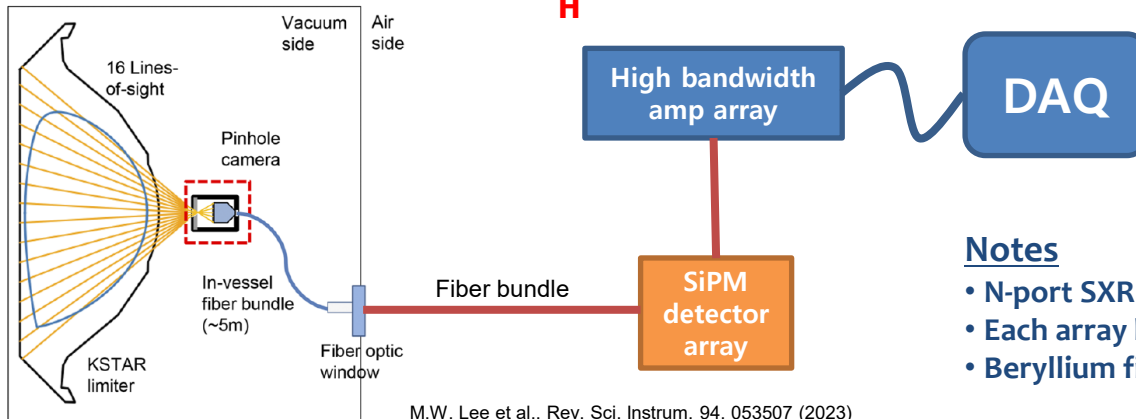
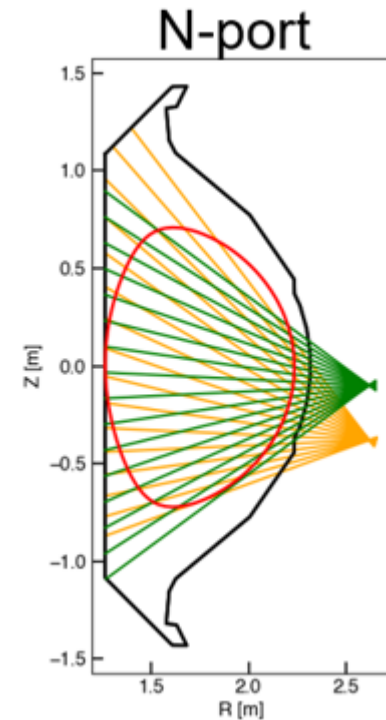
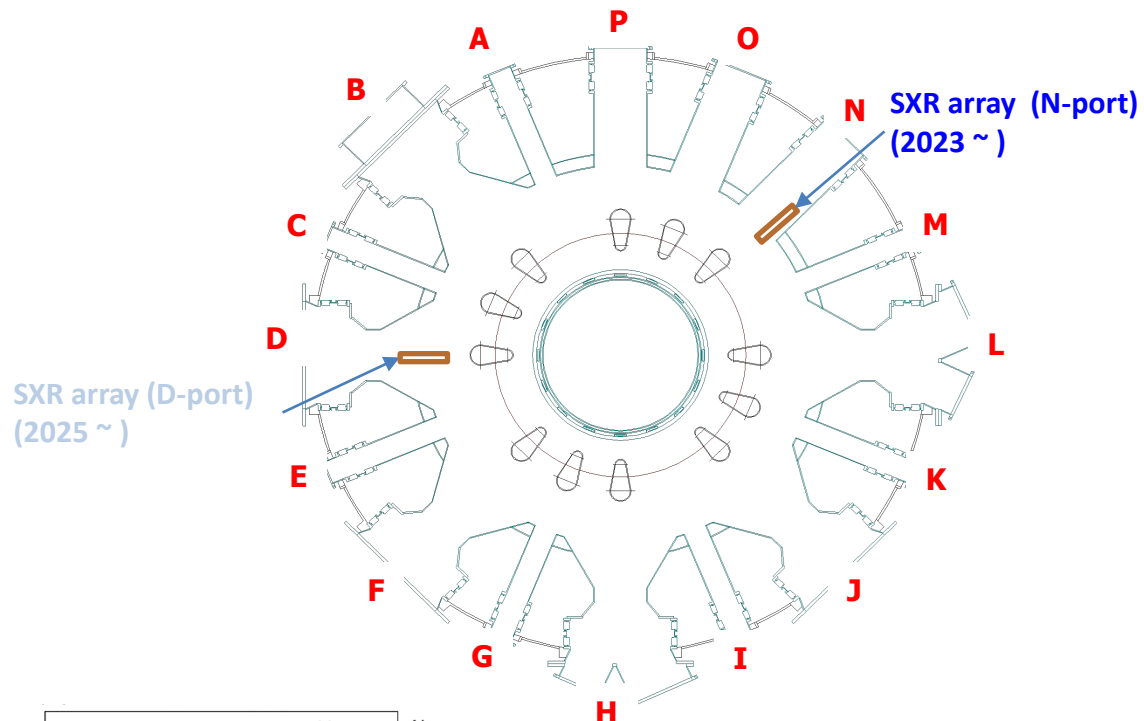
General Information

Full Name	Soft x-ray array
Contact	Junghee Kim (kimju@kfe.re.kr)
Measuring properties	• X-ray emission profile • Electron temperature/fluctuation profile • MHD mode activities • High-Z impurity contents
Port assignment	N (2023 -), N & D (2025 -)
Channel information	2 – 3 ch (2023 – Dec 2024), 16 – 64 ch (Jan 2025 –)
Time resolution	1 μ sec (1MS/s/ch)
Resolution	Spatial: 1 – 3 cm
Availability	routinely available (after Jan 2024)

- * KSTAR SXR array diagnostic system is based on the x-ray scintillation (YAG:Ce). (a.k.a. Optical SXR)
- * Nominal data type of the SXR array: HDF5
- * Data analysis tool will be provided in the OMFIT platform (OMFIT-SXR module) after Jan 2025.
- * Tomography reconstruction service will be provided in 2025 ~ .

SXR array

Diagnostics Layout



M.W. Lee et al., Rev. Sci. Instrum. 94, 053507 (2023)

Notes

- N-port SXR array: 16ch $\times$ 2 (2 arrays)
- Each array has a double layer (two-filter T_e measurement).
- Beryllium filter thickness in each array: 50 μm & 10 μm

Fusion Neutron Diagnostics

General Information

Full Name	Fusion Neutron Diagnostics
Contact	Youngseok Lee (yslee@kfe.re.kr)
Measuring properties	- Time-resolved Neutron emission rate • Micro-Fission chambers (U-235/U-238) • Micro-He3 counters • Diamond-based Thermal/Fast neutron detectors - D-D/D-T Fusion neutron Energy • Diamond-based neutron spectrometers
Port assignment	Near J-port
Channel information	Measurements at three locations (J, Lower vessel(basement), Entrance upper side) around the KSTAR tokamak
Time resolution	100 μ sec~100 msec (typical 1~10 msec.)
Resolution	-
Availability	routinely available (after Jan 2024)

Fusion Neutron Diagnostics

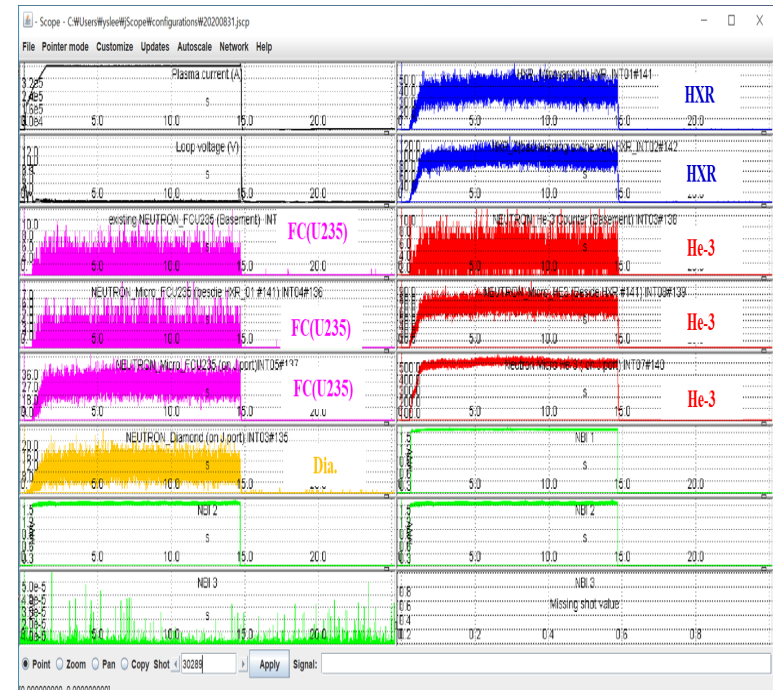
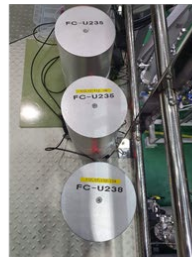
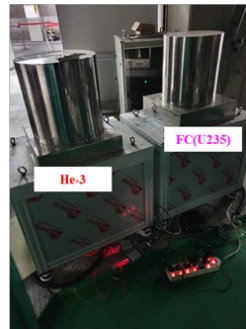
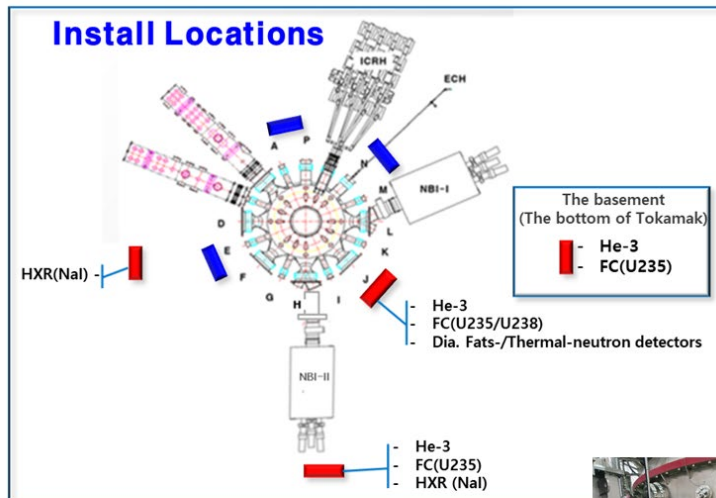
MDS+ Node Information

Name	System	Sampling	Unit	Description
\NTRN_INT01	Fission	100MHz	n/ms	Lower vessel (basement Level D2)
\NTRN_INT06	He-3	100MHz	n/ms	Lower vessel (basement Level D2)
\NTRN_INT05	Fission	100MHz	n/ms	Near J-port
\NTRN_INT08	He-3	100MHz	n/ms	Near J-port
\NTRN_INT04	Fission	100MHz	n/ms	Entrance upper side
\NTRN_INT07	He-3	100MHz	n/ms	Entrance upper side
\NTRN_INT03	Diamond-based detector	100MHz	n/ms	Near J-port

Fusion Neutron Diagnostics

Channels Layout

Measurements at three locations around the KSTAR tokamak



HXR Monitor

General Information

Full Name	Hard X-ray monitor
Contact	Youngseok Lee (yslee@kfe.re.kr)
Measuring properties	Thick target bremsstrahlung emission (Hard X-ray) & Energy spectra
Port assignment	on either side of the KSTAR tokamak
Channel information	2 system (forward & backward)
Time resolution	100 μ sec~100 msec (typical 1~10 msec.)
Spatial Resolution	-
Availability	routinely available

Notes

* Because of its data size, the HXR energy spectrum is saved only local PC.

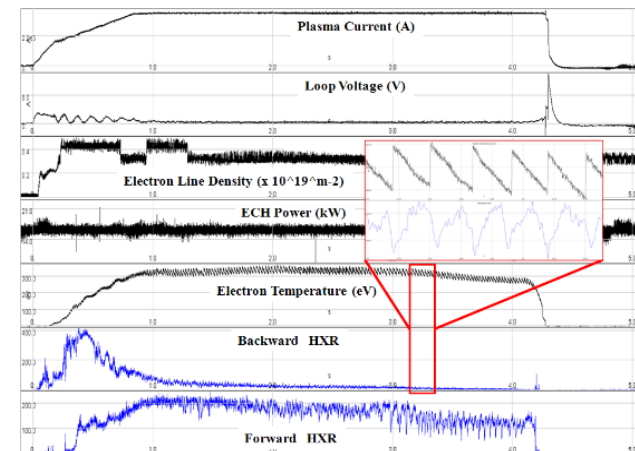
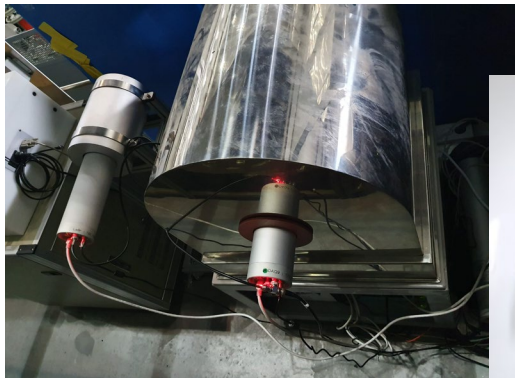
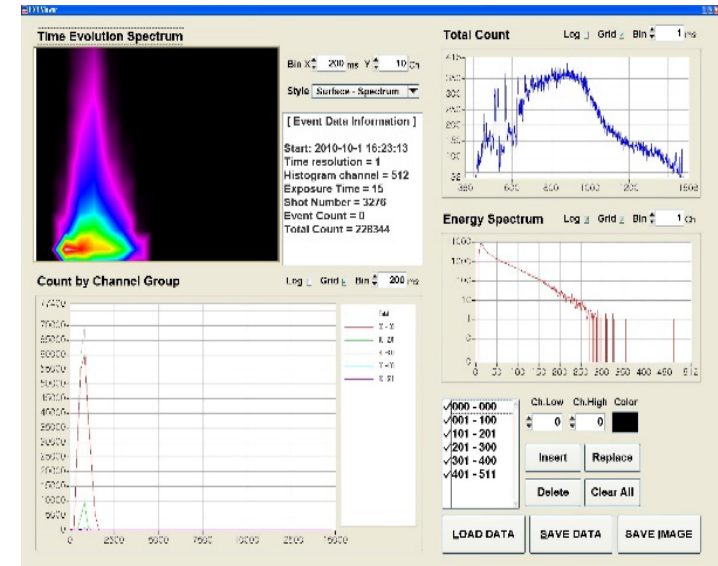
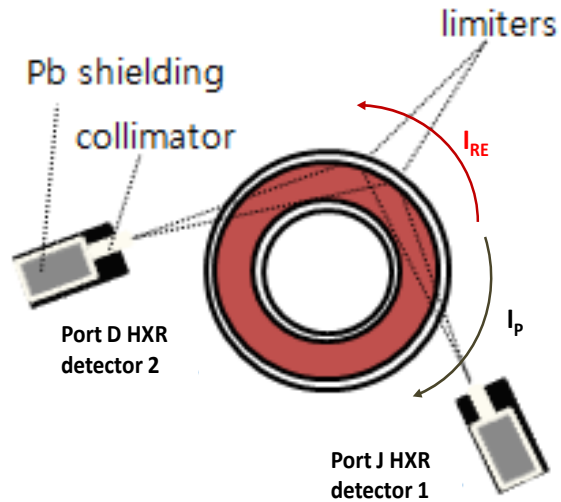
Name	Sampling	Unit	Description
\HXR_INT01			Backward HXR emission
\HXR_INT02			Forward HXR emission

Notes

- Measured HXR signals mean that runaway electrons collided with limiters.
- Measured HXR signals include the effects of neutron. Therefore, good care is needed when NBI is used.

HXR Monitor

Channels Layout

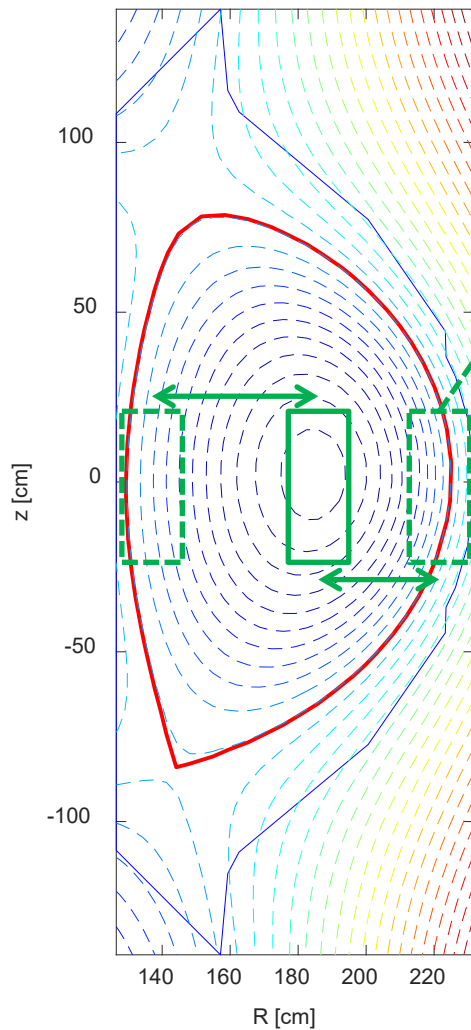


Full Name	Electron Cyclotron Emission Imaging (ECEI) system
Contact	J. Lee (jaehyun@kfe.re.kr)
Measuring properties	Electron temperature fluctuation ($\delta T_e / \langle T_e \rangle$)
Port assignment	G (1 st ECEI: dual array), H (2 nd ECEI: single array)
Channel information	24 (vertical) X 8 (radial) = 192 channels for 1 detector (ECEI consists of 3 detectors)
Measurement range	From the inboard edge to the outboard edge by changing (1) Bt, (2) LO frequencies from 78 to 103 GHz, and (3) flexible optical system
Time resolution	1 (typical) or 2 μ s
Spatial Resolution	1 X 1 ~ 2 X 2 cm ²
Availability	On request (need Bt for the system setup prior to experiment)

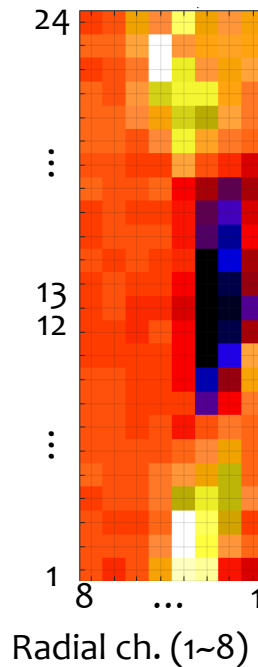
Notes

- A broadband ECE measurement system for precise structure and dynamics measurements can also operated upon request (0.5-6.5 GHz analog bandwidth, 256GB acquisition memory (~1.5 s)).
- For simple analysis, please use ‘fluctana’ software package on the iKSTAR server. Detailed image processing and spectra/statistics analysis take some time.

View position
(Poloidal cross section)



ECE image
Vertical ch. (1~24)

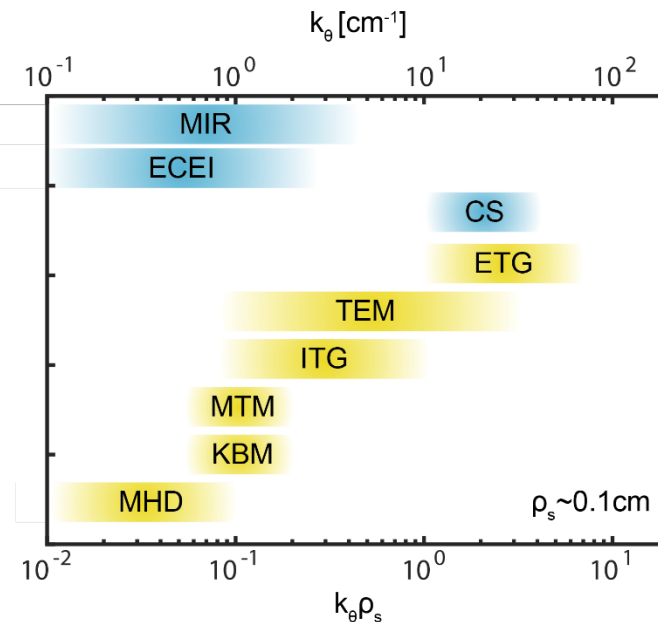
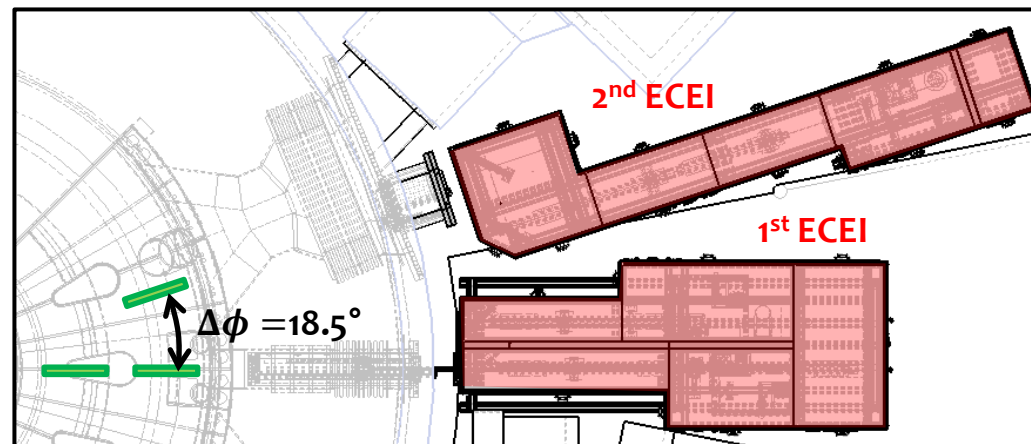


**Measurable
wave number
limits
(2018)**

$$k_r < 0.78 \text{ cm}^{-1}$$

$$k_\theta < 1.9 \text{ cm}^{-1}$$

View position
(Top view)



Name	Sampling	Unit	Description
\ECEI_GT0101:FOO ... \ECEI_GT0108:FOO	1 MHz	V	G-port Transmitted beam vertical channel 1 radial channel 1, ..., vertical channel 1 radial channel 8
...			
\ECEI_GT2401:FOO ... \ECEI_GT2408:FOO	1 MHz	V	G-port Transmitted beam vertical channel 24 radial channel 1, ..., vertical channel 24 radial channel 8
\ECEI_GR0101:FOO ... \ECEI_GR0108:FOO	1 MHz	V	G-port Reflected beam vertical channel 1 radial channel 1, ..., vertical channel 1 radial channel 8
...			
\ECEI_GR2401:FOO ... \ECEI_GR2408:FOO	1 MHz	V	G-port Reflected beam vertical channel 1 radial channel 1, ..., vertical channel 1 radial channel 8
\ECEI_HT0101:FOO ... \ECEI_HT0108:FOO	1 MHz	V	H-port Transmitted beam vertical channel 1 radial channel 1, ..., vertical channel 1 radial channel 8
...			
\ECEI_HT2401:FOO ... \ECEI_HT2408:FOO	1 MHz	V	H-port Transmitted beam vertical channel 1 radial channel 1, ..., vertical channel 1 radial channel 8

Notes

- For analysis and interpretation using ECEI data, be sure to consult a diagnostic expert.

Name	Sampling	Unit	Description
\ECEI_I_TF		kA	TF current
\GT_DPFREQ, \GR_DPFREQ, \HT_DPFREQ		GHz	Dichroic plate settings
\GT_LOFREQ, \GR_LOFREQ, \HT_LOFREQ		GHz	LO frequency settings
\GT_LENSZOOM, \GR_LENSZOOM, \HT_LENSZOOM		Cm	Zoom lens positions
\GT_LENSFOCUS, \GR_LENSFOCUS, \HT_LENSFOCUS		Cm	Focus lens positions
\GT_MODE, \GR_MODE, \HT_MODE			Harmonic modes (1: O-mode, 2: X-mode)

Notes

- For analysis and interpretation using ECEI data, be sure to consult a diagnostic expert.

RF spectrometer

General Information

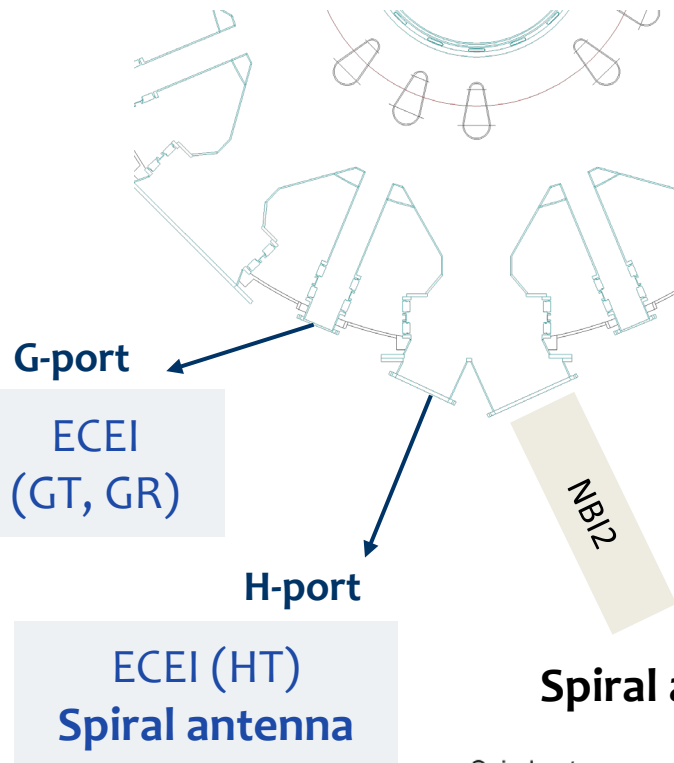
Full Name	Radio-frequency (RF) spectrometer
Contact	G.S. Yun (gunsu@postech.ac.kr), D.-K. Kim (dkwonkim@kfe.re.kr)
Measuring properties	Intensity of RF emission captured by spiral antennas
Port assignment	H (Large spiral antenna)
Channel information	Filter-bank system - Center freq (in MHz): 40,60,100,150,200,250,300,400,500,600,700,800. Bandwidth is 10% of center frequency.
Time resolution	1 MHz (filter-bank spectrometer), Up to 16GHz (fast digitizer)
Spatial Resolution	-
Availability	Filter-bank: Routinely available* Fast digitizer: On request*

*From 2024 campaign, raw signal from spiral antenna can be temporarily accessed with MDS+ nodes of ECEI: ECEI_GR2301-08:FOO and ECEI_GR2401-08:FOO.

*Fast digitizer: acquisition available up to 1.6s per shot (0.2 s per segment × 8 segments). As the sampling rate is high, the raw data may not be provided due to high file capacity; only the spectrograms may be provided.

RF spectrometer

Channels Layout



Two types of acquisition:

- **Filter-bank:** overall RF activities during the entire discharge duration.
- **Fast digitizer:** high-resolution (16 Gsa/s) information for fast events such as ELM crash. (1.6 s acquisition per shot)

Spiral antenna:

Spiral antenna
0.1–0.8 GHz
VSWR ~ 2 (Typ.)



Bandpass filter
0.1–0.85 GHz



0.1–1 GHz,
30 dB
Low noise amp.



**Filter-bank
RF spectrometer**
(16 ch., 0.03–0.8 GHz,
500 kSa/s typ.)



Fast digitizer
(5 GSa/s typ.
1 GHz bandwidth)

Full Name	Microwave Imaging Reflectometer
Contact	Woochang Lee (wclee@kfe.re.kr), Dong-jae Lee (djlee124@kfe.re.kr)
Measuring properties	Electron density fluctuations with poloidal wavenumber up to 3 cm^{-1} Poloidal rotation velocities of fluctuations Poloidal (and radial) correlation length of fluctuations
Port assignment	E-port from 2018 (G-port until 2017)
Channel information	4 x 16 (radial and poloidal)
Measurement range	Measurement radii can be positioned from the core to the outer edge by changing (1) Bt or (2) ne or (3) probing frequencies from 78 to 96 GHz.
Time resolution	Typically $1 \mu\text{s}$ (at 1 MS/s for 30 second acquisition) Minimally $0.5 \mu\text{s}$ (at 2 MS/s for 15 second acquisition)
Spatial Resolution	Radial: 1 to 10 cm (primarily depending on electron density profile) Poloidal: 0.6-0.9 cm (poloidal spot size $\sim 0.9 \text{ cm}$)
Availability	On request (need Bt and ne for system setup prior to the experiment)

Notes

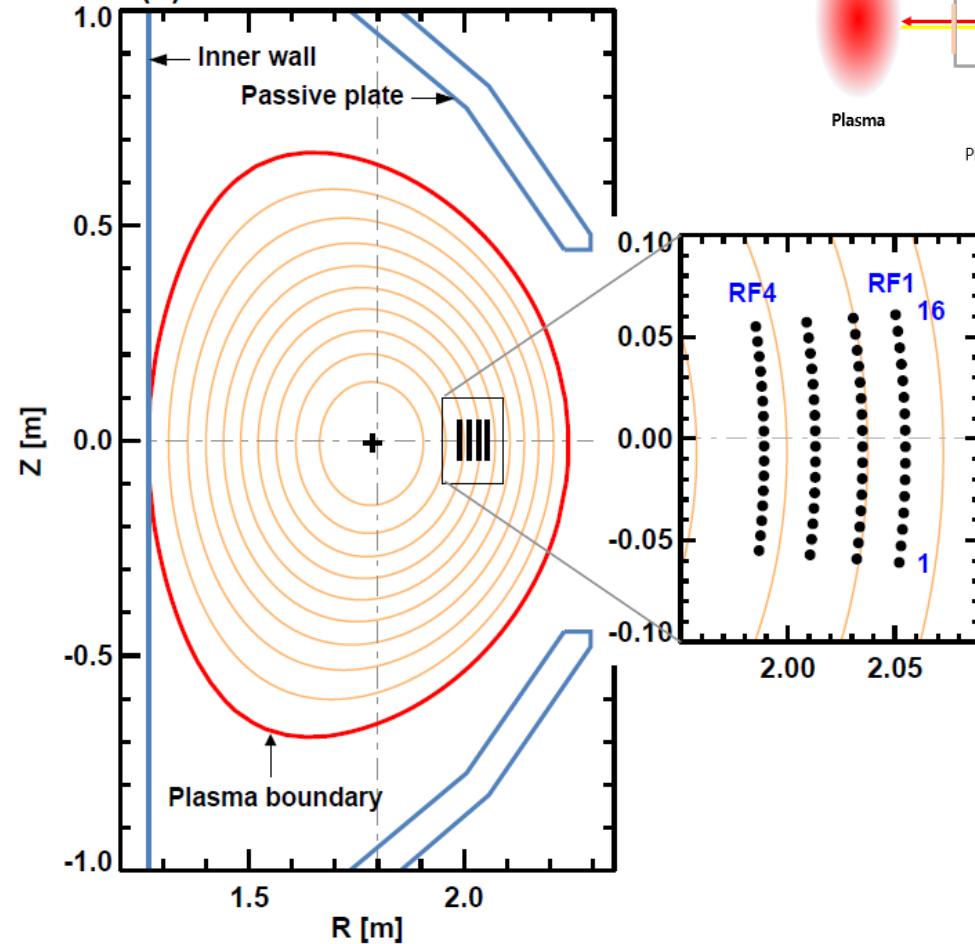
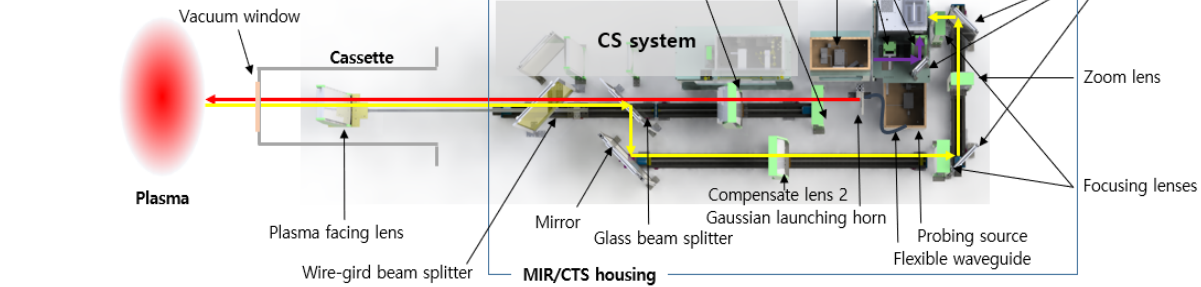
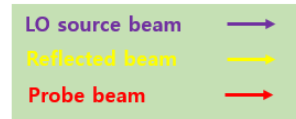
- MDS+ node is not assigned but MIR data can be delivered on your request.
- This complicated system is featured by (1) tunable 4 frequency X-mode probe beams, (2) 16-channel detector array, (3) heterodyne detection system (providing I/Q signals), and (4) imaging optical system.

MIR

Channels Layout

Example of
locations of
the MIR channels

(c) #18332 :: EFIT at 1.4 s

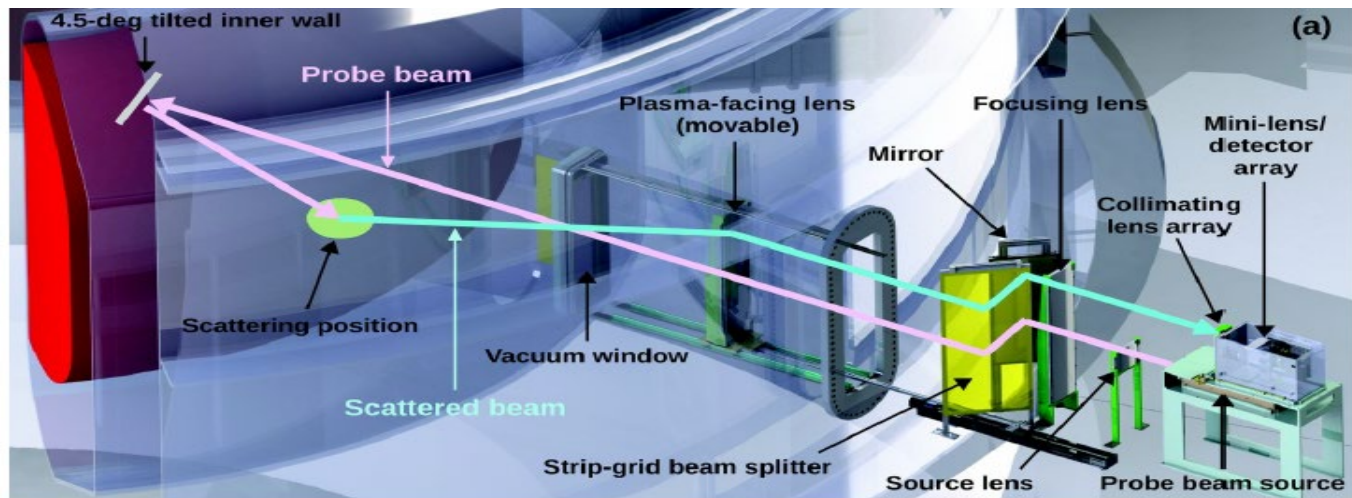
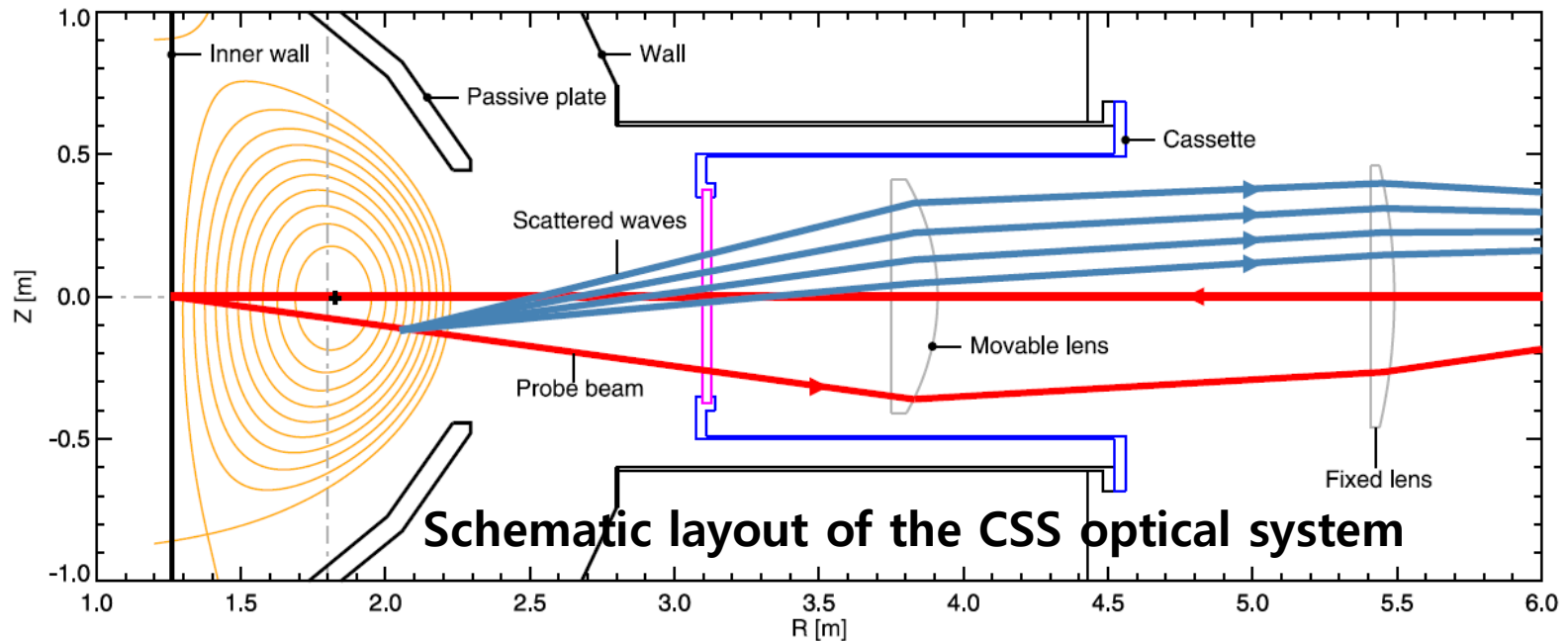


Name	Sampling	Unit	Description
\MIR_I01_01 ... \MIR_I01_16	1 MHz		In-phase signal of radial channel 1, vertical channel 1~16
\MIR_Q01_01 ... \MIR_Q01_16	1 MHz		Quadrature signal of of radial channel 1, vertical channel 1~16
\MIR_I02_01 ... \MIR_I02_16	1 MHz		In-phase signal of radial channel 2, vertical channel 1~16
\MIR_Q02_01 \MIR_Q02_16	1 MHz		Quadrature signal of of radial channel 2, vertical channel 1~16
\MIR_I03_01 ...	1 MHz		...
\MIR_Q03_01 ...	1 MHz		...
\MIR_I04_01 ... \MIR_I04_16	1 MHz		In-phase signal of radial channel 4, vertical channel 1~16
\MIR_Q04_01 ... \MIR_Q04_16	1 MHz		Quadrature signal of of radial channel 4, vertical channel 1~16

Notes

- When you make frequency spectra with MIR data, please keep in mind that the MIR signal of each channel consists of the real (in-phase signal) and imaginary (quadrature signal) components. So, you have to make the frequency spectra of complex signals.

Full Name	Collective Scattering System
Contact	Woochang Lee (wclee@kfe.re.kr), Dong-jae Lee (djlee124@kfe.re.kr)
Measuring properties	Electron density fluctuations with four discrete poloidal wavenumbers from 14 to 26 cm^{-1}
Port assignment	E-port
Channel information	4 poloidal wavenumber channels
Measurement range	Measurement radii can be positioned from the core to the outer edge
Time resolution	Typically 0.1 μs (at 10 MS/s for up to 17 seconds)
Spatial Resolution	Radial spot size: ~ 10 cm Poloidal spot size: ~ 2.3 cm
Availability	On request (need Bt and ne for system setup prior to the experiment)



Name	Sampling	Unit	Description
\CSS_I01	10 MHz		In-phase signal of channel 1
\CSS_Q01	10 MHz		Quadrature signal of channel 1
\CSS_I02	10 MHz		In-phase signal of channel 2
\CSS_Q02	10 MHz		Quadrature signal of channel 2
\CSS_I03	10 MHz		In-phase signal of channel 3
\CSS_Q03	10 MHz		Quadrature signal of channel 3
\CSS_I04	10 MHz		In-phase signal of channel 4
\CSS_Q04	10 MHz		Quadrature signal of channel 4

Notes

- When you make frequency spectra with CSS data, please keep in mind that the CSS signal of each channel consists of the real (in-phase signal) and imaginary (quadrature signal) components. So, you have to make the frequency spectra of complex signals.

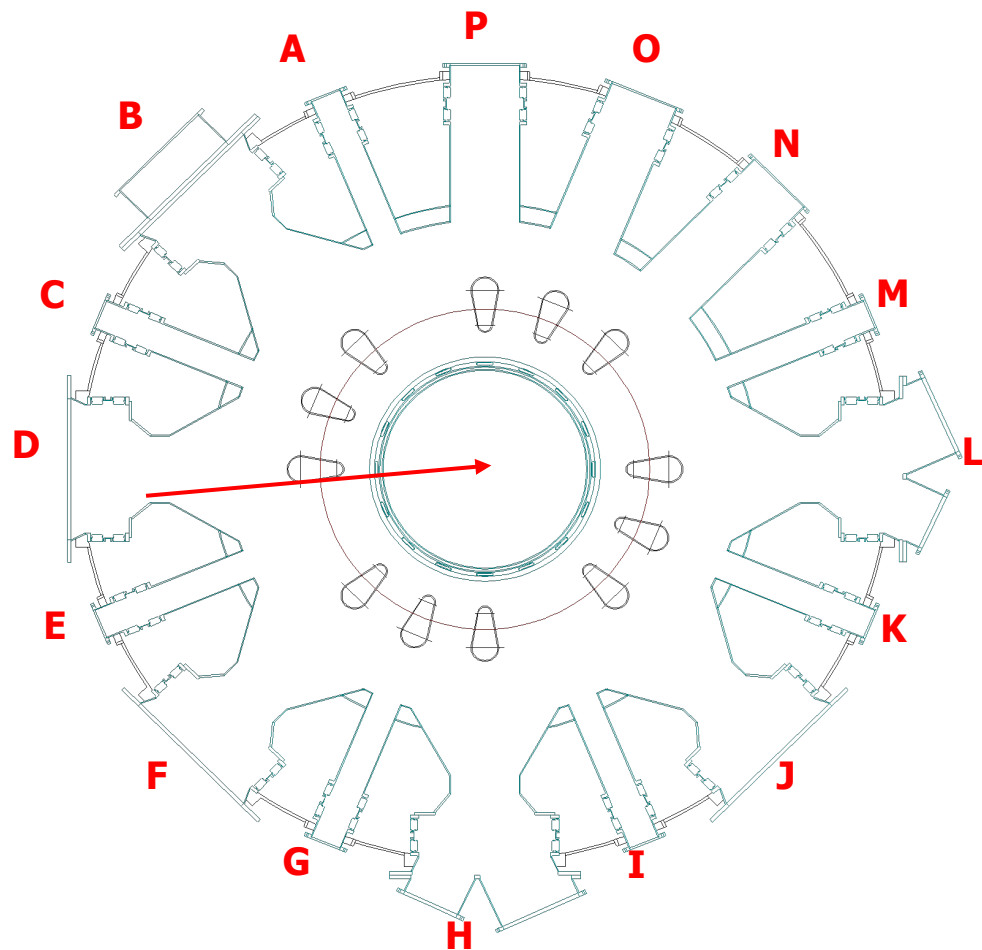
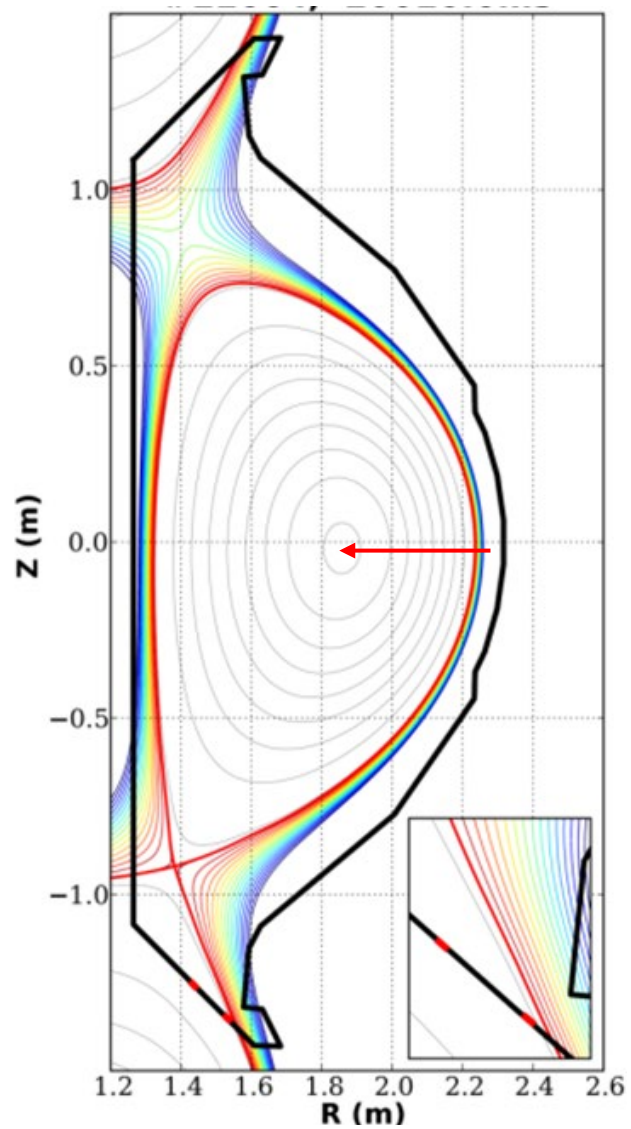
Edge reflectometer

General Information

Full Name	Edge reflectometer
Contact	Seong-Heon Seo (shseo@kfe.re.kr)
Measuring properties	Plasma density profile
Port assignment	D
Channel information	Q (30-50 GHz), V (50-75 GHz), W (75-110 GHz)
Time resolution	50 μ s for Trigger mode, 20 ms for Continuous mode
Spatial Resolution	1 mm
Operation Mode	1. Continuous : 20 ms time resolution for the whole discharge time
	2. Trigger : 50 μ s time resolution for the 8 segments of 100 ms
	Normally operated in the continuous mode. Operated in the trigger mode with request of 8 triggering times before shot

Edge reflectometer

Channels Layout



Edge reflectometer

MDS+ Node Information

Name	Sampling	Unit	Description
\r_ref		m	Radial position 2D array [radius, time index]
\ne_ref		m ⁻³	Density 2D array[density, time index]

Example

- Read data from 1 s to 1.02 s

```
IDL> s = mdsvalue('SetTimeContext($,$)', 1.0, 1.02)
```

```
IDL> t = mdsvalue('dim_of(Wr_ref)')
```

```
IDL> r = mdsvalue('Wr_ref')
```

```
IDL> n = mdsvalue('Wne_ref')
```

- Plot density profile at 1.001 s, that is 1000 μ s (= 20 x 50 μ s) after the first data

```
IDL> iplot, r[*,21], n[*,21]
```

note

- Processed data until 9 Nov. 2017

11499-11503, 11506-11510 , 19340

Edge reflectometer

MDS+ Node Information

- IDL routines for 3D plots

```
IDL> y = kstar_read2(19340, 4.09, 4.11)
```

```
IDL> plot_density, y, 19340, /D3, STEP=2, NRANGE=2.5, VERTEX=1.9
```

Options

/D3 : 3D line plot

/SURFACE : 3D surface plot

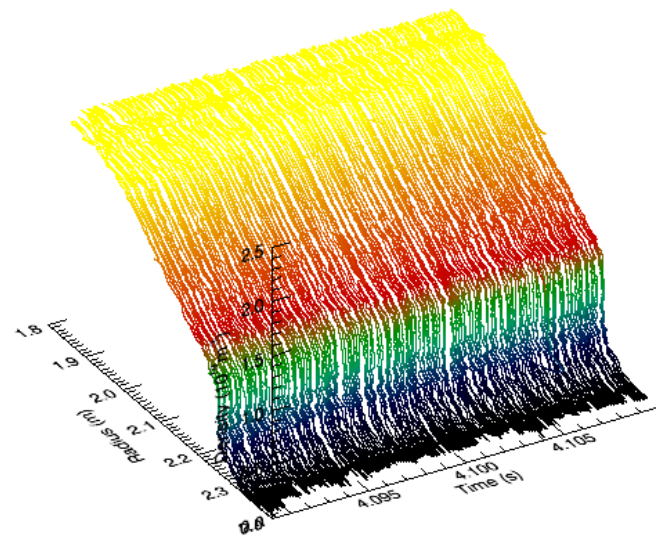
/CONTOUR : contour plot

STEP = the interval for plotting (defaults = 1, too slow)

NRANGE = the maximum value for z-axis (10^{19} m^{-3})

VERTEX = the scale factor for the color table of 'BLUE/GREEN/RED/YELLOW' (10^{19} m^{-3})

1.9 scales as <0 is BLUE, $1.9 \times 10^{19} \text{ m}^{-3}$ is YELLOW>



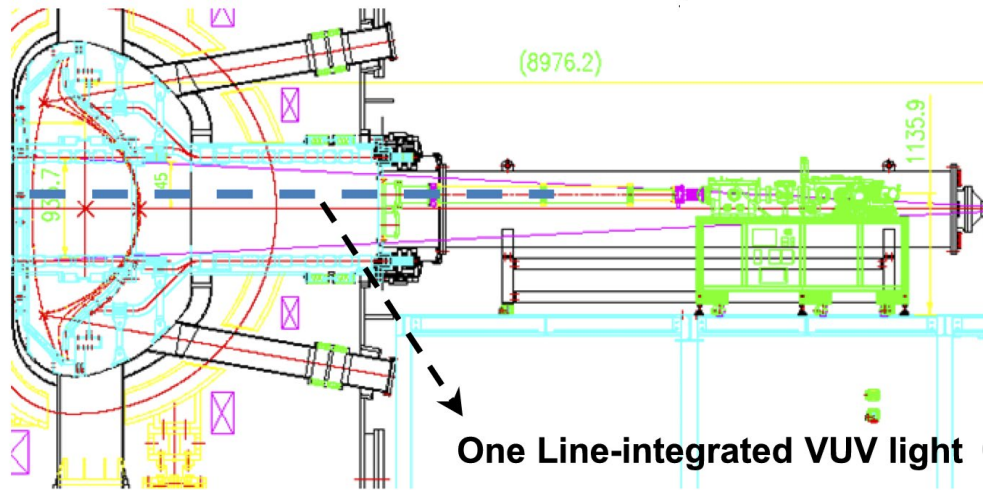
VUV core survey spectrometer

General Information

Full Name	Vacuum ultraviolet (VUV) core survey spectrometer
Contact	Changrae Seon (crseon@kfe.re.kr), Junhyeok Yoon (jhyoon97@kaist.ac.kr)
Measuring properties	Impurity line emission
Port assignment	F-port
Channel information	Channel #1 ($\lambda = 12.55 - 31.60$ nm) Channel #2 ($\lambda = 29.53 - 50.55$ nm)
Measurement range	LoS passing through a core plasma region
Time resolution	30.5 ms (channel #1), 40.8 ms (channel #2)
Spatial Resolution	-
Availability	Routinely available

VUV core survey spectrometer

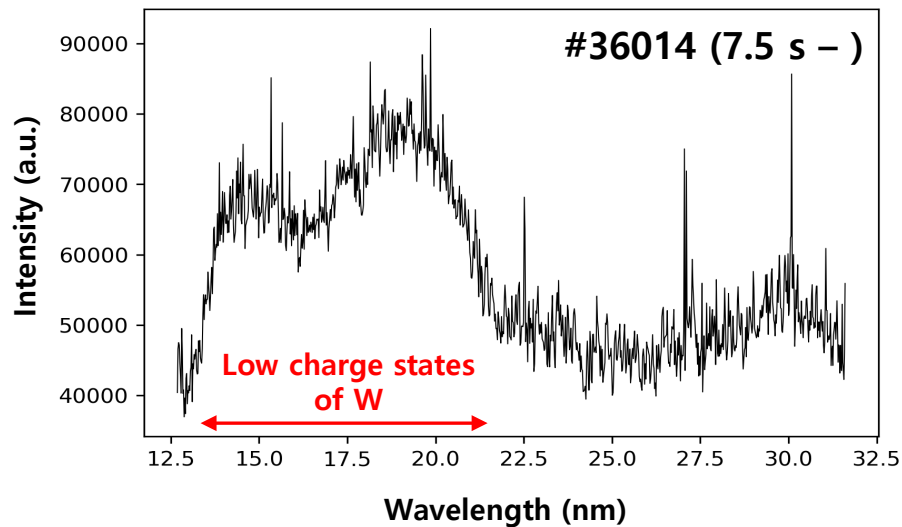
Channels Layout



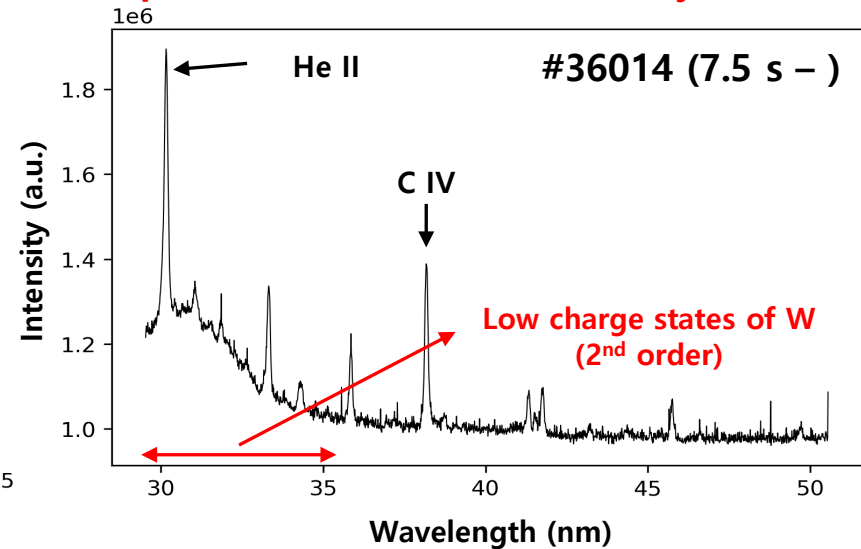
One Line-integrated VUV light (LoS of VUV core survey)

C. R. Seon *et al.*, Rev. Sci. Instrum. **88** 083511 (2017)

Spectrum from VUV core survey (ch. #1)



Spectrum from VUV core survey (ch. #2)



VUV core survey spectrometer

MDS+ Node Information

Name	Sampling	Unit	Description
\VUV_HeII		a.u.	Line-integrated He II line emission ($\lambda = 30.18$ nm)
\VUV_CIV		a.u.	Line-integrated C IV line emission ($\lambda = 38.17$ nm)
\VUV_FeXVI		a.u.	Line-integrated Fe XVI line emission ($\lambda = 35.86$ nm)

Name	Sampling	Unit	Description
\VUV_SV1_INFO:FOO			Channel #1 information → [0, 1024, 1, time resolution (s), frame #]
\VUV_SV2_INFO:FOO			Channel #2 information → [0, 2048, 1, time resolution (s), frame #]

Notes

- Additional nodes of impurity line emission will be added.
- If data is not available, please contact the contact person.

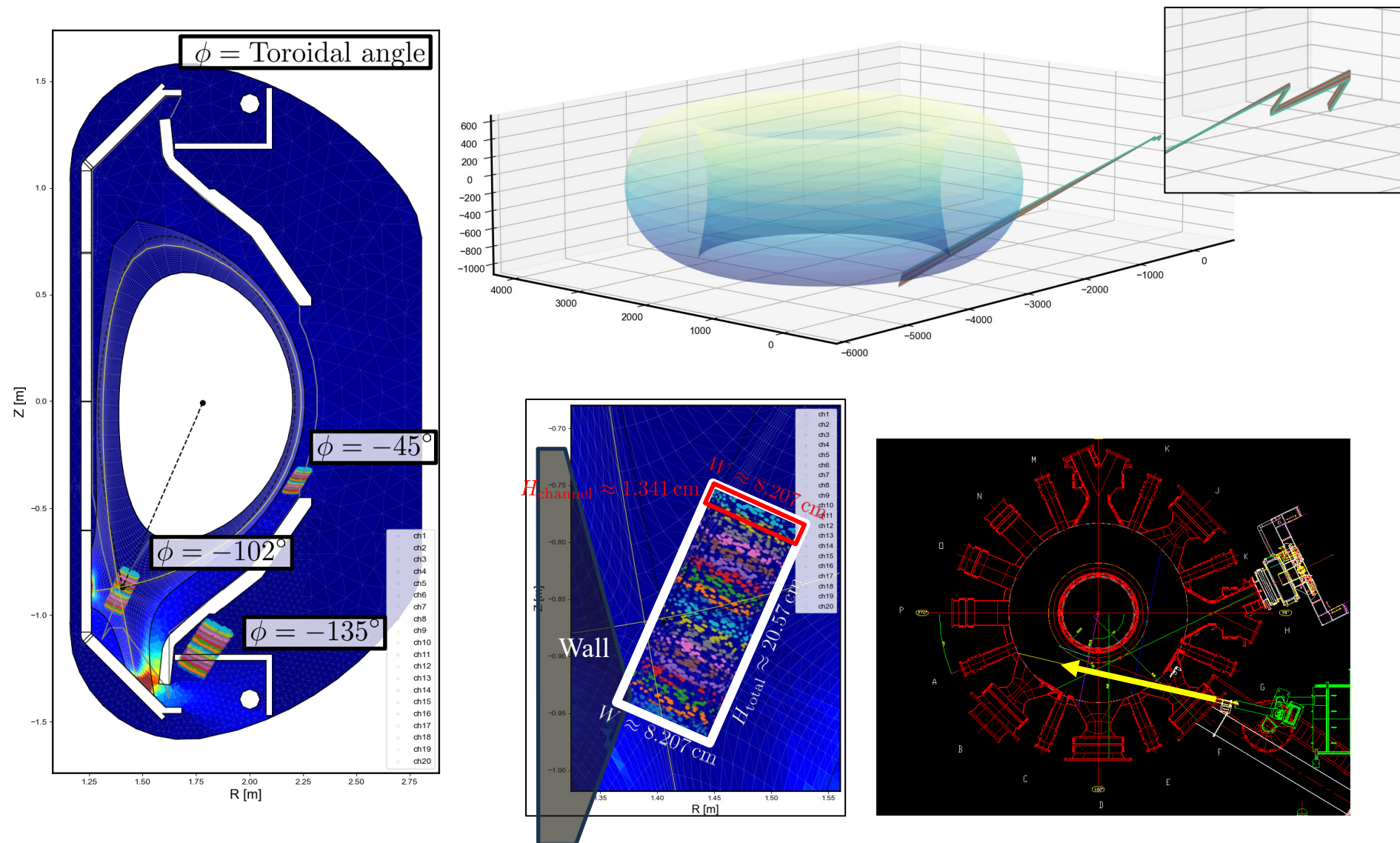
Lyman alpha diagnostics (KLAMA)

General Information

Full Name	KSTAR Lyman Alpha Measuring Apparatus (KLAMA)
Contact	Kwan-chul Lee (kclee@kfe.re.kr), Jae-seok Lee (jslee@kfe.re.kr , jaeseok.lee@kaist.ac.kr)
Measuring properties	Neutral Deuterium atom density
Port assignment	F-port
Channel information	Channel #1-20
Measurement range	LoS passing through an edge plasma region
Time resolution	Raw data: 200kHz (5 μ s)
Spatial Resolution	20 cm Around X-point along the radial direction towards the magnetic axis
Availability	Unavailable; Under commissioning

Lyman alpha diagnostics (KLAMA)

Channels Layout



Lyman alpha diagnostics (KLAMA)

MDS+ Node Information

Name	Sampling	Unit	Description
\KLAMA01:FOO ... \KLAMA20:FOO	200 kHz	V	Line-integrated Lyman alpha brightness signal from each KLAMA channel at F port
\KLAMA_NE		/m ³	Electron density profile evolution used for neutral density inference (Not available)
\KLAMA_TE		eV	Electron temperature profile evolution used for neutral density inference (Not available)
\KLAMA_NN		/m ³	Neutral deuterium atom density profile evolution (Not available)

Notes

- Only raw data is available which is measured for commissioning.
- After the commissioning, the neutral density profile will be available.